# Selling Mechanisms under Misspecified Learning

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**Abstract**

Buyers in a repeated selling mechanism do not observe its allocation rule. Each holds a possibly misspecified parametric model of his win probability, bids optimally under his current belief about the model's parameter, and learns by Bayesian updating under that model from allocation outcomes alone. The mechanism generates those outcomes, so it shapes what buyers come to believe. For a given model and belief, we ask when there exists a mechanism that sustains that belief while buyers keep learning, and how such a mechanism does it. Such a mechanism exists when the win probability predicted by the model responds strongly enough to a change in belief, relative to how bidding behaviour responds. We give sufficient conditions of this kind for convergence from every initial belief with a single buyer, and for convergence from nearby beliefs with two buyers, where feasibility ties the win probability each faces to the other's bids. To extend the basin of attraction beyond nearby beliefs, we work with discrete valuations and give a computational criterion in the form of a linear program, whose strictly feasible solutions yield mechanisms that sustain the belief from every more optimistic belief.

# 1 Introduction

Online marketplaces and advertising platforms routinely operate auctions whose details remain opaque to participants. Google's Performance Max product has been described by advertisers as "the blackest black box" they've encountered, providing no breakdown by price, ad format, media channel, or creative elements used.[1] Traditional auction houses like Christie's and Sotheby's have long kept reserve prices confidential, with bidders uncertain whether they compete against genuine bidders, the seller, or the house itself.[2]

These observations pose a fundamental challenge for mechanism design. Standard theory assumes that buyers respond to the mechanism itself, observing its rules and optimising accordingly. A buyer who does not observe the mechanism cannot respond to it directly. He responds instead to his own model of what the mechanism might be, and such a model need not be correct. When buyers respond to imagined mechanisms rather than the actual one, what the choice of mechanism accomplishes becomes an open question.

We consider a setting in which buyers who hold misspecified models of their environment interact repeatedly in a selling mechanism. Each buyer enters with a parametric model of how his bid maps into a probability of winning, a relationship that folds together the allocation rule and whatever else lies beyond his view, and after each round he observes only whether he won. The buyer acts on his model, but the feedback he learns from is generated by the actual mechanism. The mechanism therefore holds a measure of influence over what he comes to believe, exercised through outcomes alone. Experience, which might be expected to unsettle a mistaken belief, can instead confirm it.

Consider, for instance, a buyer who participates repeatedly in what he takes to be a first-price sale with an unknown reserve price. His model says the auction awards the good to the highest bid above some threshold, and the only open question is where the threshold sits. His belief is his current assessment of where the threshold sits. Each round he bids, observes whether he won, and updates his belief. The actual allocation rule need not resemble a reserve price mechanism at all. To keep the buyer convinced that the threshold sits at a given level, the rule only needs to produce wins and losses that, viewed through his model, point to that level. Every observation the buyer treats as evidence about the reserve price is an outcome the rule produced.

For a given model and belief, we ask when there exists a selling mechanism that sustains that belief while buyers keep learning, and how such a mechanism does it. Each buyer holds that model and bids optimally under whatever he currently believes, and the mechanism is the unknown.[3] Nothing is lost by posing the question in terms of beliefs rather than bidding behaviour, since a mechanism that sustains a belief also sustains the bidding that belief dictates. In an example in Section 5, the bids induced by an incorrect belief can earn the seller more than the optimal auction earns from a buyer who understands it.

The traditional responses to opacity fix beliefs by assumption. At one extreme, buyers hold rational expectations, correctly anticipating the mechanism despite not observing it, which restores

[1] "Meet Performance Max, The Blackest Black Box Of All Google Ad Products," AdExchanger, December 17, 2022.

[2] "Legislators Seek to Stop 'Chandelier Bidding' at Auction," ARTnews, August 10, 2012.

[3] The mechanism is the seller's to choose, so the relevant question is when there exists a mechanism that sustains a given belief, and which mechanism does so. The opposite exercise, characterising the beliefs that arise under an arbitrary mechanism, is generally challenging.

the link between mechanism and behaviour and reduces opacity to a loss of commitment power, as in the unobserved mechanisms of Li and Peters (2023) and the credible mechanisms of Akbarpour and Li (2020). At the other extreme is ambiguity aversion, where buyers expect the worst mechanism consistent with what they observe, as in the ambiguous mechanisms of Di Tillio et al. (2017). The latter can be read as a misspecification adopted by assumption, fixed at the pessimistic extreme before play. In practice, misspecification is unlikely to take either extreme form. Buyers hold imperfect models of their strategic environment and refine their beliefs through repeated interaction, and we leave the form of the misspecification free and let learning discipline it.[4]

A mechanism that sustains a given belief exists when the win probability predicted by the buyer's model responds strongly enough to a change in belief, relative to how bidding behaviour responds to the same change. Consider a mechanism under which buyers who hold the belief win, at every bid, exactly as often as they predict, so that their experience confirms the belief. Suppose now that buyers share a belief more optimistic than this one, and that a more optimistic buyer bids less aggressively. At any given bid, each buyer then faces weaker bids from his opponent, and wins at least as often as he would at the belief being sustained. If his model's prediction at that bid has risen by more than this change in bidding has raised his win probability, he wins less often than he expects and revises back toward the belief being sustained. When the same comparison holds with the directions reversed, buyers who share a more pessimistic belief win more often than they expect and revise back as well. We call the feedback a mechanism generates corrective when it has both properties, and Proposition 1 shows that a mechanism generating corrective feedback sustains the belief.

With a single buyer the win probability he faces depends on his own bid alone, and no belief moves it. It is then enough that his predicted win probability be higher at more optimistic beliefs and lower at more pessimistic ones. Proposition 2 shows that under this condition on the model a mechanism determined by the model and the belief being sustained draws every initial belief to it.

With two buyers, one buyer's win can be the other's loss, so the win probability each faces moves with the other's bids and therefore with the other's belief. Proposition 3 shows that the belief can be sustained locally when the response of a buyer's predicted win probability to a change in belief is large relative to that of his opponent's bids. With discrete valuations, Proposition 4 gives a finite linear program which, when strictly feasible, yields a mechanism that sustains the belief from every more optimistic belief.

A growing literature examines asymptotic belief convergence when agents learn from experience using misspecified models, including Esponda and Pouzo (2016), Esponda et al. (2021), Fudenberg et al. (2021), and Heidhues et al. (2021), with Bohren and Hauser (2025) providing a review. That literature treats objective reality as an exogenous primitive and seeks joint conditions on the agent's model and the environment under which beliefs converge. Proposition 1 is a condition of this kind, but we do not take the whole environment as given. The mechanism shapes what the buyer observes and learns from, and we ask when there exists a mechanism that generates corrective feedback. The convergence arguments build on the martingale framework of Frick et al. (2023), similar in spirit and modified in technique. The appendix delimits exactly which of their steps carry over and which are re-derived. We take the buyer's model as given and do not address model selection, the subject of He and Libgober (2025) and Ba (2025). A mechanism that

[4] Jehiel and Mierendorff (2024) analyse a different channel, design under data-driven misspecification, where buyers draw incorrect inferences from limited data on correlated values.

generates corrective feedback produces outcomes that match the model's predictions exactly at the belief it sustains, and therefore satisfies Ba's criterion for robust model selection.

Section 2 sets out the model and the notions of stability. Section 3 contains the main analysis, treating first a single buyer and then two buyers learning simultaneously. Section 4 exhibits convergence and its failure at computed instances. Section 5 takes up revenue, robustness, and the limits of the approach, and Section 6 concludes. Proofs are collected in the appendix.

# 2 Model and Preliminaries

Two buyers, indexed $i \in \{1,2\}$, participate in a repeated selling mechanism over an infinite horizon. Each period, each buyer draws a private valuation, submits a bid, observes whether he wins, and updates his belief.

Buyers know only the payment rule, which is first-price, so that a buyer pays his bid if he wins and nothing otherwise. A buyer does not observe the allocation rule. Instead, each buyer holds a subjective model of his win probability, which he treats as correctly specified, and updates his belief through misspecified Bayesian learning on his sequence of outcomes. The true allocation rule bears no necessary relation to that model. For a given model and belief, we ask when there exists an allocation rule that sustains that belief. All results are statements about this selling environment, with the payment rule fixed throughout, and such generality as the framework carries enters through the flexibility of the subjective model.

## 2.1 Objective Environment

Buyer $i$ has private valuation $v_i \in V_i = [\underline{v}_i, \bar{v}_i]$ drawn each period according to distribution $F_i$ with continuous density $f_i > 0$, independently across periods and buyers. The payment rule is first-price. The true allocation rule is a mapping from the profile of submitted bids $\boldsymbol{b} = (b_1, b_2)$ to a distribution over allocation outcomes $\boldsymbol{y} = (y_1, y_2) \in \{0,1\}^2$, and is fixed at the start of time. We write $\boldsymbol{q}(\boldsymbol{y} \mid \boldsymbol{b})$ for the probability of outcome $\boldsymbol{y}$ under bid profile $\boldsymbol{b}$, and $q^i(b_i, b_{-i})$ for the marginal probability that buyer $i$ obtains the good. We call $\boldsymbol{q}$ feasible if $\boldsymbol{q}((1,1) \mid \boldsymbol{b}) = 0$ at every bid profile. The marginals of a feasible rule satisfy

$$q^i(b_i, b_{-i}) + q^{-i}(b_{-i}, b_i) \leq 1 \qquad \text{for all bid profiles } (b_i, b_{-i}), \tag{1}$$

and any marginals satisfying (1) are the marginals of some feasible rule. Feasibility is also where the two buyers' learning problems meet. Because the good is scarce, one buyer's win forecloses the other's, coupling the outcomes the two buyers learn from. The feedback the mechanism can supply to one buyer therefore constrains what it can supply to the other.

## 2.2 Subjective Environment

Rather than knowing the true allocation rule, each buyer holds a parametrised family of subjective win probabilities. We denote buyer $i$'s subjective win probability by $g^i(b, v, \theta)$, where $b$ is his bid, $v$ his valuation, and $\theta$ a parameter lying in a compact set $\Theta_i \subseteq \mathbb{R}$. The family $\{g^i(\cdot, \cdot, \theta)\}_{\theta \in \Theta_i}$ constitutes buyer $i$'s *model*, and a distribution $\mu_i \in \Delta\Theta_i$ over parameters constitutes his *belief*. The parameter is ordered, with a higher $\theta$ describing a systematically tougher environment in which

the model predicts fewer wins at every bid and valuation. (R2) below imposes this ordering. We denote the degenerate belief on $\theta$ by $\delta_\theta$ or $\theta$ when there is no risk of confusion. We write $\theta_i^*$ for buyer $i$'s target, the parameter value on which his belief is to settle.

The family $g^i$ is a reduced-form model of the relationship between the buyer's own bid and his outcome, folding into a single function the allocation rule, the opponent's bidding, and whatever else mediates between the two. Nothing requires it to derive from an explicit account of the environment, and the analysis treats $g^i$ as an arbitrary function that may bear no relation to the true environment. Among the buyer's informational constraints, the essential one is that he does not observe the mechanism. A buyer who observed the mechanism would hold correct beliefs about it, and the model through which he interprets his outcomes would adjust with the mechanism itself. Because the mechanism instead reaches the buyer only through the outcomes it generates, its properties shape the evidence he encounters without affecting how he interprets that evidence.[5]

We assume throughout that a buyer updates his belief $\mu_i$ over parameter values but does not revise the model family itself. A buyer uncertain about a reserve price revises his estimate of its value, but does not consider the possibility that the mechanism operates on different principles entirely.

The parameter $\theta$ captures the one dimension of uncertainty the buyer recognises, and it need not represent a feature of the mechanism itself. In the example of Section 2.4.1 and the further models of Appendix 7.4, it parametrises whatever aspect of the strategic environment the buyer treats as unknown, from a reserve price to the opponent's value distribution to the aggressiveness of the pricing rule, while $g$ embeds his maintained hypotheses about everything else.

## 2.3 The Learning Process

Each period $t$ proceeds as follows. Buyer $i$ enters holding belief $\mu_{i,t} \in \Delta\Theta_i$, observes his valuation $v_{i,t} \sim F_i$, and chooses a bid $b_{i,t}$ to myopically maximise expected utility,[6]

$$b_{i,t} \in \arg\max_b \int_{\theta\in\Theta_i} (v_{i,t} - b)\, g^i(b, v_{i,t}, \theta)\, d\mu_{i,t}(\theta). \tag{2}$$

We denote the set of bids consistent with optimal bidding under belief $\mu$ and valuation $v$ for buyer $i$ by $\beta^i(v,\mu)$ and its inverse with respect to $v$, when it exists, by $(\beta^i)^{-1}(b,\mu)$. When $(\beta^i)^{-1}(b,\mu)$ is well defined, denote the CDF of buyer $i$'s bid distribution under belief $\mu$ by $\Phi^i(b \mid \mu)$ with density $\phi^i(b \mid \mu)$, where $\Phi^i(b \mid \mu) = F_i((\beta^i)^{-1}(b,\mu))$.

The buyer observes only his own allocation outcome $y_{i,t} \in \{0,1\}$, equal to 1 with true probability $q^i(b_{i,t}, b_{-i,t})$. In particular he observes neither the opponent's bid nor the opponent's allocation outcome, and updates via a misspecified Bayes' rule,

$$\mu_{i,t+1}(\Theta_i') \propto \int_{\theta\in\Theta_i'} g^i(b_{i,t}, v_{i,t}, \theta)^{y_{i,t}} \left[1 - g^i(b_{i,t}, v_{i,t}, \theta)\right]^{1-y_{i,t}} d\mu_{i,t}(\theta). \tag{3}$$

Although outcomes are generated by the true allocation rule $\boldsymbol{q}$, buyer $i$ updates as though they were generated by his subjective model $g^i$. Beyond his own valuation and the payment rule, the

[5] In our framework the mechanism also reflects the buyer's model. A buyer who observed the mechanism would hold a model that adjusts to the very mechanism it informs, a fixed-point problem we do not pursue.

[6] Myopic optimisation is the standard assumption in models of misspecified learning. Allowing forward-looking experimentation would complicate the bid stage without, we expect, meaningfully altering the long-run analysis.

environment enters the buyer's bid and his update only through $g^i$. Two models that agree as functions therefore generate the same bidding and the same learning, however each was arrived at. The analysis accordingly requires no account of where $g^i$ comes from, and we take no position on it.

## 2.4 Symmetry and Regularity

For the presentation of our results we impose symmetry on the buyers and mechanism, deferring the asymmetric case to the appendix.

**Assumption 1** (Symmetry)**.** *Both buyers share the same value distribution $F$ on $V = [\underline{v}, \bar{v}]$, the same model $g$ on a common parameter space $\Theta$, a common target $\theta^* = \theta_1^* = \theta_2^*$, and we restrict attention to symmetric allocation rules satisfying $q^1(b_1, b_2) = q^2(b_2, b_1)$.*

Under symmetry, we suppress the buyer index on $g^i$, $\Theta_i$, $F_i$, $V_i$, and $\beta^i$ wherever a statement concerns a representative buyer. We write $q(b_1, b_2)$ for $q^1(b_1, b_2)$, the probability that buyer 1 obtains the good under profile $(b_1, b_2)$, retaining the buyer index only where both buyers appear together. The question of sustaining the target reduces to sustaining a common $\theta^*$ on a common set $\Theta'$.

The buyer's optimisation problem under a general model $g$ imposes little structure on optimal bids, and the conditions below supply what the analysis uses. The economic content is carried by the orderings and the technical content by the rest. The model's predicted win probability is to rise with the bid and fall with the parameter. Optimal behaviour is to be ordered in turn, higher valuations and higher parameters inducing more aggressive bids. Learning is to be non-degenerate, no single observation eliminating a parameter value. The following conditions formalise these requirements.

**Assumption 2** (Regularity)**.** *The function $g$ on $\mathbb{R} \times V \times \Theta$ and the buyers' initial beliefs satisfy:*

*(R1)* *$g(b, v, \theta)$ takes values in $(0, 1)$ and is twice continuously differentiable in $(b, v, \theta)$;*

*(R2)* *$g(b, v, \theta)$ is strictly increasing in b, weakly decreasing in v, and strictly decreasing in $\theta$;*

*(R3)* *$\dfrac{g(b, v, \theta)}{g_b(b, v, \theta)}$ is strictly increasing in b, weakly decreasing in v, and strictly decreasing in $\theta$;*

*(R4)* *initial beliefs have full support, assigning positive mass to every non-empty relatively open subset of $\Theta$.*

The regularity conditions are maintained throughout Sections 3 and 5, together with the requirement that the target $\theta^*$ lie in the relative interior of $\Theta$[7] and the restriction to mechanisms non-increasing in the opponent's bid.[8] Additional result-specific conditions are assumed as needed.

[7] Targets on the boundary of $\Theta$ are excluded only because the straddling compacts quantified over in Condition 1 would not exist, not because they are hard to sustain. A rule that always allocates produces more wins than the model predicts at every belief above $\underline{\theta}$, the one-sided margin holds automatically since $h$ is continuous and smaller than 1, and beliefs are driven to $\underline{\theta}$. A rule that never allocates drives beliefs to $\bar{\theta}$ by the symmetric argument, and remains feasible with any number of buyers. Feasibility limits the always-allocate rule across buyers, but that constraint is orthogonal to the learning problem.

[8] Proposition 5 is the one exception. Its impossibility is stated without (R1)–(R3), for every target in $\Theta$, and with no restriction on the allocation rule beyond feasibility.

The economic content sits in (R2) and (R4). Within (R2), monotonicity in $b$ is inherent to any model of a win probability, and monotonicity in $\theta$ is a normalisation. The weak monotonicity in $v$ accommodates models in which the buyer treats his own valuation as informative about the strength of competition, as in the taste projection setting of Gagnon-Bartsch et al. (2021). In the appendix it extends the comparisons to bids outside the range of the bid function. (R4) requires initial beliefs of full support, so that no parameter is excluded at the outset.

The technical content sits in (R1) and (R3). (R3) delivers the unique ordered bid.[9] Strict interiority in (R1) keeps the likelihood ratio finite, so that no single observation eliminates a parameter value.

Under regularity, a buyer holding the degenerate belief $\theta$ submits a unique optimal bid $\beta(v,\theta)$ at each valuation $v$. For sets $A \subseteq V$ and $B \subseteq \Theta$ we write $\beta(A,B)$ for the set of bids $\{\beta(v,\theta) \mid v \in A, \theta \in B\}$, with a point argument understood as the corresponding singleton.

### 2.4.1 Correlated Reserve Price

The conditions also admit models derived from an explicit account of the environment. Consider a buyer who thinks he faces an unknown reserve price, which may incorporate partial information about his private valuation $v$, so that price and valuation are correlated. Nothing requires the mechanism he actually faces to have a reserve price. His model describes only how his own bid maps to his outcome, and he may be holding it while bidding against a rival in a first-price auction in reality. He believes the reserve price equals

$$r = \psi(v,\theta) + \varepsilon,$$

where $\varepsilon$ is mean-zero noise whose CDF $G$ is twice continuously differentiable with full support $\mathbb{R}$, and $\psi(v,\theta)$ is twice continuously differentiable, increasing in $v$, and strictly increasing in $\theta$. The parameter $\theta$ describes the aggressiveness of the pricing rule. The buyer's model is therefore $g(b,v,\theta) = G(b - \psi(v,\theta))$, which satisfies (R1) and (R2) directly. Under strict log-concavity of $G$ ($\frac{G'(x)}{G(x)}$ strictly decreasing), condition (R3) is also satisfied. Two further models in Appendix 7.4 satisfy the same conditions.

## 2.5 Convergence

Characterising bidding behaviour and beliefs along arbitrary sample paths is intractable, as $\mu_{i,t}$ is a measure-valued process depending on the full history through the endogenous, potentially set-valued bidding correspondence. We therefore focus on the long-run outcome. A buyer $i$ whose belief converges to $\delta_{\theta_i^*}$ eventually bids according to $\beta^i(v,\theta_i^*)$ at every valuation $v$, so the limit belief determines long-run behaviour.

Because $\beta^i(v,\mu)$ may not be single valued, the belief dynamics are fully specified only once a rule selects among optimal bids. We place no restriction on the rule. It may depend on the entire history and may randomise. Any selection rule, together with mechanism $\boldsymbol{q}$ and an initial belief profile $\boldsymbol{\mu}$, induces a distribution over period-$t$ belief profiles, and we denote by $\gamma_{t,t}(\cdot \mid \boldsymbol{\mu}, \boldsymbol{q})$ the

[9] Under (R3), the first-order condition for optimal bidding has a unique solution, and the resulting bid function $\beta(v,\theta)$ is strictly increasing in $(v,\theta)$.

marginal distribution of buyer $i$'s period-$t$ belief, leaving the selection rule implicit.[10] We say that mechanism $\boldsymbol{q}$ *sustains* $\boldsymbol{\theta}^* = (\theta_1^*, \theta_2^*)$ on $\Theta_1' \times \Theta_2'$ if, whenever buyers begin with beliefs sufficiently concentrated on their respective $\Theta_i'$, each buyer $i$'s beliefs converge to $\delta_{\theta_i^*}$.

**Definition 1** (Convergence on a Set)**.** *For given intervals $\Theta_i' \subseteq \Theta_i$ with $\theta_i^* \in \Theta_i'$ for $i = 1,2$, let $\bar{\Theta}_i'$ denote the closure of $\Theta_i'$. Mechanism $\boldsymbol{q}$ sustains $\boldsymbol{\theta}^* = (\theta_1^*, \theta_2^*)$ on $\Theta_1' \times \Theta_2'$ if for every neighbourhood $U_i$ of $\delta_{\theta_i^*}$ and every $\varepsilon > 0$, there is a neighbourhood $D_i(U_i, \varepsilon)$ of $\Delta\bar{\Theta}_i'$[11] such that for every selection rule and all $(\mu_1, \mu_2) \in D_1(U_1, \varepsilon) \times D_2(U_2, \varepsilon)$*

$$\lim_{t \to \infty} \gamma_{i,t}(U_i \mid \boldsymbol{\mu}, \boldsymbol{q}) \geq 1 - \varepsilon \qquad \text{for } i \in \{1,2\}.$$

Definition 1 makes precise the sense in which *sustains* has been used so far. When beliefs converge to $\theta^*$ on $\Theta'$ under $\boldsymbol{q}$, we say interchangeably that $\theta^*$ is stable on $\Theta'$ under $\boldsymbol{q}$ and that $\boldsymbol{q}$ sustains $\theta^*$ on $\Theta'$.

The definition demands convergence under every selection from the bidding correspondence, so no tie-breaking rule can defeat it, and the initial concentration it requires on $\Theta_i'$ grows with the probability of convergence demanded. Mechanism $\boldsymbol{q}$ sustains $\boldsymbol{\theta}^*$ *globally* if it sustains $\boldsymbol{\theta}^*$ on $\Theta_1 \times \Theta_2$, in which case no restriction on initial beliefs is needed. It sustains $\boldsymbol{\theta}^*$ *locally* if it sustains $\boldsymbol{\theta}^*$ on $\{\theta_1^*\} \times \{\theta_2^*\}$, in which case beliefs sufficiently close to the degenerate belief on the target converge to it. When $\boldsymbol{q}$ sustains $\boldsymbol{\theta}^*$ on $\Theta_1' \times \Theta_2'$, we call $\Theta_1' \times \Theta_2'$ a *basin of attraction* of $\boldsymbol{\theta}^*$ under $\boldsymbol{q}$, and extending the basin means sustaining $\boldsymbol{\theta}^*$ on a larger set. Because initial beliefs are required only to lie near the simplex over $\bar{\Theta}_i'$, they may place positive mass outside $\Theta_i'$ even in the local case, and convergence must drive that mass to zero. Intermediate cases, with $\Theta_i'$ strictly between a singleton and the whole space, arise in the two-buyer analysis, and Proposition 1 accordingly gives sufficient conditions for a mechanism to sustain the target on an arbitrary set.

# 3 Sustaining a Target Belief

We ask whether, for the buyers' model $g$, value distribution $F$, and a target $\theta^*$, some feasible mechanism sustains $\theta^*$, and what structure such a mechanism must have. The demand is strong, since a mechanism that sustains $\theta^*$ makes each buyer certain of it, whether or not his model at $\theta^*$ describes the environment he faces. Meeting it requires that a buyer's learning receive the right feedback, expressed as a comparison between the win probability he faces and the win probability his model predicts, and that a feasible mechanism supply that feedback. The analysis separates these two questions, addressing the former through Proposition 1 and the latter first in a single-buyer benchmark and then in the two-buyer environment. In the benchmark, the mechanism alone stands between the buyer's bid and his outcome. In the two-buyer environment, the case of principal interest, the opponent's bidding intervenes within each period, and, across periods, each buyer's learning shifts the evidence from which the other learns.

[10] Appendix 7.1 records the formal selection machinery, defining adapted selection rules and constructing the distributions over period-$t$ belief profiles that a rule induces.

[11] Neighbourhoods in $\Delta\Theta_i$, of beliefs and of sets of beliefs alike, are taken in the Prokhorov metric, as in Frick et al. (2023).

## 3.1 Convergence and Self-Confirmation

The convergence analysis compares two objects. The first is the win probability the buyer's model predicts at a given bid under optimal bidding,

$$h(b,\theta) = g(b,\beta^{-1}(b,\theta),\theta). \tag{4}$$

The function $h$ evaluates the buyer's model at the valuation for which $b$ is optimal under $\theta$, eliminating the valuation by imposing optimality and leaving a function of bid and parameter alone. For bids outside the range of $\beta(\cdot,\theta)$ we adopt the convention that $\beta^{-1}(b,\theta) = \underline{v}$ when $b$ lies below the range and $\beta^{-1}(b,\theta) = \bar{v}$ when $b$ lies above it, so that $h$ is defined at every bid.

The second is the objective win probability of a bid $b$ against an opponent who bids according to belief $\mu$,

$$Q(b \mid \mu) = \int q(b,\beta(v,\mu))\, f(v)\, dv. \tag{5}$$

This is the win probability the buyer actually faces, as opposed to the win probability $h(b,\theta)$ he believes he faces. For non-degenerate $\mu$ the correspondence $\beta(v,\mu)$ need not be single valued. In that case $Q(b \mid \mu)$ is evaluated at an arbitrary selection, and conditions stated in terms of $Q$ are required to hold for every selection. With a slight abuse of notation, for any set $\Theta' \subseteq \Theta$, we write $Q(b \mid \Theta') = \{Q(b \mid \mu) \mid \mu \in \Delta\Theta'\}$ for the range of objective win probabilities as the opponent's belief varies over $\Delta\Theta'$.

The comparison between $h$ and $Q$ governs the buyer's learning. The mechanism $q$ determines $Q$ given the opponent's bidding, while the buyer's model $g$ determines $h$. Fix a bid $b$ and a parameter $\theta \neq \theta^*$. The *subjective responsiveness* at $b$ and $\theta$ is the amount by which $h(b,\theta)$ exceeds $h(b,\theta^*)$ when $\theta < \theta^*$, and falls short of it when $\theta > \theta^*$. It measures how far the buyer's prediction moves when his belief moves from $\theta^*$ to $\theta$, signed to be positive when the prediction moves in the direction of the belief, up for a more optimistic belief and down for a more pessimistic one, and it depends on the model alone, with optimal bidding folded into $h$. The *objective responsiveness* at $b$ and $\theta$ is the same quantity for $Q(b \mid \theta)$, measuring how far the win probability the buyer faces moves when his opponent's belief makes the same move. It depends on the mechanism and the opponent's bidding.

**Definition 2** (Adverse Ordering)**.** *The model is adversely ordered if $h(b,\theta)$ is strictly decreasing in $\theta$, so that higher parameters predict lower win probabilities at every bid.*

A buyer updates by comparing how often he wins against how often his model predicts he should win. If his belief about $\theta$ is too high, he is too pessimistic about his chances, and winning more often than his model predicts leads him to revise downward. If his belief about $\theta$ is too low, he is too optimistic, and winning less often than his model predicts leads him to revise upward. The mechanism determines how often the buyer actually wins and can therefore be arranged so that this self-correcting dynamic always points toward $\theta^*$. We call the feedback generated by a mechanism corrective on $\Theta'$ if these prediction errors point toward the target at every bid the buyer might submit and for every belief the opponent might hold, with margins uniform over beliefs bounded away from $\theta^*$, so that a buyer holding any belief other than $\theta^*$ receives systematic evidence against it. Condition 1 is the formal version of this requirement.

**Condition 1** (Corrective Feedback)**.** *Fix a compact interval $\Theta' \subseteq \Theta$ with $\theta^* \in \Theta'$ and a mechanism $q$.*[12] *There exists an interval $W$, open relative to $\Theta$, with $\Theta' \subseteq W \subseteq \Theta$, such that for every $c > 0$:*

*(a)* $\inf\{\, Q(b \mid \bar{\theta}'') - h(b, \bar{\theta}'') \;:\; b \in \overline{\beta(V,W)},\ \bar{\theta}'' \in W,\ \bar{\theta}'' \geq \theta^* + c \,\} > 0$

*(b)* $\inf\{\, h(b, \underline{\theta}'') - Q(b \mid \underline{\theta}'') \;:\; b \in \overline{\beta(V,W)},\ \underline{\theta}'' \in W,\ \underline{\theta}'' \leq \theta^* - c \,\} > 0$

*where $h(b, \theta)$ is as defined in (4).*

**Proposition 1** (Convergence)**.** *Every mechanism generating corrective feedback on $\Theta'$, as formalised in Condition 1, sustains $\theta^*$ on $\Theta'$.*

Part (a) formalises the pessimistic side, where $\theta$ is above $\theta^*$. At the upper endpoint, a buyer who entertains $\bar{\theta}''$ while his opponent bids under the same parameter wins more often than his model predicts, since the objective win probability $Q(b \mid \bar{\theta}'')$ exceeds his subjective win probability $h(b, \bar{\theta}'')$ by a margin uniform over bids, and the surplus of wins pushes his belief down toward the target. Monotonicity in the opponent's bid extends the comparison from the endpoint parameters to every opponent belief supported between them. Part (b) is the symmetric statement on the optimistic side, where a buyer facing an opponent bidding under the same parameter wins less often than his model predicts and revises upward. In each case the prediction error at a shared parameter points back toward the target, and this tends to happen when subjective responsiveness exceeds objective responsiveness.

The comparison at endpoint parameters also controls updating at every belief the buyer himself may hold. A belief affects what he observes only through the bids it induces, and each such bid lies between the bids induced at the endpoint parameters. The proof of Proposition 1 carries the comparison the rest of the way, from bids and predictions evaluated one parameter at a time to updating from every belief supported between the endpoints.

The margins do no more than absorb the discontinuities that a step mechanism induces in $Q$. They replace the belief continuity that the arguments of Frick et al. (2023) require, which fails at these discontinuities. When $Q$ is continuous, strict versions of the comparisons supply them automatically, a positive continuous gap over compact bids and endpoints being bounded away from zero, and they may vanish as the endpoints approach $\theta^*$, where self-confirmation closes the gap. Because $W$ can extend beyond $\Theta'$, the conditions remain nontrivial even when $\Theta'$ is small or when $\theta^*$ lies at its boundary.

The conditions are sufficient but not necessary for convergence. Necessary and sufficient conditions for convergence under misspecification remain an open problem in the learning literature, and Proposition 1 represents one of the tightest tractable conditions available. The full derivation appears in Appendix 7. Sustaining a belief is a property of a single comparison between the win probability the buyer's model predicts and the one the mechanism delivers, evaluated bid by bid away from the target.

[12] When $q$ may increase in the opponent's bid, the natural form of the conditions below replaces the endpoint comparisons with $\min_{\mu \in \Delta\Theta''} Q(b \mid \mu)$ and $\max_{\mu \in \Delta\Theta''} Q(b \mid \mu)$, over compacts $\Theta'' \subseteq W$ straddling the target. The result continues to hold in that form when $q$ is continuous in the opponent's bid or valuations are drawn from a continuum, since belief continuity in the sense of Frick et al. (2023) holds in either case and their convergence framework applies directly. We do not pursue this. With a single buyer, the restriction is vacuous, the mechanism depending on his bid alone.

Condition 1 does not determine what form the mechanism must take, nor does it identify which model primitives admit a mechanism satisfying it. It does, however, pin the objective win probability at the target to the model's prediction there.

**Definition 3** (Self-Confirming Mechanism)**.** *A mechanism is self-confirming with respect to $\theta^*$ on $\Theta'$, for an interval $\Theta' \subseteq \Theta$ with $\theta^* \in \Theta'$, if $Q(b \mid \theta^*) = h(b, \theta^*)$ for all $b \in \overline{\beta(V, \Theta')}$. Where no set is named, $\Theta' = \{\theta^*\}$ and the bid range is $[\beta(\underline{v}, \theta^*), \beta(\bar{v}, \theta^*)]$.*

**Corollary 1.** *If Condition 1 holds for $\theta^*$ on some $\Theta' \ni \theta^*$ under mechanism $q$, then $q$ is self-confirming with respect to $\theta^*$ on $\Theta'$.*

A self-confirming mechanism exactly confirms the buyer's model at the target belief, so that a buyer holding belief $\delta_{\theta^*}$ observes outcomes fully consistent with his model and has no reason to revise. Self-confirmation also turns corrective feedback into a comparison of subjective and objective responsiveness. Wherever the mechanism is self-confirming, $Q(b \mid \theta^*) = h(b, \theta^*)$, so the prediction error at a shared parameter, measured in the direction of the target, is subjective responsiveness less objective responsiveness, and parts (a) and (b) of Condition 1 ask that the first exceed the second, by a margin uniform over bids and over parameters bounded away from the target.

## 3.2 The Single Buyer

With a single buyer, the objective win probability $Q(b \mid \mu)$ reduces to $q(b)$ for all $\mu$, since there is no opponent whose behaviour could affect it. Corollary 1 then pins the mechanism to $q(b) = h(b, \theta^*)$ at every bid it covers, and we take $q(b) = h(b, \theta^*)$ at all bids as the canonical choice.[13]

Substituting $q(b) = h(b, \theta^*)$ into Condition 1 reduces it to a requirement on $h$ alone.

**Proposition 2** (Single-Buyer Convergence)**.** *Fix a compact interval $\Theta' \subseteq \Theta$ with $\theta^*$ in the interior of $\Theta'$ relative to $\Theta$. The mechanism $q(b) = h(b, \theta^*)$ sustains $\theta^*$ on $\Theta'$ if for all $b \in \overline{\beta(V, \Theta')}$ and all $\theta \in \Theta'$:*

*(a) $h(b, \theta) < h(b, \theta^*)$ if $\theta > \theta^*$;*

*(b) $h(b, \theta) > h(b, \theta^*)$ if $\theta < \theta^*$.*

Strict inequality at each bid does not suffice for Condition 1 in general, but in the present benchmark it does. The pointwise inequalities yield margins that are positive, attained on the compact bid range by continuity of $h$, and uniform over parameters bounded away from $\theta^*$. The proof extends them beyond $\Theta'$ to the slightly larger set that Condition 1 quantifies over. Taking $\Theta'$ to be all of $\Theta$ delivers global convergence, with no restriction on initial beliefs. The two inequalities of Proposition 2 compare, at the canonical mechanism itself, the win probability a buyer expects with the one he faces. Under $q(b) = h(b, \theta^*)$, a buyer bidding optimally under any $\theta' < \theta^*$ in $\Theta'$ expects to obtain the good with probability $h(b, \theta')$, above the $q(b) = h(b, \theta^*)$ he in fact faces, so he wins less often than his model predicts, concludes that the true $\theta$ must be larger, and revises upward. A buyer bidding optimally under a $\theta' > \theta^*$ wins more often than his model predicts and revises downward. We call the two inequalities the responsiveness condition. No belief moves the

[13] Bids outside $\beta(V, W)$ never enter Condition 1, so the choice of mechanism there is unconstrained by it.

win probability a single buyer faces, so objective responsiveness is zero, and the condition asks only that subjective responsiveness be positive. With a single buyer, whether the target can be sustained is decided by the buyer's model alone, and self-confirmation determines the mechanism that does it.

A natural case where the responsiveness condition holds is when the model is adversely ordered.

**Lemma 1.** *Suppose $g_v < 0$.*[14] *Then the model is adversely ordered if*

$$\frac{g_{b\theta}}{g_\theta g_b} > \frac{g_{bv}}{g_b g_v} + \frac{g_b}{g g_v}.$$

The condition can be read as requiring the direct effect of $\theta$ on the win probability the buyer's model predicts to dominate an indirect effect operating through the bidding strategy. Since $h(b,\theta) = g(b,\beta^{-1}(b,\theta),\theta)$, a change in $\theta$ affects $h$ through two channels. The first is direct, through $g$'s dependence on $\theta$, which is negative by (R2). The second is indirect, through the change in $\beta^{-1}(b,\theta)$, which decreases the valuation associated with any given bid as $\theta$ increases. When the bidding strategy moves little with $\theta$, the indirect channel is weak and $h$ is decreasing in $\theta$. When bidding moves sharply with $\theta$, the indirect channel can offset or reverse the direct effect. Subjective responsiveness combines the two channels, so bidding that moves sharply with the belief weakens it. The cross-derivative condition of the lemma is the algebraic form of the requirement that the direct effect dominate.

The correlated reserve price model of Section 2.4.1 is adversely ordered, the appendix verifying this from the first-order condition and the implicit function theorem, which give $\frac{\partial}{\partial\theta}\psi(\beta^{-1}(b,\theta),\theta) > 0$. Proposition 2 therefore applies.

## 3.3 Two Buyers

The buyer's problem carries over unchanged, his model giving an opponent no separate role, so his bid function and his belief updating are those of the single-buyer case. The objective win probability, however, now depends on the opponent's bids and therefore moves as the opponent's beliefs evolve. The buyer's model reads every such movement as evidence about $\theta$, and this cross-contamination of the evidence is the source of objective responsiveness when there are two buyers. Whether objective responsiveness must be present depends on feasibility. The mechanism $q(b_1,b_2) = h(b_1,\theta^*)$, which ignores the opponent's bid, is self-confirming and has zero objective responsiveness, and it is feasible on the bid range at the target if and only if $h(b,\theta^*) \leq 1/2$ at every bid in that range. When this holds, the opponent's learning can be kept out of the evidence the buyer receives. When some self-confirmed win probability exceeds one half, no feasible mechanism can ignore the opponent's bid there, and the opponent's learning necessarily reaches the buyer's objective win probability.

Consider, for instance, buyers with two valuations drawn with equal probability, so that at the target each buyer submits one of two bids. Self-confirmation fixes, at each bid, the buyer's average win probability across the opponent's two bids, and leaves free how that average divides between the opponent's low bid and his high bid. Against an opponent submitting the same bid the buyer

[14] When $g$ does not depend on $v$, so that $g_v = 0$, the indirect channel vanishes and $h_\theta = g_\theta < 0$ by (R2), so the model is adversely ordered without further conditions.

can win at most half the time, so once the average at the high bid exceeds one half, he must win more often against the low bid than against the high one. As the opponent grows optimistic, his high type's bid falls, shifting mass toward the bid the buyer beats more often, and the buyer's win rate rises by that difference times the mass that shifts, without his own belief moving.

Feasibility has a second consequence that is independent of learning. A single good may be unable to meet both buyers' self-confirmed expectations at once. By Corollary 1, a mechanism generating corrective feedback must deliver $h(b,\theta^*)$ at every bid in $[\beta(\underline{v},\theta^*),\beta(\bar{v},\theta^*)]$ to a buyer whose opponent holds the target belief. With a single buyer any such schedule of win probabilities can be met, since $h$ takes values in $(0,1)$. With two buyers the good must be divided between them, so the self-confirming win probabilities must arise as the marginals of a single feasible allocation, and whether such an allocation exists is characterised by the following condition, due to Border (1991).

**Condition 2** (Border's Condition)**.** *The win probabilities $h(b,\theta^*)$ can be matched by a feasible symmetric allocation if and only if, for all measurable $\hat{B} \subseteq [\beta(\underline{v},\theta^*),\beta(\bar{v},\theta^*)]$,*

$$2\int_{\hat{B}} h(b,\theta^*)\,\phi(b \mid \theta^*)\,db \leq 1-\left(1-\int_{\hat{B}} \phi(b \mid \theta^*)\,db\right)^2.$$

Even when Border's condition is satisfied, the allocation attaining the self-confirming marginal is not unique, and Border's condition alone does not guarantee the existence of a mechanism in the maintained class. Under the maintained symmetry assumptions (Assumption 1) and the additional regularity condition

(Bid Monotonicity) $\frac{\partial h}{\partial b}(b,\theta) > 0$ for all $b,\theta$,

the self-confirming allocation can be taken to be symmetric and non-increasing in $b_2$, following the constructive approach of Maskin and Riley (1984) and Matthews (1984). Like the responsiveness condition of the single-buyer analysis, bid monotonicity weighs a direct effect against an indirect one, requiring the direct effect of a higher bid in the model to dominate the indirect effect of the higher valuation that the bid reveals. We maintain bid monotonicity throughout the multi-buyer analysis. All existence claims that follow quantify within the maintained class of mechanisms, symmetric across buyers and non-increasing in the opponent's bid.

Near the target, the effect of the opponent's learning can be bounded by its largest value over self-confirming mechanisms, and when the subjective responsiveness of the buyer's model exceeds this bound, every feasible self-confirming mechanism sustains the target locally. Further from the target the bound is generally not met, and a mechanism that sustains the target there must contend with the opponent's learning.

### 3.3.1 Local Stability

Checking separately whether each mechanism generates corrective feedback is impractical, since the class of feasible self-confirming mechanisms is large. Under a regularity condition on the bid distribution, however, a single mechanism has the largest objective responsiveness in the class, and corrective feedback under that mechanism implies corrective feedback under every other mechanism in the class.

**Definition 4** (Monotone Likelihood Ratio Property)**.** *The bid distribution $\Phi(b \mid \theta)$ satisfies MLRP if the density ratio $\phi(b \mid \theta')/\phi(b \mid \theta)$ is increasing in $b$ for all $\theta' > \theta$.*

Under MLRP, the mechanism with the largest objective responsiveness awards the good with certainty when the opponent bids below a threshold $\alpha(b)$ and withholds it otherwise, where $\alpha(b) = \Phi^{-1}(h(b,\theta^*) \mid \theta^*)$ is chosen so that an opponent holding belief $\delta_{\theta^*}$ bids below it with probability exactly $h(b,\theta^*)$. Because it places all of the buyer's win probability on the opponent's lowest bids, its objective win probability moves with the opponent's belief as much as self-confirmation permits. It need not itself be feasible, and it serves only to bound the objective responsiveness of the mechanisms that are.

**Proposition 3** (Local Stability)**.** *Suppose MLRP holds, Border's condition is satisfied for $h(b,\theta^*)$, and $h(b,\theta)$ is strictly decreasing in $\theta$ at $\theta^*$. A feasible mechanism self-confirming with respect to $\theta^*$ on a neighbourhood of $\theta^*$ exists, and every such mechanism sustains $\theta^*$ locally, if for all $b \in [\beta(\underline{v},\theta^*), \beta(\bar{v},\theta^*)]$*

$$\left| \frac{\partial}{\partial \theta} \Phi\big(\alpha(b) \mid \theta\big) \Big|_{\theta=\theta^*} \right| < |h_\theta(b,\theta^*)|\,.$$

The displayed condition compares objective and subjective responsiveness at $\theta^*$, in derivative form. The left-hand side is the objective responsiveness of the threshold mechanism, the rate at which the opponent's bid distribution moves mass across $\alpha(b)$ as the opponent's belief changes, and the right-hand side is the buyer's subjective responsiveness. The condition therefore requires the subjective responsiveness of the buyer's model to exceed the largest objective responsiveness that any self-confirming mechanism can have. Since the threshold mechanism attains this maximum, verifying the condition for it suffices for the entire class (Appendix 7.3.1).

In the single-buyer setting, the corresponding local condition is $h_\theta(b,\theta^*) < 0$, which is the displayed condition with its left-hand side set to zero. The left-hand side therefore measures the additional requirement imposed by the second buyer. Because this term is a maximum over both feasible and infeasible self-confirming mechanisms, it constitutes a conservative bound on that requirement.

### 3.3.2 One-Sided Convergence

Proposition 3 guarantees convergence only from beliefs close to the target, and it is natural to ask how far the basin of attraction can be extended. Away from the target, the objective responsiveness of the threshold mechanism can exceed the buyer's subjective responsiveness, so extending the basin requires a mechanism whose objective responsiveness stays below it. The two directions of extension differ in difficulty. Extending the basin above $\theta^*$ is governed by part (a) of Condition 1, which requires pessimistic buyers to win more often than they expect, and feasibility works against this, since the two buyers cannot both win. Extending it below $\theta^*$ is governed by part (b), which requires optimistic buyers to win less often than they expect, and feasibility works in its favour, since it suffices that they frequently lose to each other. We therefore extend the basin on the optimistic side of $\theta^*$. Whether it can also be extended above $\theta^*$ remains open, and Section 5.3 examines why the method developed here does not settle that question.

With discrete valuations, whether the basin extends over every belief below the target is decided by a finite computation. Let $V = \{v_1, \ldots, v_K\}$ with $v_1 < v_2 < \cdots < v_K$, $\underline{v} = v_1$, and $\bar{v} = v_K$, where

valuation $v_k$ occurs with probability $p_k$. For any $\theta \in \Theta$, write $B(\theta) = \{\beta(v_k, \theta) \mid v_k \in V\}$ for the set of optimal bids, and call the elements of $B(\theta^*)$ the grid bids. As an opponent's belief falls below $\theta^*$, his bids fall. The beliefs at which one of them reaches a grid bid form the set $\tilde{\Theta} = \{\tilde{\theta} < \theta^* \mid B(\tilde{\theta}) \cap B(\theta^*) \neq \emptyset\}$, whose elements we call the overlap beliefs.

We say that $\theta^*$ is *one-sided convergent* under $q$ if $q$ sustains $\theta^*$ on $[\underline{\theta}, \theta^*]$. Initial beliefs need not be supported below $\theta^*$, only close to beliefs that are, so they may place a small mass above the target. Near the target, one-sided convergence requires only local stability, which with discrete valuations imposes no further requirement. When $h$ is strictly decreasing in $\theta$, a mechanism that is constant in the opponent's bid near each grid bid has zero objective responsiveness near the target while subjective responsiveness is positive, so prediction errors on both sides of $\theta^*$ point back toward it. The work of one-sided convergence therefore lies further below the target.

Beyond the assumptions maintained throughout, the analysis requires one additional condition on $h(b, \theta)$.

(Fading Sensitivity) $\frac{\partial h}{\partial b}(b, \theta)$ is weakly increasing in $\theta$.

Fading sensitivity makes the gap in $h$ between a lower parameter and a higher one weakly decreasing in the bid, so that under adverse ordering subjective responsiveness is weakly larger at lower bids. It allows corrective feedback established at a grid bid to extend to the bids below it.

**Proposition 4** (Finite Reduction)**.** *Suppose valuations are discrete and the model has fading sensitivity and adverse ordering. There is a linear program in $K^2$ variables, with a number of constraints polynomial in K, such that if it has a strictly feasible solution, some mechanism sustains $\theta^*$ on $[\underline{\theta}, \theta^*]$.*

The variables of the program are the win probabilities that a mechanism assigns to buyer 1 at each pair of grid bids, and its constraints are feasibility, self-confirmation, monotonicity in the opponent's bid, and corrective feedback at each grid bid against each overlap belief. A solution is strictly feasible if the feasibility and corrective-feedback constraints hold with slack. There are at most $K(K-1)/2$ overlap beliefs, so the corrective-feedback constraints, the most numerous, number at most $K^2(K-1)/2$, and the program can be solved in time polynomial in $K$.

Discretising valuations does not by itself make the problem finite. A buyer whose belief differs from $\theta^*$ bids between grid bids, so a mechanism must still specify win probabilities at a continuum of bids, and corrective feedback must hold against a continuum of opponent beliefs below the target.

The reduction along the belief dimension, from a continuum of opponent beliefs below the target to the finitely many overlap beliefs, rests on a restriction of the mechanism at opponent bids off the grid. For $b_2 < \beta(v_K, \theta^*)$ with $b_2 \notin B(\theta^*)$, it is without loss to set $q(b_1, b_2) = q(b_1, \bar{b}(b_2))$, where $\bar{b}(b_2)$ is the smallest grid bid above $b_2$. Monotonicity makes this the lowest value $q(b_1, b_2)$ can take, feasibility and corrective feedback on the optimistic side both favour low values of $q(b_1, b_2)$ at such bids, and self-confirmation does not constrain them, since an opponent holding the target belief bids only at grid bids. Under this step form, $Q(b_1 \mid \tilde{\theta})$ is constant in $\tilde{\theta}$ between consecutive overlap beliefs, since the interval between grid bids into which each of the opponent's bids falls changes only at an overlap belief. Under adverse ordering $h(b_1, \tilde{\theta})$ rises as $\tilde{\theta}$ falls, so corrective feedback at the finitely many overlap beliefs implies corrective feedback at the continuum of beliefs $\tilde{\theta} < \theta^*$. The treatment of own bids $b_1 \notin B(\theta^*)$, and further details, are in Appendix 7.3.3.

# 4 Failure of Convergence

With two buyers, feasibility and corrective feedback can conflict. A target's self-confirming win probabilities may be deliverable by a feasible mechanism and yet by no feasible mechanism that generates corrective feedback. This section describes such win probabilities in a setting with two valuations and explains how they arise. It then simulates learning in two instances of this setting. The first avoids this conflict and the second exhibits it.

Consider a setting with two valuations $v_1 < v_2$, where $v_1$ occurs with probability $p$. The grid bids are $b_1 = \beta(v_1, \theta^*)$ and $b_2 = \beta(v_2, \theta^*)$. Write $h_1 = h(b_1, \theta^*)$ and $h_2 = h(b_2, \theta^*)$ for the self-confirming win probabilities. With two valuations there is a single overlap belief $\tilde{\theta}$, at which the opponent's high type's bid falls to $b_1$. Write $\tilde{h}_1 = h(b_1, \tilde{\theta})$ and $\tilde{h}_2 = h(b_2, \tilde{\theta})$ for the corresponding subjective win probabilities.

Figure 1 compares two sets of pairs $(h_1, h_2)$ at $p = 1/2$, for four values of $(\tilde{h}_1, \tilde{h}_2)$. The Border region, bounded by the Border frontier, consists of the pairs that some feasible self-confirming mechanism delivers. Only the pairs consistent with the model are shaded, since bid monotonicity requires $h_1 < h_2$ and adverse ordering requires $h_j < \tilde{h}_j$. Among them, the convergence region, shaded orange, consists of the pairs at which the program of Proposition 4 has a solution, so that some feasible self-confirming mechanism also generates corrective feedback on the optimistic side. The gap, shaded blue, consists of the pairs at which a feasible self-confirming mechanism exists but none generates corrective feedback. In panel (a) the two regions coincide. In the other panels the gap is a band along the top of the Border region, where corrective feedback caps $h_2$ below the bound that feasibility alone imposes, and in panel (d) a second cap also removes a sliver along the diagonal boundary. More generally, the gap is nonempty if and only if $\tilde{h}_2 > 1/2$, and the sliver appears only when $\tilde{h}_1 < 1/2$. Appendix 8.1 gives both regions in closed form.

The gap arises where feasibility forces more objective responsiveness than the model's subjective responsiveness can absorb. Self-confirmation fixes the buyer's average win probability at each bid, and objective responsiveness is created by dividing it unevenly between the opponent's two bids. When $\tilde{h}_2 \leq 1/2$, every pair consistent with the model has $h_2 < 1/2$, and the mechanism that awards each bid its self-confirming win probability whatever the opponent bids is feasible. Its objective responsiveness is zero while subjective responsiveness is positive, so there is no gap. Once $h_2$ exceeds one half this mechanism is no longer feasible, since a buyer can win at most half the time against an opponent submitting the same bid. The excess must then be delivered against the opponent's low bid, and the objective responsiveness this forces at $b_2$ grows as $h_2$ rises, while subjective responsiveness at $b_2$, which is $\tilde{h}_2 - h_2$, shrinks. Near the top of the Border region the forced objective responsiveness therefore exceeds subjective responsiveness, and these pairs form the band. The sliver arises in the same way at $b_1$. Near the diagonal boundary feasibility requires the good to be allocated almost always, so two buyers who both bid $b_1$ must split it nearly evenly, and a buyer bidding $b_1$ wins close to half the time against the opponent's low bid. At the overlap belief this is his objective win probability, and when $\tilde{h}_1 < 1/2$ it exceeds his model's prediction $\tilde{h}_1$. The band is narrower the larger the subjective responsiveness at $b_2$, and bids that respond little to $\theta$ push the overlap belief further below $\theta^*$, which enlarges that responsiveness. The band is therefore narrowest when predictions respond strongly to beliefs and bids respond weakly.

We now turn to the learning dynamics at two instances and examine to what extent beliefs can fail to converge. The first instance's self-confirming win probabilities $(h_1, h_2)$ lie inside the convergence region, and the second's lie at the corner of the Border frontier and so in the gap,

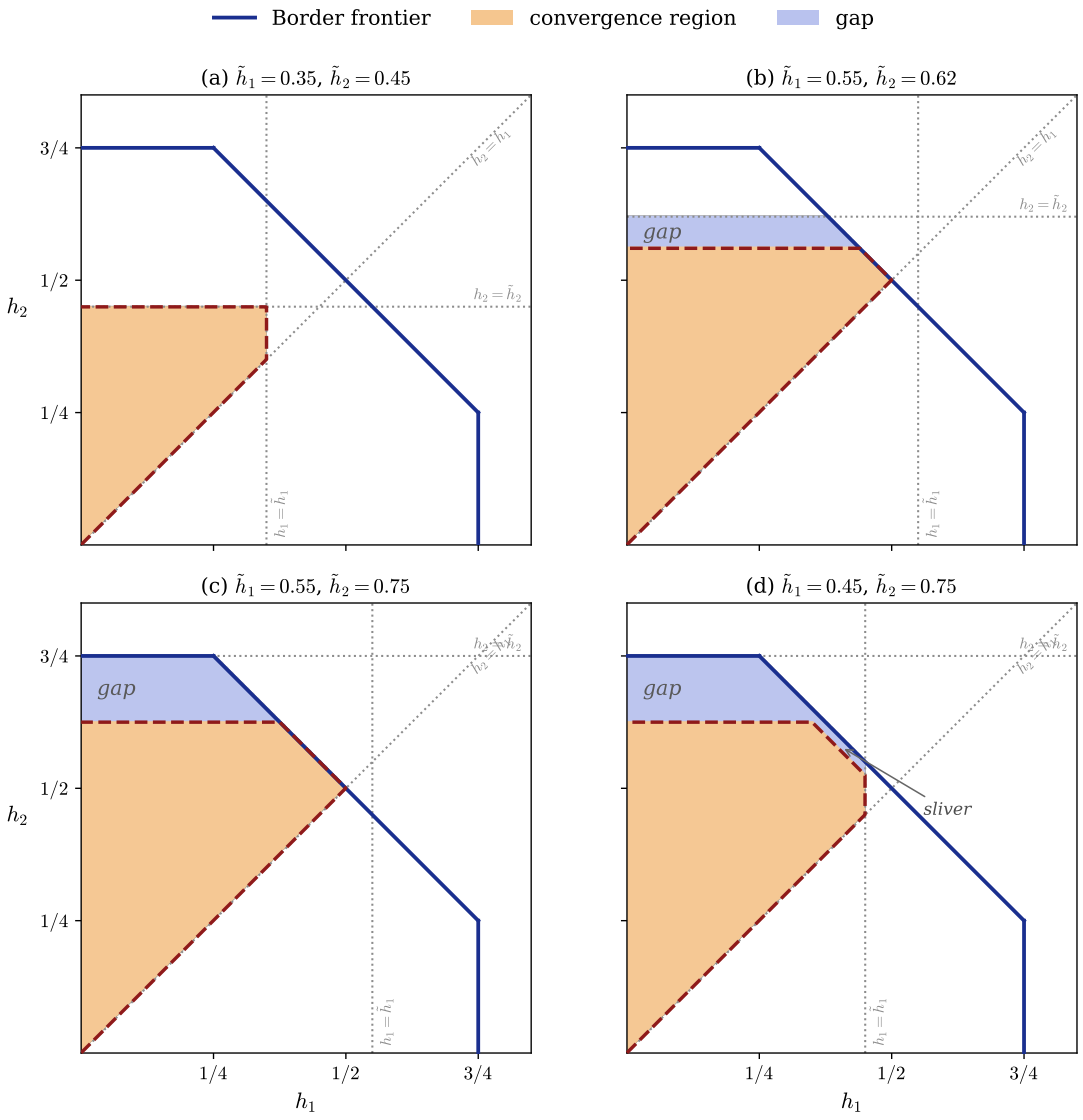



Figure 1: Border region vs. convergence region in $(h_1, h_2)$ space for $p = 0.5$, with $(\tilde{h}_1, \tilde{h}_2)$ varying by panel. Shading is restricted to pairs consistent with the model ($h_1 < h_2$, $h_i < \tilde{h}_i$), within which the Border region is the union of the convergence region and the gap. The regions coincide in panel (a), the band is widest in panel (c), and the sliver of panel (d) reflects the cap at $b_1$.

as Figure 2 shows. Failure, throughout, means that beliefs do not converge to $\theta^*$ from some full-support prior concentrated below it. At the first instance the program of Proposition 4 has a strictly feasible solution and some mechanism sustains the target, but mechanisms drawn from the same feasible self-confirming class nonetheless fail, so membership of the feasible self-confirming class does not by itself sustain the target. At the second, no feasible self-confirming mechanism generates corrective feedback and beliefs do not converge in simulation, while the same model converges when the opponent is removed, so although corrective feedback is only sufficient for convergence, a violation of it can prevent convergence. Both instances are value-independent, so

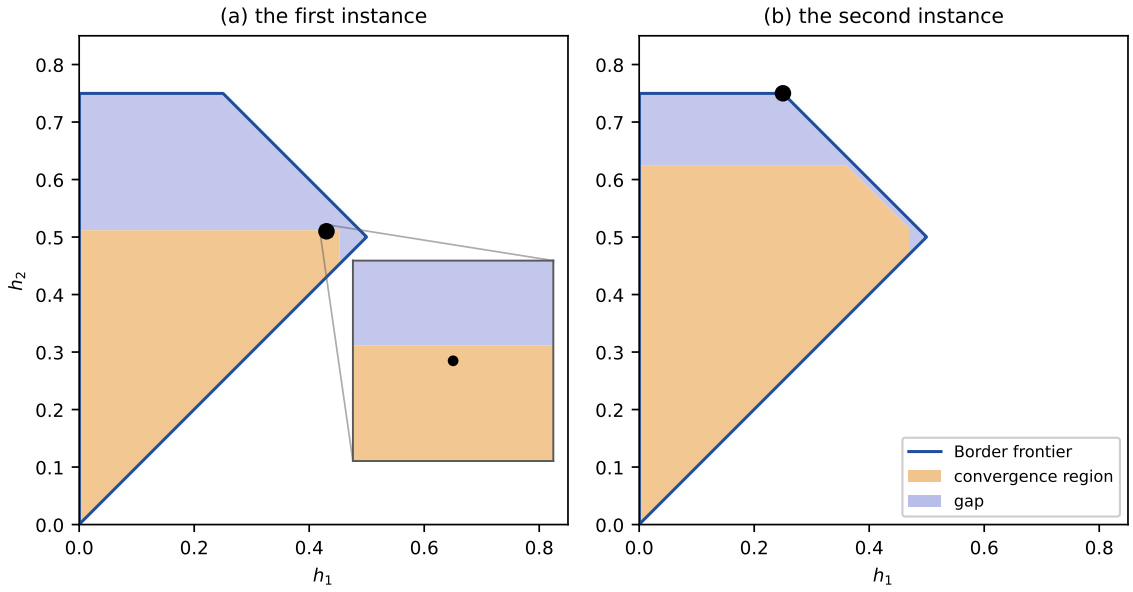



Figure 2: The self-confirming win probabilities of the two instances in $(h_1, h_2)$ space at $p = 1/2$, each panel drawn at its own $(\tilde{h}_1, \tilde{h}_2)$, geometry as in Figure 1. In panel (a) the first instance's lie inside the convergence region and against its boundary. In panel (b) the second's lie at the corner of the Border frontier, where both binding inequalities hold with equality. Both panels shade only pairs with $h_1 < h_2$.

$g(b, v, \theta) = h(b, \theta)$ and each is given by the function $h$ alone. Structural properties, (R1) to (R3), adverse ordering, and fading sensitivity, are established analytically where the construction permits and numerically otherwise. Every claim below about whether beliefs converge rests on simulation, and the settings of every run are recorded in Appendix 8.3.

## 4.1 Inside the Convergence Region

The first instance has two valuations drawn with equal probability, with $h_1 = 0.43$ and $h_2 = 0.51$ (Appendix 8.2). Since $h_2$ exceeds one half, every mechanism in the class has positive objective responsiveness at $b_2$.

A mechanism's win probabilities at the grid bids are four numbers $\hat{q}_{jk}$, where $\hat{q}_{jk}$ is the probability that buyer 1, bidding $b_j$, wins against buyer 2 bidding $b_k$. Since self-confirmation fixes their average at each own bid, the grid values of the strictly feasible, self-confirming, and monotone mechanisms form a polytope in the plane of $\hat{q}_{11}$ and $\hat{q}_{21}$. Corrective feedback is not imposed on it. We draw sixty sets of grid values uniformly from that polytope, extend each to all bids by the construction of Appendix 7.3.3, and simulate each from two priors concentrated near the overlap belief.

Twenty-seven of the sixty fail, a simulated failure fraction of 0.45, while the mechanism built from the program's solution in Appendix 7.3.3 converges from every prior and every seed.

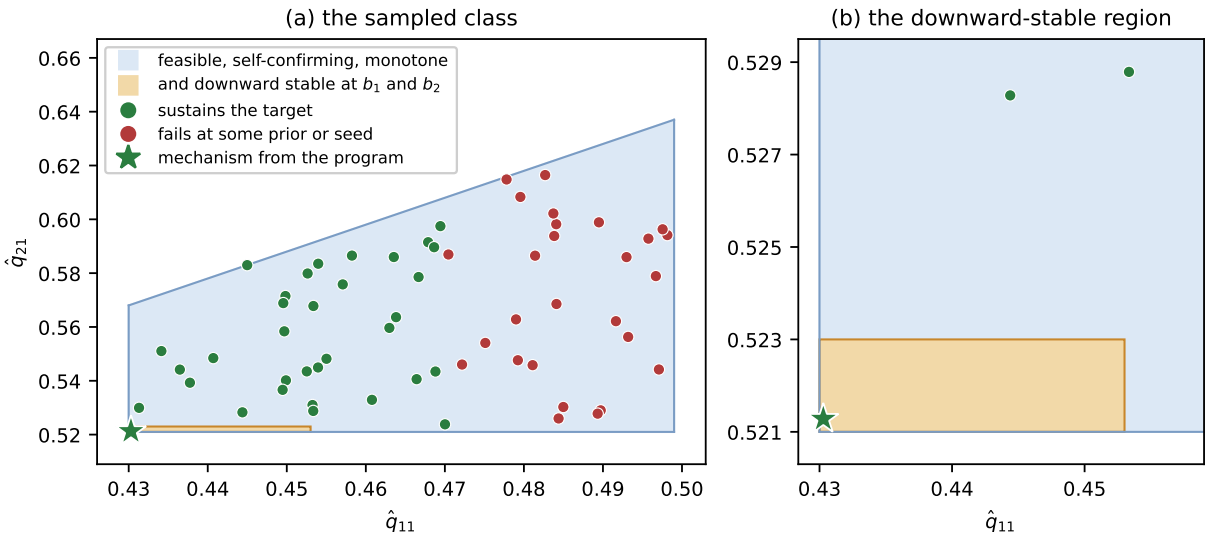



Figure 3: Mechanisms drawn from the class at the first instance, where the program of Proposition 4 is strictly feasible. Blue shading is the sampled polytope of grid values that are strictly feasible, self-confirming, and monotone. Amber is the part of it that also generates corrective feedback on the optimistic side at both bids. Points are the sixty draws, coloured by simulated outcome, with the mechanism built from the program's solution starred. Panel (b) magnifies the amber region.

As panel (a) of Figure 3 shows, the outcome is ordered by $\hat{q}_{11}$ and not by $\hat{q}_{21}$. Every failing draw carries a larger $\hat{q}_{11}$ than every converging draw, while failing and converging draws are mixed across the range of $\hat{q}_{21}$. Every draw fails to generate corrective feedback at $b_2$, and Appendix 8.4 explains why this failure does not affect the outcome. Since self-confirmation fixes $h_1$, a larger $\hat{q}_{11}$ means that the buyer wins more often against the low bid than against the high one, and so a larger objective responsiveness at $b_1$. No draw generates corrective feedback, and nearly half of them fail to converge, so at this instance mechanisms that do not generate corrective feedback fail at a substantial rate.

## 4.2 In the Gap

The second instance has two valuations drawn with equal probability, and its belief space has the target as its top, so no belief above $\theta^*$ is entertained and convergence is one-sided by construction. Here $h_1 = p/2$ and $h_2 = (1+p)/2$ (Appendix 8.2), so the pair lies at the corner of the Border frontier, where both of Border's binding inequalities hold with equality.

Consider first a single buyer with the same model. The win probability he faces depends on no one else's behaviour, so the mechanism $q(b) = h(b, \theta^*)$ is self-confirming at every bid and its objective responsiveness is zero. Since $h$ is decreasing in $\theta$ throughout the belief space, Proposition 2 applies and delivers convergence from every initial belief. Panel (b) of Figure 4 shows the single buyer converging in simulation from six priors.

A second buyer changes the problem, because feasibility now constrains the mechanism. At

the corner exactly one assignment of the four grid values is both feasible and self-confirming,

$$\hat{q}_{11} = \tfrac{1}{2}, \qquad \hat{q}_{12} = 0, \qquad \hat{q}_{21} = 1, \qquad \hat{q}_{22} = \tfrac{1}{2},$$

and every feasible self-confirming mechanism carries these values at the grid bids, whatever it does between them. None of these mechanisms generates corrective feedback. The failure at $b_2$ is forced by the corner, since $\hat{q}_{21} = 1$ exceeds every interior prediction, and the failure at $b_1$ results from the choice of parameters (Appendix 8.2). The runs use one such mechanism, in the step form of Section 3.3.2, described in full in Appendix 7.3.3.

With two buyers, beliefs are trapped from three of the six priors, all concentrated below the overlap belief, and settle at a common level for the rest of the horizon, as panel (a) of Figure 4 shows.[15] The model is the same as with a single buyer, and so is each buyer's problem. The difference is the objective responsiveness that the second buyer creates, which every feasible self-confirming mechanism carries at the grid bids. Once both beliefs lie below the overlap belief, every bid either buyer makes lies at or below $b_1$. In the mechanism used in the runs, a bid below the grid inherits the win probabilities of $b_1$, so it wins nothing against an opponent bidding above $b_1$. Self-confirmation then obliges the mechanism to deliver $h(b, \theta^*)/p$ against an opponent bidding at or below $b_1$, which is twice the self-confirming level. The opponent's optimism therefore raises the buyer's win probability above what his model predicts, and with $h$ decreasing in $\theta$ he reads the surplus as evidence of a low $\theta$. Each buyer's optimism thus feeds the other's, and beliefs fall until the model's predictions at the bids made have grown by the factor $1/p$. The low type's prediction does so almost at once and the high type's only further down, and the chains settle between the two.

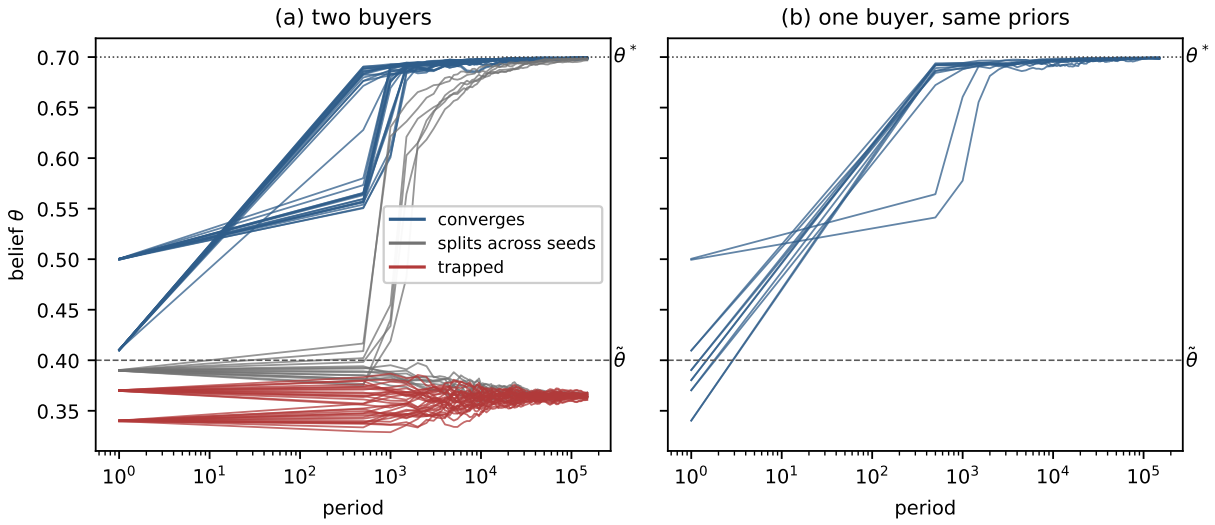



Figure 4: The same six priors under two buyers and under one. In panel (a) the two-buyer runs divide into three fates, and the trapped chains stay below the overlap belief $\tilde{\theta}$. In panel (b) the single buyer converges from all six. Time is on a logarithmic scale.

[15] From the two priors concentrated above the overlap belief and from the uniform prior, beliefs converge to $\theta^*$. From the remaining prior, the one closest to the overlap belief, the chains divide between the two fates. Appendix 8.4 explains why trapping requires both beliefs to lie below the overlap belief.

The two instances differ in how much objective responsiveness feasibility forces. At the first, $h_2$ barely exceeds one half, so feasibility forces little of it, and some mechanisms in the class sustain the target and others do not. At the second, $h_2 = (1+p)/2$ pins the grid values to a single assignment, whose objective responsiveness exceeds subjective responsiveness at both bids. Approaching the Border frontier shrinks the class of feasible self-confirming mechanisms and at the same time raises the objective responsiveness that every member of the class carries.

# 5 Discussion

Sections 3 and 4 say when a target can be sustained, by what mechanism, and where sustaining fails. This section considers what follows, for the seller who would act on these results and for the conditions that produce them.

## 5.1 Revenue

What do the convergence results imply for revenue? The answer depends on whether the buyer's model is value-dependent. When the model is value-independent, meaning $g(b,v,\theta) = \tilde{g}(b,\theta)$ for some function $\tilde{g}$, self-confirmation forces each buyer to optimise against his objective interim win probability, and revenue equivalence precludes exceeding Myerson's optimal auction. Value-independent models encompass settings where the buyer is uncertain about aspects of the mechanism that do not interact with his private value, such as a fixed reserve price or the number of competing bidders. When the model is value-dependent, this link between the buyer's subjective and objective win probabilities breaks down, and the expected revenue bears no definitive relation to Myerson's benchmark.

**Proposition 5** (Revenue Bound)**.** *If $g(b,v,\theta)$ is constant over $v$, then there is no target $\theta^*$ for which a feasible self-confirming mechanism can simultaneously sustain $\theta^*$ and generate expected revenue exceeding Myerson's optimal auction.*

Intuitively, under value-independent beliefs, $h(b,\theta) = \tilde{g}(b,\theta)$ for all $b$ and $\theta$, since the $v$-dependence that would ordinarily enter through $\beta^{-1}(b,\theta)$ drops out. Self-confirmation then gives $Q(b \mid \theta^*) = \tilde{g}(b,\theta^*)$ for all on-path bids, where $Q(b \mid \theta^*)$ is the objective interim win probability when the opponent also holds belief $\delta_{\theta^*}$. Once beliefs have converged to $\theta^*$, each buyer optimises against $\tilde{g}(b,\theta^*)$ and therefore optimises against his objective interim win probability. His on-path behaviour coincides with that of a correctly specified buyer facing the true allocation rule. The Bayesian incentive compatibility conditions are therefore satisfied, and revenue equivalence implies the seller cannot exceed Myerson's benchmark.

When the model is value-dependent, this coincidence no longer obtains. Even with both buyers holding the target belief $\delta_{\theta^*}$, a buyer with value $v$ best-responds to his subjective win probability $g(b,v,\theta^*)$, which differs from the objective interim win probability $Q(b \mid \theta^*)$ at bids other than $\beta(v,\theta^*)$. Self-confirmation ensures agreement at $\beta(v,\theta^*)$, but not globally, so the incentive compatibility conditions need not hold and revenue equivalence does not apply. We illustrate using the single-buyer correlated reserve price model of Section 2.4.1, where the revenue gains from belief steering can be substantial.

### 5.1.1 Revenue in the Correlated Reserve Price Model

Consider the correlated reserve price model with $\psi(v,\theta) = v\theta$, noise $\varepsilon \sim N(0,\sigma^2)$, and $v \sim \text{Uniform}[0,1]$. The buyer's model is $g(b,v,\theta) = \Phi\left(\frac{b-v\theta}{\sigma}\right)$, where $\Phi$ denotes the standard normal CDF. As established in Section 2.4.1, this model satisfies conditions (R1)–(R3), and Proposition 2 applies. The seller chooses $\theta^* \in [0,1]$ to maximise expected revenue under the self-confirming mechanism.

A buyer with value $v$ and degenerate belief $\delta_{\theta^*}$ maximises $(v-b)\Phi\left(\frac{b-v\theta^*}{\sigma}\right)$. The first-order condition yields

$$(v-b)\cdot\frac{\varphi(z)}{\sigma} = \Phi(z), \text{ where } z = \frac{b-v\theta^*}{\sigma}, \tag{6}$$

and $\varphi$ denotes the standard normal PDF. Rearranging the terms, we obtain

$$v(1-\theta^*) = \sigma\cdot\rho(z), \text{ where } \rho(z) \equiv z + \frac{\Phi(z)}{\varphi(z)}. \tag{7}$$

The function $\rho$ is strictly increasing and unbounded, so the bidding strategy is uniquely determined by

$$\frac{\beta(v,\theta^*)-v\theta^*}{\sigma} = \rho^{-1}\left(\frac{v(1-\theta^*)}{\sigma}\right) \iff \beta(v,\theta^*) = v\theta^* + \sigma\rho^{-1}\left(\frac{v(1-\theta^*)}{\sigma}\right).$$

Under the self-confirming mechanism, the buyer with value $v$ wins with probability $\Phi\left(\frac{\beta(v,\theta^*)-v\theta^*}{\sigma}\right)$, and the seller's expected revenue is

$$\text{Rev}(\theta^*) = \int_0^1 \left[v\theta^* + \sigma\rho^{-1}\left(\frac{v(1-\theta^*)}{\sigma}\right)\right]\Phi\left(\frac{\beta(v,\theta^*)-v\theta^*}{\sigma}\right)dv. \tag{8}$$

The revenue expression has a transparent structure. The term in brackets is the buyer's bid, which is his perceived fair price $v\theta^*$ plus a noise-dependent markup. The second term is his win probability under the self-confirming mechanism. The seller's revenue is therefore the expected winning bid, weighted by the probability of sale.

Figure 5 plots $\text{Rev}(\theta^*)$ for three values of $\sigma$. In each case shown, the seller's optimisation problem has a unique interior maximum; its location and height depend on $\sigma$. However, exceeding Myerson's benchmark is not guaranteed even under value-dependent beliefs. For $\sigma = 0.10$, the optimal revenue falls below the Myerson benchmark. The key tension is between inducing aggressive bids and satisfying the self-confirming constraint. A higher $\theta^*$ causes the buyer to bid more aggressively, but the self-confirming mechanism must deliver a win probability matching the buyer's expectations. When $\sigma$ is large, even a buyer who bids aggressively perceives substantial uncertainty about whether his bid exceeds the reserve price, and therefore expects to win with only moderate probability. The self-confirming mechanism must match this expectation by withholding the good frequently, which reduces revenue.

As $\sigma$ decreases, this tension dissipates. For $\sigma = 0.05$, the optimal $\theta^* \approx 0.72$ yields approximately 1.19 times Myerson revenue. For $\sigma = 0.02$, the optimal $\theta^* \approx 0.82$ yields approximately 1.47 times Myerson revenue. As $\sigma$ decreases further, the optimal $\theta^*$ shifts toward 1, and the ex-

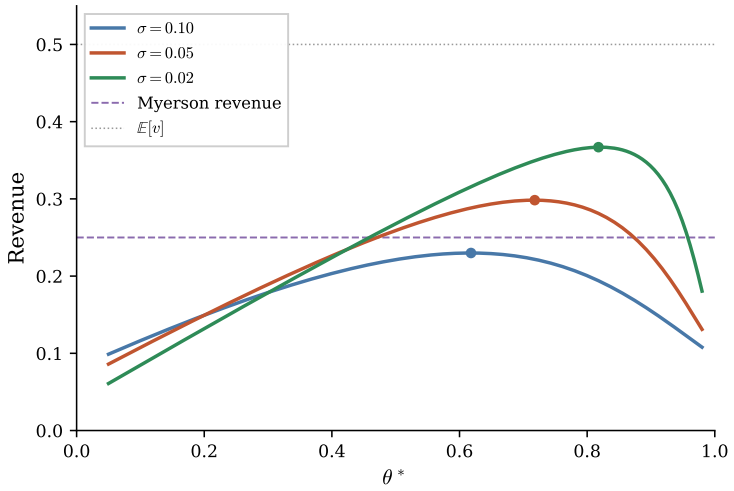



Figure 5: Revenue as a function of $\theta^*$ in the correlated reserve price model with $\psi(v,\theta) = v\theta$ and $v \sim \text{Uniform}[0,1]$. Each curve corresponds to a different noise level $\sigma$. Dots indicate the revenue-maximising $\theta^*$. The dashed line marks Myerson's optimal auction revenue; the dotted line marks $\mathbb{E}[v] = 1/2$.

pected revenue approaches full surplus extraction.[16] Thus, while Proposition 5 shows that value-independence makes super-Myerson revenue impossible, value-dependence makes it possible but not automatic.

## 5.2 Robustness

These results assume the seller knows the buyer's model $g$ perfectly. In practice, this is unrealistic, since the model is an object that exists entirely in the buyer's mind. This raises the concern that errors in the seller's knowledge of $g$ could undermine both the convergence and revenue conclusions. The following result shows this concern is limited in the single-buyer setting. Approximate knowledge suffices for approximate convergence.

**Proposition 6** (Robustness)**.** *Suppose the model is adversely ordered, and let $q_{\theta^*}(b) = h(b,\theta^*)$ be the self-confirming allocation rule for $g$. For any $\delta > 0$ and any open $B \subseteq \Delta\Theta$ with $\Delta([\theta^* - \delta, \theta^* + \delta] \cap \Theta) \subseteq B$, there exists $\varepsilon > 0$ such that if the buyer's true model $\tilde{g}(b,v,\theta)$ is continuous with $\sup_{b,v,\theta} |\tilde{g}(b,v,\theta) - g(b,v,\theta)| \leq \varepsilon$, then for every initial belief $\mu_0 \in \Delta\Theta$ and every selection rule the induced belief distributions satisfy*

$$\lim_{t\to\infty} \gamma_t(B \mid \mu_0, q_{\theta^*}(\cdot)) = 1.$$

The proposition says that if the seller designs the self-confirming mechanism for a model $g$ that is $\varepsilon$-close to the buyer's true model $\tilde{g}$, beliefs still converge to a neighbourhood of $\theta^*$ whose size is

[16] Fix $\theta^* < 1$. As $\sigma \to 0$, $\frac{v(1-\theta^*)}{\sigma} \to \infty$ for all $v > 0$, so the win probability $\Phi\left(\frac{\beta(v,\theta^*) - v\theta^*}{\sigma}\right) \to 1$. Since the buyer optimises, $v\theta^* < \beta(v,\theta^*) < v$, so revenue satisfies $\theta^*/2 < \text{Rev}(\theta^*) < 1/2$ for $\sigma$ sufficiently small. Choosing $\theta^*$ arbitrarily close to 1 gives $\sup_{\theta^*} \text{Rev}(\theta^*) \to \mathbb{E}[v] = 1/2$. The argument extends to any log-concave noise distribution, requiring only that $\rho(z) = z + G(z)/G'(z)$ is strictly increasing and unbounded.

controlled by the approximation error $\varepsilon$. Beyond continuity, the true model $\tilde{g}$ need not satisfy any of the regularity conditions imposed on $g$. This means the convergence results are robust not just to misspecification of the model parameters, but also to small violations of the regularity conditions themselves.

From a revenue perspective, Proposition 6 implies that the seller does not need to know the buyer's model exactly to capture the revenue benefits of belief steering. If she has approximate knowledge of $g$ and designs a self-confirming mechanism accordingly, the buyer's limiting belief will be close to the target $\theta^*$, and his limiting bidding strategy will be close to the desired one. Small errors in the seller's knowledge of $g$ therefore can lead to only small losses in expected revenue relative to the full-information benchmark.

### 5.3 Beyond One-Sided Stability

The step form underlies the one-sided results. Within the maintained class, imposing it is without loss for the one-sided question, and the boundary of those results lies where it gives way.

The reduction of the one-sided convergence problem to the linear program of Proposition 4 relies on the step form, which makes the mechanism a step function in the opponent's bid $b_2$ with jumps only at grid bids $B(\theta^*)$. Under this structure, the objective win probability $Q(b_1 \mid \tilde{\theta})$ depends on the opponent's bid distribution only through the probabilities that his bid falls between consecutive grid bids, and the infinite-dimensional verification over all $\tilde{\theta} < \theta^*$ collapses to a finite system of inequalities in the grid values.

As Section 3.3.2 explains, the step form is without loss on the optimistic side. Rounding $b_2$ up to the nearest grid bid binds the monotonicity constraint, making $q$ as small as possible between grid bids. This is the right direction for every $\tilde{\theta} < \theta^*$ simultaneously, so the step-function form is without loss of generality.

Extending this to the pessimistic side fails for two related reasons. First, part (a) of Condition 1 can require $q(b_1, b_2)$ to be large at off-grid opponent bids for $\tilde{\theta} > \theta^*$, so rounding up unconditionally is no longer valid, and the off-grid component of the mechanism can no longer be read off from the on-grid values by a single rule. Second, and more fundamentally, making $q(b_1, b_2)$ large at off-grid $b_2$ means either binding the monotonicity constraint downward or binding the feasibility constraint. The former would still produce a step function, but the latter does not. When feasibility binds, $q(b_1, b_2) = 1 - q(b_2, b_1)$, and $q(b_2, b_1)$ at off-grid $b_2$ varies continuously in $b_2$ through the cap that restores self-confirmation at an off-grid own bid, which depends on $h(b_2, \theta^*)$. The step-function structure in $b_2$ is destroyed. Without it, $Q(b_1 \mid \tilde{\theta})$ depends on the full shape of $q$ between grid bids rather than just the probabilities of the intervals between grid bids, and the reduction from a continuum of $\tilde{\theta}$'s to the finite set $\tilde{\Theta}$ no longer holds.

## 6 Conclusion

This paper studies buyers who hold misspecified models of their environment and learn from repeated participation in a fixed selling mechanism. For a given model and belief, it asks when there exists a mechanism that sustains that belief while buyers keep learning, and how such a mechanism does it. The mechanism partially determines the data from which the buyers learn, so the seller, in choosing it, chooses among the long-run conclusions that learning can reach, even when

the models drawing those conclusions are fundamentally incorrect. To sustain a belief in this setting is to supply outcomes under which learning itself holds it in place. A mechanism that does so exists when the win probability predicted by the buyer's model responds strongly enough to a change in belief, relative to how bidding behaviour responds. A self-confirming mechanism can then keep objective responsiveness below subjective responsiveness, uniformly over bids. Under such a mechanism, buyers who share a belief more optimistic than the one being sustained win less often than they expect, and buyers who share a more pessimistic belief win more often, so that the errors on either side push beliefs back toward it. Feedback of this kind is corrective, and a mechanism generating corrective feedback sustains the belief (Proposition 1).

With a single buyer no belief moves the win probability he faces, so objective responsiveness is zero, and conditions on his model alone make convergence global, with beliefs reaching the target from every full-support prior. With two buyers, each buyer's evidence depends also on how the other bids, and the analysis restricts to symmetric mechanisms monotone in the opponent's bid. Within this class our conditions guarantee local stability, with beliefs that begin near the target remaining near it and converging to it. The central condition is that subjective responsiveness at the target exceed the largest objective responsiveness that a self-confirming mechanism can have. Sustaining the target against all beliefs below it demands more structure. With discrete valuations and under further conditions on the model, a strictly feasible solution of a finite linear program yields a mechanism that does so. The conditions are sufficient but not necessary, and Section 4 exhibits both convergence and its failure at computed instances.

These findings carry implications for the regulation of opaque marketplaces. If a platform can steer buyer beliefs through the choice of mechanism alone, then opacity is not merely a loss of commitment power or a source of buyer ambiguity, but an active instrument for shaping market outcomes. What belief steering achieves for revenue depends on whether the buyer's model is value-dependent. Under value-independent models no feasible self-confirming mechanism that sustains a target generates revenue exceeding Myerson's optimal auction (Proposition 5). Under value-dependent models the single-buyer correlated reserve price example shows what becomes possible, revenue at the sustained belief exceeding that benchmark at small noise and approaching full surplus extraction as the noise vanishes, while falling short of it when the noise is large. The robustness result (Proposition 6) implies that in the single-buyer setting the seller need not know the buyer's model exactly to capture such gains, since approximate knowledge suffices.

The analysis restricts the buyer's observation to a binary outcome, whether he won. The natural next step is to let him observe continuous signals. Such an extension opens the door to settings beyond selling mechanisms, where continuous feedback is the norm. Insurance contracts, employment agreements, and platform matching algorithms may not be literally hidden from participants, but their complexity can render them effectively opaque. Agents interact repeatedly with rules they do not fully understand, forming and revising beliefs based on experienced outcomes. Our approach would provide a natural framework for studying the design of such rules. Which of these rules can sustain a belief, and how, would likely again come down to where they make prediction errors corrective.

# 7 Appendix A

## 7.1 Convergence Framework

### 7.1.1 Base Notions

The following notions specialize the learning framework of Frick et al. (2023) to the notation of this paper. The definitions of KL-Dominance and of a strict B-NE set are the un-margined baseline anchoring their margined counterparts and are not themselves used in the arguments. Throughout, $\Delta\Theta$ carries the Prokhorov metric, and neighbourhoods of beliefs and of sets of beliefs, here and in Definition 1, are taken with respect to it, as in Frick et al. (2023).

Denote the outcome observed by the agent by $z \in Z$. For any $\mu \in \Delta\Theta$, let

$$\mathbb{P}_\mu = \left\{ (p^*_\mu(z), \{\hat{p}_\mu(z|\theta)\}_{\theta\in\Theta}) \right\}$$

be the set of all pairs of objective and subjective outcome distributions that arise under belief $\mu$. We denote a specific element of $\mathbb{P}_\mu$ by $p_\mu = (p^*_\mu(z), \{\hat{p}_\mu(z|\theta)\}_{\theta\in\Theta})$, where the first element denotes the objective outcome distribution and the second element denotes the family of subjective outcome distributions under various $\theta$.

The mapping from beliefs to outcome distributions is a correspondence rather than a function, reflecting possible non-uniqueness of optimal bids. The margin-based framework below accommodates the resulting multiplicity through adapted selections, as recorded in Remark 6.

A selection rule chooses, in each period, a bid profile from the bidding correspondence given the realised valuations and current beliefs. A selection rule is *adapted* if its period-$t$ choice depends only on the history realised through period $t$; since bids are chosen sequentially, every selection rule in the model is adapted, and the term makes the measurability explicit rather than imposing a restriction. To construct the induced belief processes, fix an allocation rule $\boldsymbol{q}$ and a current belief profile $\boldsymbol{\mu}$. A belief profile $\boldsymbol{\mu}'$ is *consistent with* $(\boldsymbol{q}, \boldsymbol{\mu})$ if it can arise as buyers' posteriors following some realisation of $(\boldsymbol{y}, \boldsymbol{b}, \boldsymbol{v})$ with $b_i \in \beta^i(v_i, \mu_i)$ for $i = 1,2$ and $\boldsymbol{q}(\boldsymbol{y} \mid \boldsymbol{b}) > 0$. A period-$t$ belief profile $\boldsymbol{\mu}_t$ is consistent with $(\boldsymbol{q}, \boldsymbol{\mu}_0)$ if there exist belief profiles $\{\boldsymbol{\mu}_\tau\}_{\tau=0,\ldots,t}$ such that $\boldsymbol{\mu}_\tau$ is consistent with $(\boldsymbol{q}, \boldsymbol{\mu}_{\tau-1})$ for all $\tau = 1,\ldots,t$. We denote by $\Gamma_t(\boldsymbol{\mu}, \boldsymbol{q})$ the set of distributions over period-$t$ belief pairs induced by some adapted selection rule under mechanism $\boldsymbol{q}$ and initial belief $\boldsymbol{\mu}$; the marginals $\gamma_{i,t}(\cdot \mid \boldsymbol{\mu}, \boldsymbol{q})$ of its elements are the distributions quantified over in Definition 1, whose conclusion accordingly reads

$$\lim_{t\to\infty} \gamma_{i,t}(U_i \mid \boldsymbol{\mu}, \boldsymbol{q}) \geq 1 - \varepsilon \qquad \text{for all } \gamma_t \in \Gamma_t(\boldsymbol{\mu}, \boldsymbol{q}) \text{ and } i \in \{1,2\},$$

the form used in the arguments below. By Remark 6, every bound in the supporting lemmas holds uniformly over $\mathbb{P}_\mu$ at every history, so the conclusions hold under every adapted selection rule.

**Definition 5** (KL-Dominance)**.** *Let $\theta', \theta'' \in \Theta$, and let $\mu \in \Delta\Theta$. We say that $\theta' \succsim^{KL}_\mu \theta''$ if $\forall p_\mu \in \mathbb{P}_\mu$:*

$$\int_z \log\left(\frac{\hat{p}_\mu(z|\theta'')}{\hat{p}_\mu(z|\theta')}\right) p^*_\mu(z)dz < 0$$

Intuitively, if $\theta'$ KL dominates $\theta''$ under belief $\mu$, then $\theta'$ is a better explanation of any objective outcome distribution associated with $\mu$ than $\theta''$.

**Definition 6** (Locally Stable Set)**.** *$M \subseteq \Delta\Theta$ is a locally stable set if for any $\gamma < 1$, there exists a neighbourhood $B$ of $M$ such that $\forall \mu_0 \in B$:*

$$\mathbb{P}(\inf_{\nu \in M} d(\mu_t, \nu) \to 0) \geq \gamma$$

**Definition 7** (Globally Stable Set)**.** *$M \subseteq \Delta\Theta$ is a globally stable set if for every initial belief $\mu_0 \in \Delta\Theta$:*

$$\mathbb{P}(\inf_{\nu \in M} d(\mu_t, \nu) \to 0) = 1$$

**Definition 8** (Strict B-NE Set)**.** *For each non-empty measurable $\Theta' \subseteq \Theta$, $\Delta\Theta'$ is a strict B-NE set if for all $\theta \notin \Theta'$, $\exists \theta' \in \bar{\Theta}'$ such that:*

$$\theta' \succsim_{\mu}^{KL} \theta \; \forall \mu \in \Delta\bar{\Theta}'$$

**Remark 1.** *Intuitively, a strict B-NE set is one whose complement is uniformly dominated in fit, in that every parameter outside the set is a strictly worse explanation than some parameter in its closure at every belief supported on the closure.*

### 7.1.2 Uniform-Margin Framework

The following definitions and results replace the belief-continuity-dependent framework of Frick, Iijima, and Ishii with a belief-continuity-free alternative. The outcome space, notation $\mathbb{P}_\mu$, KL-Dominance, locally stable sets, and strict B-NE sets are as defined above.

**Definition 9** (Margined KL-Dominance)**.** *Let $M \subseteq \Delta\Theta$, $\theta', \theta'' \in \Theta$, and $\kappa > 0$. We say $\theta'$ KL-dominates $\theta''$ with margin $\kappa$ on $M$, written $\theta' \succsim_{M}^{KL(\kappa)} \theta''$, if for all $\mu \in M$ and all $p_\mu \in \mathbb{P}_\mu$:*

$$\int_z \log\left(\frac{\hat{p}_\mu(z|\theta'')}{\hat{p}_\mu(z|\theta')}\right) p_\mu^*(z)\, d\nu(z) \leq -\kappa$$

**Definition 10** (Bounded Likelihood Ratios)**.** *Let $M \subseteq \Delta\Theta$ and $L < \infty$. We say likelihood ratios are bounded by $L$ on $M$ if for all $\mu \in M$, all $p_\mu \in \mathbb{P}_\mu$, and all $\theta', \theta'' \in \Theta$:*

$$\left|\log \frac{\hat{p}_\mu(z|\theta'')}{\hat{p}_\mu(z|\theta')}\right| \leq L \quad \text{for } p_\mu^*\text{-almost every } z$$

**Lemma 2** (Uniform $q$-Dominance)**.** *Let $M \subseteq \Delta\Theta$ and suppose likelihood ratios are bounded by $L$ on $M$. If $\theta' \succsim_{M}^{KL(\kappa)} \theta''$ for some $\kappa > 0$, then, setting*

$$\bar{q} = \min\left\{1, \frac{\kappa}{L^2 e^L}\right\}$$

*every $q \in (0, \bar{q}]$ satisfies, for all $\mu \in M$ and all $p_\mu \in \mathbb{P}_\mu$:*

$$\int_z \left(\frac{\hat{p}_\mu(z|\theta'')}{\hat{p}_\mu(z|\theta')}\right)^q p_\mu^*(z)\, d\nu(z) \leq 1 - \frac{q\kappa}{2}$$

*Proof.* Fix $\mu \in M$, $p_\mu \in \mathbb{P}_\mu$, and $q \in (0, \bar{q}]$, and write $X(z) = \log \frac{\hat{p}_\mu(z|\theta'')}{\hat{p}_\mu(z|\theta')}$, so that $|X(z)| \le L$ for $p^*_\mu$-almost every $z$ and $\int X\, dP^*_\mu \le -\kappa$, where $dP^*_\mu = p^*_\mu\, d\nu$. For any $y$ with $|y| \le L$, the Taylor expansion of the exponential gives

$$e^y - 1 - y = \sum_{k\ge 2} \frac{y^k}{k!}, \qquad \left|\sum_{k\ge 2} \frac{y^k}{k!}\right| \le \frac{y^2}{2} \sum_{j\ge 0} \frac{|y|^j}{j!} = \frac{y^2}{2} e^{|y|}$$

using $k! \ge 2(k-2)!$ for $k \ge 2$. Applying this with $y = qX(z)$, and using $q \le 1$ and $|X(z)| \le L$:

$$e^{qX(z)} \le 1 + qX(z) + \frac{q^2 L^2}{2} e^L$$

Integrating against $P^*_\mu$ and using the margin:

$$\int e^{qX}\, dP^*_\mu \le 1 + q\int X\, dP^*_\mu + \frac{q^2L^2}{2} e^L \le 1 - q\kappa + \frac{q^2L^2}{2} e^L$$

Because $q \le \bar{q} \le \frac{\kappa}{L^2 e^L}$, the last term is at most $\frac{q\kappa}{2}$, and the claim follows. □

**Remark 2.** *Bounded likelihood ratios make the integrability condition of Assumption 1.2 of Frick et al. (2023) hold automatically, so the framework below carries no separate integrability hypothesis. Equicontinuity of likelihoods takes over the role of their Assumption 2, continuity of likelihoods in the state, whose every use lies within the steps re-derived here.*

**Remark 3.** *In the selling environments of this paper, bounded likelihood ratios hold with an explicit constant. On any compact domain of $(b, v, \theta)$, regularity (R1) gives $c \le g(b, v, \theta) \le 1 - c$ for some $c > 0$. In the single-buyer setting the outcome is $z = (y, v)$ with the $v$-marginal independent of $\theta$, so every likelihood ratio is a ratio of $g$ values or $1 - g$ values and $L = \log \frac{1-c}{c}$ suffices. In the two-buyer setting the ratio is a product of two such terms and $L = 2\log \frac{1-c}{c}$ suffices.*

**Definition 11** (Equicontinuous Likelihoods)**.** *Let $M \subseteq \Delta\Theta$. We say likelihoods are equicontinuous on $M$ if for every $\varepsilon > 0$ there exists $\delta > 0$ such that for all $\mu \in M$ and all $p_\mu \in \mathbb{P}_\mu$, for $p^*_\mu$-almost every $z$, all $\theta_1, \theta_2 \in \Theta$ with $|\theta_1 - \theta_2| \le \delta$ satisfy*

$$\left|\log \frac{\hat{p}_\mu(z|\theta_1)}{\hat{p}_\mu(z|\theta_2)}\right| \le \varepsilon$$

*the exceptional set of $z$ not depending on the pair $(\theta_1, \theta_2)$.*

**Remark 4.** *In the selling environments of this paper the condition holds on all of $\Delta\Theta$. Regularity (R1) makes $\log g$ and $\log(1-g)$ Lipschitz in $\theta$ with a common constant on the compact reachable domain, and the modulus carries over to the embedded likelihoods, in the two-buyer case with twice the constant.*

**Definition 12** (Margined Eliminator)**.** *For any $\Theta' \subseteq \Theta$, define $S_m(\Theta')$ as:*

$$S_m(\Theta') = \left\{ \theta \in \bar{\Theta}' \;\middle|\; \nexists\, \theta' \in \bar{\Theta}',\ \kappa > 0,\ \text{open } M \supseteq \Delta\bar{\Theta}' \text{ s.t. } \theta' \succsim^{KL(\kappa)}_M \theta \right\}$$

*where $\bar{\Theta}'$ is the closure of $\Theta'$, openness of $M$ is in $\Delta\Theta$ under the topology of weak convergence, and $S^k_m$, $S^\infty_m$ denote the iterates and their intersection.*

**Theorem 1** (Global Stability under Margined Elimination)**.** *Suppose $\Theta$ is compact, likelihood ratios are bounded by some $L < \infty$ on $\Delta\Theta$, and likelihoods are equicontinuous on $\Delta\Theta$. Then $\Delta S_m^\infty(\Theta)$ is globally stable.*

**Definition 13** (Margined Strict B-NE Set)**.** *For each non-empty measurable $\Theta' \subseteq \Theta$, $\Delta\Theta'$ is a margined strict B-NE set if for all $\theta \notin \Theta'$, there exist $\theta' \in \bar{\Theta}'$, $\kappa > 0$, and an open $M \supseteq \Delta\bar{\Theta}'$ such that $\theta' \succsim_M^{KL(\kappa)} \theta$.*

**Theorem 2** (Local Stability under Margined Dominance)**.** *Suppose $\Theta' \subseteq \Theta$ is open, and likelihood ratios are bounded by some $L < \infty$ and likelihoods equicontinuous on some open $M_0 \supseteq \Delta\bar{\Theta}'$. If $\Delta\Theta'$ is a margined strict B-NE set, then $\Delta S_m^k(\Theta')$ is a locally stable set for all $k = 0, 1, \ldots, \infty$.*

**Remark 5.** *Let $\mathscr{P} \subseteq \Delta\Theta$ be closed and invariant under the Bayesian updating dynamics. Theorems 1 and 2 hold verbatim with every belief quantifier relativized to $\mathscr{P}$: the margins in Definitions 12 and 13 are required only for beliefs on $M \cap \mathscr{P}$, the regularity conditions only on $\mathscr{P}$, and the stability conclusions apply to processes started in $\mathscr{P}$.*

The stability theorems were stated for the iterates of the eliminator $S_m$, which removes every parameter admitting a witness at each stage; their proofs consume only per-stage witnesses and compactness, never this maximality. The following corollary records the resulting flexibility.

**Corollary 2.** *Let $\mathscr{P} \subseteq \Delta\Theta$ be closed and invariant under the Bayesian updating dynamics, in the sense that $\mu_0 \in \mathscr{P}$ implies $\mu_t \in \mathscr{P}$ for all $t$ almost surely. Suppose likelihood ratios are bounded and likelihoods equicontinuous on $\mathscr{P}$. Let $R^0 \supseteq R^1 \supseteq \cdots$ be nested non-empty compact subsets of $\Theta$ such that every $\theta \in R^{k-1} \setminus R^k$ admits a witness $(\varphi(\theta), \kappa_\theta, M_\theta)$ with $\varphi(\theta) \in R^{k-1}$, $\kappa_\theta > 0$, and $M_\theta$ a relatively open subset of $\mathscr{P}$ containing $\Delta R^{k-1} \cap \mathscr{P}$, such that $\varphi(\theta) \succsim_{M_\theta}^{KL(\kappa_\theta)} \theta$. Then:*

*(i) If $R^0 = \Theta$, then $\Delta \bigcap_k R^k$ is globally stable for processes in $\mathscr{P}$.*

*(ii) If $R^0 = \bar{\Theta}'$ for $\Theta' \subseteq \Theta$ open and $\Delta\Theta'$ is a margined strict B-NE set relative to $\mathscr{P}$, then for every $\gamma < 1$ there is a neighbourhood $B_0$ of $\Delta\Theta' \cap \mathscr{P}$ such that, for every $k = 0, 1, \ldots, \infty$, every process started in $B_0 \cap \mathscr{P}$ satisfies $\mathbb{P}(\inf_{\nu \in \Delta R^k} d(\mu_t, \nu) \to 0) \geq \gamma$. In particular each $\Delta R^k$ is a locally stable set for processes in $\mathscr{P}$.*

### 7.1.3 Framework Proofs and Supporting Lemmas

**Remark 6.** *The belief process realizes, in each period, some $p_{\mu_t} \in \mathbb{P}_{\mu_t}$, and we require only that the realized selection process is adapted. Every bound in the supporting lemmas below, in particular the conclusion of Lemma 5(ii), holds uniformly over $\mathbb{P}_\mu$, so each conditional one-step inequality holds at every history regardless of which selection the period realizes. Since the arguments of Frick et al. (2023) consume only these conditional one-step inequalities together with pathwise properties of the resulting processes, all conclusions hold for every adapted selection process. Under the full-support assumption, Lemma 3 below keeps every set of positive prior mass charged at all times, so the ratio processes are finite and the positivity qualifier of Lemma 5(ii) holds along every trajectory.*

**Lemma 3** (Support Persistence)**.** *Suppose likelihood ratios are bounded by $L < \infty$ on $\Delta\Theta$. Then for every adapted selection and every measurable $O \subseteq \Theta$ with $\mu_0(O) > 0$, almost surely $\mu_t(O) \geq e^{-Lt}\mu_0(O) > 0$ for all $t$.*

*Proof.* For almost every signal $z$, $\hat{p}_\mu(z \mid \theta) \geq e^{-L}\hat{p}_\mu(z \mid \theta')$ for all $\theta, \theta'$, so $\int_O \hat{p}_{\mu_t}(z \mid \theta)\, d\mu_t \geq e^{-L}\mu_t(O)\int_\Theta \hat{p}_{\mu_t}(z \mid \theta)\, d\mu_t$, and Bayes' rule gives $\mu_{t+1}(O) \geq e^{-L}\mu_t(O)$ at every history. Induction over the countably many periods completes the proof. □

**Remark 7.** *The module consumes the arguments of Frick et al. (2023) from precisely delimited points. In the proof of their Lemma 15, everything from the final paragraph, the one opening with the words "For the 'moreover' part", is adopted, namely the derivation of their (B.3) from their (B.2) through the stopped processes $\ell_t^i$, Doob convergence, Markov's inequality, and the conditional contraction; their Claims 1 and 2 and the finite subcover preceding that paragraph, where belief continuity enters, are replaced by Lemma 5(ii). The correspondence of notation reads $\Omega \mapsto \Theta$, $\omega \mapsto \theta$, $\Omega' \mapsto E$, $D \mapsto D$, $p_\mu \mapsto p^*_\mu$, and $\hat{p}_\mu(z \mid \omega) \mapsto \hat{p}_\mu(z \mid \theta)$; their family $\{A_i\}, \{A'_i\}, q_i, \varepsilon$ with bound $1-\varepsilon$ in (B.2) maps to the family of Lemma 5(ii) with the common exponent $\bar{q}$ in place of each $q_i$, the bound $1-\bar{q}\zeta$ in place of $1-\varepsilon$, and the radius $\varepsilon$ unchanged. The two roles their single $\varepsilon$ plays, contraction slack and neighbourhood radius, are separated here into $\zeta$ and $\varepsilon$. The proofs of their Lemmas 16 and 17 are adopted in full under the same dictionary, with the substitutions of Lemma 5(i) for their Lemma 14 and Lemma 5(ii) for their invocation of Lemma 15, applied to the eliminated compact and the stage simplex each indicates. From the proof of their Theorem 4, the anchoring of the basin before the induction over stages and the passage sending the per-stage Lyapunov weight to one are adopted as they stand. Every step whose hypotheses differ from theirs is re-derived, namely the construction of the dominating family and the compactness and non-emptiness of the survivor sets.*

*Proof of Theorem 1.* Write $\Theta^k = S_m^k(\Theta)$ for $k = 0, 1, \ldots$, a nested sequence of non-empty compact sets by Lemma 5(i) applied inductively with $M_0 = \Delta\Theta$, so that $S_m^\infty(\Theta)$ is non-empty and compact by Cantor's intersection theorem.

Following the proof of Theorem 3 of Frick et al. (2023), it suffices to show inductively that $\mathbb{P}_{\mu_0}[\mu_t(\Theta^k) \to 1] = 1$ for every initial belief and every $k$, the case $k = 0$ being trivial. For the inductive step at stage $\kappa$, take $N$ with $\Theta \setminus B_{1/N}(\Theta^\kappa)$ non-empty and set $E = \Theta \setminus B_{1/N}(\Theta^\kappa)$ and $D = \Delta\Theta^{\kappa-1}$. Every $\hat{\theta} \in E$ lies outside $\Theta^\kappa$, hence lies in $\Theta^k \setminus \Theta^{k+1}$ for some $k \leq \kappa - 1$, and therefore admits a witness by Definition 12 whose neighbourhood contains $\Delta\Theta^k \supseteq \Delta\Theta^{\kappa-1} = D$. Lemma 5(ii) thus yields the family $\{(A_i, A'_i)\}_{i=1}^I$, constants $\bar{q}, \zeta$, and $\varepsilon > 0$ with the one-step bound $1 - \bar{q}\zeta$ on $B_\varepsilon(D)$.

This family is exactly the input that the remainder of the argument of Frick et al. (2023) consumes. Their derivation within the proof of their Lemma 15 uses only the stopped processes $\ell_t^i = (\mu_{\min\{t,\tau\}}(A_i)/\mu_{\min\{t,\tau\}}(A'_i))^{\bar{q}}$ with $\tau$ the exit time from $B_\varepsilon(D)$, the supermartingale property from the one-step bound, Doob convergence, Markov's inequality, and the conditional contraction with factor $1 - \bar{q}\zeta$; no property of $\mu \mapsto \mathbb{P}_\mu$ enters, and by Remark 6 the supermartingale property holds for every adapted selection. Lyapunov stability of each $\Delta\Theta^k$ follows by induction exactly as in their Lemma 16. The probability chaining and the passage $k \to \infty$ are verbatim as in their proof of Theorem 3. Hence $\Delta S_m^\infty(\Theta)$ is globally stable. □

*Proof of Theorem 2.* Following the proof of Theorem 4 of Frick et al. (2023), the first step is the analogue of their Lemma 17: set $D = \Delta\bar{\Theta}'$ and $E = \Theta \setminus \Theta'$, which is compact since $\Theta'$ is open in the compact $\Theta$. Definition 13 supplies, for every $\hat{\theta} \in E$, a witness with dominator in $\bar{\Theta}'$, positive margin, and open neighbourhood containing $D$, so Lemma 5(ii) yields the family $\{(A_i, A'_i)\}_{i=1}^I$, constants $\bar{q}, \zeta$, and $\varepsilon > 0$ with $\bigcup_i A_i = \Theta \setminus \Theta'$ and the one-step bound on $B_\varepsilon(D)$. The Lyapunov

stability argument of their Lemma 17 and the local stability argument that follows it consume only this family and the derived supermartingale bound; both carry over verbatim for every adapted selection by Remark 6.

For the iterates, $S_m^k(\Theta') \subseteq \bar{\Theta}'$ for every $k$, so $\Delta S_m^k(\Theta') \subseteq \Delta\bar{\Theta}' \subseteq M_0$, and the regularity hypotheses hold on a common open neighbourhood of every iterate's simplex. Lyapunov stability of each $\Delta S_m^k(\Theta')$ follows by the induction of their Lemma 16, with the family at each stage supplied by the special case of Lemma 5(ii) applied to the eliminated compact $S_m^k(\Theta') \setminus B_{1/N}(S_m^{k+1}(\Theta'))$ and $D = \Delta S_m^k(\Theta')$. The inductive chaining and the passage to $k = \infty$ are verbatim as in their proof of Theorem 4. Hence $\Delta S_m^k(\Theta')$ is locally stable for all $k = 0, 1, \ldots, \infty$.

□

*Proof of Corollary 2.* The proofs of Theorems 1 and 2 use the iterates $S_m^k$ through exactly three properties, namely that each is compact and non-empty, the sequence is nested, and every point removed at stage $k$ carries a margined witness with dominator in the stage-$(k-1)$ set and neighbourhood containing the stage-$(k-1)$ simplex. The first is hypothesized here, the second and third likewise. Maximality of the eliminator is invoked nowhere, and Remark 5 supplies the relativization. The inductions therefore apply verbatim to $(R^k)$. The basin in the local case is the neighbourhood their Lemma 17 constructs for $\Delta\Theta'$, fixed before the induction over stages, so the conclusion at every $k$ holds on that common neighbourhood. □

**Lemma 4** (Exponential Moment Bound)**.** *Let $\mu \in \Delta\Theta$, $p_\mu \in \mathbb{P}_\mu$, and let $X$ be measurable with $|X(z)| \le L$ for $P_\mu^*$-almost every $z$ and $\int X\, dP_\mu^* \le -\kappa$ for some $\kappa > 0$. Then for every $q \in \left(0, \min\left\{1, \frac{\kappa}{L^2 e^L}\right\}\right]$:*

$$\int e^{qX}\, dP_\mu^* \le 1 - \frac{q\kappa}{2}$$

*Proof.* Identical to the proof of Lemma 2, which uses only the two displayed properties of $X$. □

**Lemma 5** (Elimination Structure)**.** *Suppose likelihood ratios are bounded by $L < \infty$ and likelihoods are equicontinuous on an open $M_0 \subseteq \Delta\Theta$.*

*(i) For every non-empty $\Theta' \subseteq \Theta$ with $\Delta\bar{\Theta}' \subseteq M_0$, $S_m(\Theta')$ is compact and non-empty.*

*(ii) Let $D \subseteq M_0$ be non-empty and compact, and let $E \subseteq \Theta$ be compact such that every $\hat{\theta} \in E$ admits a witness $(\varphi(\hat{\theta}), \kappa_{\hat{\theta}}, M_{\hat{\theta}})$ with $\varphi(\hat{\theta}) \in \Theta$, $\kappa_{\hat{\theta}} > 0$, and $M_{\hat{\theta}} \supseteq D$ open, such that $\varphi(\hat{\theta}) \succsim_{M_{\hat{\theta}}}^{KL(\kappa_{\hat{\theta}})} \hat{\theta}$. Then there exist $I \in \mathbb{N}$, radii $\rho_i > 0$, centres $\theta_i \in E$ and dominators $\theta_i''' = \varphi(\theta_i)$ for $i = 1, \ldots, I$, constants $\bar{q} \in (0, 1]$, $\zeta > 0$, and $\varepsilon > 0$, such that the sets $A_i = B_{\rho_i}(\theta_i) \cap E$ satisfy $\bigcup_i A_i = E$, and for every $i$, every $\mu \in B_\varepsilon(D)$ with $\mu(A_i) > 0$ and $\mu(A_i') > 0$ where $A_i' = B_{\rho_i}(\theta_i''')$, every $p_\mu \in \mathbb{P}_\mu$, and every $q \in (0, \bar{q}]$:*

$$\int_z \left( \frac{\frac{1}{\mu(A_i)} \int_{A_i} \hat{p}_\mu(z|\theta)\, \mu(d\theta)}{\frac{1}{\mu(A_i')} \int_{A_i'} \hat{p}_\mu(z|\theta)\, \mu(d\theta)} \right)^q p_\mu^*(z)\, d\nu(z) \le 1 - q\zeta$$

*Proof.* Throughout, $\omega(\cdot)$ denotes a modulus for equicontinuity on $M_0$. We prove part (ii) first.

**Step 1: ball extension of margins.** Let $\hat{\theta} \in E$ with witness $(\varphi(\hat{\theta}), \kappa_{\hat{\theta}}, M_{\hat{\theta}})$. Replacing $M_{\hat{\theta}}$ by $M_{\hat{\theta}} \cap M_0$, we may assume $M_{\hat{\theta}} \subseteq M_0$. Choose $\rho_{\hat{\theta}} > 0$ with $\omega(\rho_{\hat{\theta}}) \le \min\{\kappa_{\hat{\theta}}/8, 1/2\}$. Fix any

$\mu \in M_{\hat{\theta}}$ and $p_\mu \in \mathbb{P}_\mu$, and write $X(z) = \log \frac{\hat{p}_\mu(z|\hat{\theta})}{\hat{p}_\mu(z|\varphi(\hat{\theta}))}$, so that $\int X\, dP_\mu^* \le -\kappa_{\hat{\theta}}$ and $|X| \le L$ almost everywhere.

For any probability measures $\hat{\lambda}$ supported on $B_{\rho_{\hat{\theta}}}(\hat{\theta})$ and $\check{\lambda}$ supported on $B_{\rho_{\hat{\theta}}}(\varphi(\hat{\theta}))$, the mixture log-ratio $\tilde{X}(z) = \log\left[\int \hat{p}_\mu(z|\theta)\,\hat{\lambda}(d\theta) \Big/ \int \hat{p}_\mu(z|\theta)\,\check{\lambda}(d\theta)\right]$ satisfies $\tilde{X}(z) \le X(z) + 2\omega(\rho_{\hat{\theta}})$ and $|\tilde{X}(z)| \le L+1 =: L'$ almost everywhere, so:

$$\int \tilde{X}\, dP_\mu^* \le -\kappa_{\hat{\theta}} + 2\omega(\rho_{\hat{\theta}}) \le -\frac{3\kappa_{\hat{\theta}}}{4}$$

By Lemma 4 applied to $\tilde{X}$ with constants $(L', 3\kappa_{\hat{\theta}}/4)$, for every $q \in (0, q_{\hat{\theta}}]$ where $q_{\hat{\theta}} = \min\{1, \frac{3\kappa_{\hat{\theta}}}{4L'^2 e^{L'}}\}$:

$$\int e^{q\tilde{X}}\, dP_\mu^* \le 1 - \frac{3q\kappa_{\hat{\theta}}}{8}$$

**Step 2: part (ii).** The balls $\{B_{\rho_{\hat{\theta}}}(\hat{\theta})\}_{\hat{\theta}\in E}$ cover the compact $E$; extract a finite subcover with centres $\theta_1, \dots, \theta_I$ and set $\theta_i''' = \varphi(\theta_i)$, $\rho_i = \rho_{\theta_i}$, $\bar{q} = \min_i q_{\theta_i}$, and $\zeta = \frac{3}{8}\min_i \kappa_{\theta_i}$. The sets $A_i = B_{\rho_i}(\theta_i) \cap E$ satisfy $\bigcup_i A_i = E$. The set $M = \bigcap_{i=1}^I M_{\theta_i}$ is open and contains $D$; since $D$ is compact, there exists $\varepsilon > 0$ with $B_\varepsilon(D) \subseteq M$. For $\mu \in B_\varepsilon(D)$ with $\mu(A_i), \mu(A_i') > 0$, the conditional distributions of $\mu$ on $A_i$ and $A_i'$ are probability measures supported on $B_{\rho_i}(\theta_i)$ and $B_{\rho_i}(\theta_i''')$ respectively, so Step 1 applies and yields the displayed bound for every $q \in (0, \bar{q}]$.

**Step 3: compactness.** Let $\Theta'$ be as in part (i). If $\hat{\theta} \in \bar{\Theta}'$ is eliminated with witness $(\varphi(\hat{\theta}), \kappa_{\hat{\theta}}, M_{\hat{\theta}})$ per Definition 12, then for any $\theta \in B_{\rho_{\hat{\theta}}}(\hat{\theta}) \cap \bar{\Theta}'$, integrating the modulus bound over $P_\mu^*$ gives, for all $\mu \in M_{\hat{\theta}} \cap M_0$ and all selections,

$$\int_z \log \frac{\hat{p}_\mu(z|\theta)}{\hat{p}_\mu(z|\varphi(\hat{\theta}))}\, p_\mu^*(z)\, d\nu(z) \le -\kappa_{\hat{\theta}} + \omega(\rho_{\hat{\theta}}) \le -\frac{\kappa_{\hat{\theta}}}{2}$$

Thus $\theta$ is eliminated with witness $(\varphi(\hat{\theta}), \kappa_{\hat{\theta}}/2, M_{\hat{\theta}} \cap M_0)$, so $\bar{\Theta}' \setminus S_m(\Theta')$ is relatively open, $S_m(\Theta')$ is closed in $\bar{\Theta}'$, and hence compact.

**Step 4: non-emptiness.** Suppose $S_m(\Theta') = \emptyset$. Apply Step 2 with $E = \bar{\Theta}'$ and $D = \Delta\bar{\Theta}'$; witnesses exist by Definition 12. Set $\kappa_* = \min_i \kappa_{\theta_i} > 0$. For any $\mu \in \Delta\bar{\Theta}'$, the telescoping argument summing the margin $-\kappa_*/2$ per step over $N > 2L/\kappa_*$ steps gives a contradiction with the bounded likelihood ratio bound. Hence $S_m(\Theta')$ is non-empty. □

## 7.2 Contraction and Convergence

This subsection reduces the two-buyer process to the uniform-margin framework through a product-model embedding. We define:

1. Outcome: $z = (y, v) \in \{0,1\} \times V$ where $y$ is the allocation outcome
2. Outcome Distribution:

$$\mathbb{P}_\mu = \{(yq(b_\mu(v))f(v) + (1-y)(1-q(b_\mu(v)))f(v), \\ \{yg(b_\mu(v), v, \theta)f(v) + (1-y)(1-g(b_\mu(v), v, \theta))f(v)\}_{\theta\in\Theta}) | b_\mu(v) \in \beta(v, \mu)\ \forall v\}$$

where the first term corresponds to the objective outcome distribution and the second term corresponds to the subjective one.

**Remark 8.** *The multiplicity of outcome distribution is introduced through the multiplicity of the optimal bidding correspondence $\beta(v,\mu)$. Note that the subjective marginal distribution over $v$ is independent of $\theta$ and always agrees with the objective distribution over $v$.*

### 7.2.1 Componentwise Machinery

In this subsection we establish the convergence results for the general asymmetric two-agent setting, allowing agents to have distinct models $g^i$, parameter spaces $\Theta_i$, and value distributions $F_i$. The symmetric results in Section 3.3 follow as a special case. To handle this generality, we use bold font to denote a profile of objects: $\boldsymbol{b} = (b_1, b_2)$, $\boldsymbol{\theta} = (\theta_1, \theta_2)$ etc.

As in Section 2, agent $i \in \{1,2\}$ holds model $\{g^i(b_i, v_i, \theta_i)\}_{\theta_i \in \Theta_i}$ and belief $\mu_i \in \Delta\Theta_i$, valuations are drawn from $F_i$ with density $f_i$, and the allocation rule $\boldsymbol{q}(\boldsymbol{y} \mid \boldsymbol{b})$ has marginals $q^i(b_i, b_{-i})$.

We first show that this two-agent learning process can be written as a single-agent learning about a two-dimensional parameter ($\boldsymbol{\theta} = (\theta_1, \theta_2)$) from the observed outcomes of both agents ($\boldsymbol{b} = (b_1, b_2), \boldsymbol{y} = (y_1, y_2), \boldsymbol{v} = (v_1, v_2)$). Specifically, consider an agent with the following model:

$$\tilde{g}(\boldsymbol{y}, \boldsymbol{b}, \boldsymbol{v}, \boldsymbol{\theta}) = \begin{cases} g^1(b_1, v_1, \theta_1) g^2(b_2, v_2, \theta_2) & \text{if } y_1 = y_2 = 1 \\ (1 - g^1(b_1, v_1, \theta_1))(1 - g^2(b_2, v_2, \theta_2)) & \text{if } y_1 = y_2 = 0 \\ (1 - g^1(b_1, v_1, \theta_1)) g^2(b_2, v_2, \theta_2) & \text{if } y_1 = 0, y_2 = 1 \\ g^1(b_1, v_1, \theta_1)(1 - g^2(b_2, v_2, \theta_2)) & \text{if } y_1 = 1, y_2 = 0 \end{cases}$$

**Lemma 6** (Product Updating)**.** *Given any belief of agents 1 and 2, $\mu_1 \in \Delta\Theta_1, \mu_2 \in \Delta\Theta_2$, let $\tilde{\mu}(\theta_1, \theta_2) = \mu_1(\theta_1)\mu_2(\theta_2) \in \Delta(\Theta_1 \times \Theta_2)$. For any observation $(\boldsymbol{y}, \boldsymbol{b}, \boldsymbol{v}) = ((y_1, y_2), (b_1, b_2), (v_1, v_2))$, let $\{\mu_i'(\theta_i | y_i, b_i, v_i)\}_{i=1,2}$ denote the posterior belief from updating $\mu_i$ independently according to $\{\{g^i(b, v, \theta)\}_{\theta \in \Theta}\}_{i=1,2}$, and let $\tilde{\boldsymbol{\mu}}'(\boldsymbol{\theta} | \boldsymbol{y}, \boldsymbol{b}, \boldsymbol{v})$ denote the posterior belief from updating $\tilde{\mu}$ jointly according to $\{\tilde{g}(\boldsymbol{y}, \boldsymbol{b}, \boldsymbol{v}, \boldsymbol{\theta})\}_{\boldsymbol{\theta} \in \Theta_1 \times \Theta_2}$; then*

$$\tilde{\boldsymbol{\mu}}'(\theta_1, \theta_2 | \boldsymbol{y}, \boldsymbol{b}, \boldsymbol{v}) = \mu_1'(\theta_1 | y_1, b_1, v_1) \mu_2'(\theta_2 | y_2, b_2, v_2)$$

*Proof.* This can be shown by straightforward application of Bayes' Rule. First observe that

$$\mu_i'(\theta_i | y_i, b_i, v_i) = \begin{cases} \frac{g^i(b_i, v_i, \theta_i)}{\int_{x \in \Theta_i} g^i(b_i, v_i, x) \mu_i(x) dx} \mu_i(\theta_i) & \text{if } y_i = 1 \\ \frac{1 - g^i(b_i, v_i, \theta_i)}{\int_{x \in \Theta_i} [1 - g^i(b_i, v_i, x)] \mu_i(x) dx} \mu_i(\theta_i) & \text{if } y_i = 0 \end{cases}$$

Now consider $\tilde{\mu}'$ when $y_1 = y_2 = 1$

$$\begin{aligned} \tilde{\boldsymbol{\mu}}'(\theta_1, \theta_2 | (1,1), \boldsymbol{b}, \boldsymbol{v}) &= \frac{g^1(b_1, v_1, \theta_1) g^2(b_2, v_2, \theta_2)}{\int_{(x,y) \in \Theta_1 \times \Theta_2} \tilde{g}((1,1), \boldsymbol{b}, \boldsymbol{v}, (x,y)) \tilde{\mu}(x,y) d(x,y)} \tilde{\boldsymbol{\mu}}(\theta_1, \theta_2) \\ &= \frac{g^1(b_1, v_1, \theta_1) g^2(b_2, v_2, \theta_2)}{\int_{y \in \Theta_2} \int_{x \in \Theta_1} g^1(b_1, v_1, x) g^2(b_2, v_2, y) \mu_1(x) dx \mu_2(y) dy} \\ &= \frac{g^1(b_1, v_1, \theta_1)}{\int_{x \in \Theta_1} g^1(b_1, v_1, x) \mu_1(x) dx} \frac{g^2(b_2, v_2, \theta_2)}{\int_{x \in \Theta_2} g^2(b_2, v_2, x) \mu_2(x) dx} \\ &= \mu_1'(\theta_1 | 1, b_1, v_1) \mu_2'(\theta_2 | 1, b_2, v_2) \end{aligned}$$

Similar algebra shows that

$$\tilde{\boldsymbol{\mu}}'(\theta_1,\theta_2|\boldsymbol{y},\boldsymbol{b},\boldsymbol{v}) = \mu_1'(\theta_1|y_1,b_1,v_1)\mu_2'(\theta_2|y_2,b_2,v_2)$$

For every $(y_1,y_2) \in \{0,1\}^2$. □

Lemma 6 means that we can easily adapt the two-agent learning problem into the framework of Frick et al. (2023) by reformulating it as a single-agent one. Specifically, we augment the single-agent definition into the following:

1. $\boldsymbol{\theta} = (\theta_1,\theta_2) \in \boldsymbol{\Theta} = \Theta_1 \times \Theta_2$
2. Outcome: $z = ((y_1,y_2),(v_1,v_2)) \in \{0,1\}^2 \times V_1 \times V_2$
3. Probability Distributions: Let $\boldsymbol{v} = (v_1,v_2)$ and $\boldsymbol{\mu} \in \Delta\boldsymbol{\Theta}$. Define $\mu_i(\theta_i) = \int_{\theta_{-i}\in\Theta_{-i}} \mu(\theta_1,\theta_2)d\theta_{-i}$ as the marginal belief over $\theta_i$. For a given $b_{\boldsymbol{\mu}}(v) = (b^1_{\mu_1}(v_1), b^2_{\mu_2}(v_2))$ where $b^i_{\mu_i}(v_i) \in \beta^i(v_i,\mu_i)$

   (a) The objective outcome distribution is given by:

$$p^*_{\boldsymbol{\mu}}(\boldsymbol{y},\boldsymbol{v}) = \begin{cases} q(1,1|b^1_{\mu_1}(v_1),b^2_{\mu_2}(v_2))f_1(v_1)f_2(v_2) & \text{if } y_1 = y_2 = 1 \\ q(0,0|b^1_{\mu_1}(v_1),b^2_{\mu_2}(v_2))f_1(v_1)f_2(v_2) & \text{if } y_1 = y_2 = 0 \\ q(0,1|b^1_{\mu_1}(v_1),b^2_{\mu_2}(v_2))f_1(v_1)f_2(v_2) & \text{if } y_1 = 0, y_2 = 1 \\ q(1,0|b^1_{\mu_1}(v_1),b^2_{\mu_2}(v_2))f_1(v_1)f_2(v_2) & \text{if } y_1 = 1, y_2 = 0 \end{cases}$$

   (b) The subjective outcome distribution is given by:

$$\hat{p}_{\boldsymbol{\mu}}(\boldsymbol{y},\boldsymbol{v}|\boldsymbol{\theta}) = \begin{cases} g^1(b^1_{\mu_1}(v_1),v_1,\theta_1)g^2(b^2_{\mu_2}(v_2),v_2,\theta_2)f_1(v_1)f_2(v_2) & \text{if } y_1 = y_2 = 1 \\ [1-g^1(b^1_{\mu_1}(v_1),v_1,\theta_1)][1-g^2(b^2_{\mu_2}(v_2),v_2,\theta_2)]f_1(v_1)f_2(v_2) & \text{if } y_1 = y_2 = 0 \\ [1-g^1(b^1_{\mu_1}(v_1),v_1,\theta_1)]g^2(b^2_{\mu_2}(v_2),v_2,\theta_2)f_1(v_1)f_2(v_2) & \text{if } y_1 = 0, y_2 = 1 \\ g^1(b^1_{\mu_1}(v_1),v_1,\theta_1)[1-g^2(b^2_{\mu_2}(v_2),v_2,\theta_2)]f_1(v_1)f_2(v_2) & \text{if } y_1 = 1, y_2 = 0 \end{cases}$$

**Remark 9.** *The parameter $\boldsymbol{\theta} = (\theta_1,\theta_2)$ of the single-agent reformulation is two-dimensional, and margined dominance over all beliefs on a subset of $\boldsymbol{\Theta}$ is correspondingly demanding. Lemma 6 shows that a belief process started from any product prior remains in the set $\mathscr{P}$ of product beliefs, so $\mathscr{P}$ is closed and invariant, and by Remark 5 Theorems 1 and 2 and Corollary 2 apply with every belief quantifier relativized to $\mathscr{P}$. On $\mathscr{P}$, margined dominance between elements of $\boldsymbol{\Theta}$ reduces to margined dominance for each agent's parameter separately by Lemma 7, so witnesses for the elimination arguments can be constructed one agent at a time.*

For a product belief $\boldsymbol{\mu}$ with marginals $(\mu_1,\mu_2)$, the outcome distributions displayed in the reformulation above are determined by the marginals and the selection pair, and

$$\mathbb{P}_{\boldsymbol{\mu}} = \left\{ \left(p^*_{\boldsymbol{\mu}},\ \{\hat{p}_{\boldsymbol{\mu}}(\cdot \mid \boldsymbol{\theta})\}_{\boldsymbol{\theta}\in\boldsymbol{\Theta}}\right) \,\middle|\, b^i_{\mu_i}(v_i) \in \beta^i(v_i,\mu_i)\ \forall v_i \in V_i,\ i = 1,2 \right\}$$

This specifies the data that Definition 9 consumes in the embedded environment, with parameters ranging over $\boldsymbol{\Theta}$ and beliefs over $\mathscr{P}$.

**Definition 14** (Margined Componentwise KL-Dominance)**.** *Let $M \subseteq \Delta\Theta_1 \times \Delta\Theta_2$, $i \in \{1,2\}$, $\theta_i', \theta_i'' \in \Theta_i$, and $\kappa > 0$. We say $\theta_i'$ componentwise KL-dominates $\theta_i''$ with margin $\kappa$ on $M$, written $\theta_i' \succsim_M^{KL\text{-}C(\kappa)} \theta_i''$, if for all $(\mu_1,\mu_2) \in M$ and all selections $b^j_{\mu_j}(v_j) \in \beta^j(v_j,\mu_j)$ for $j = 1,2$,*

$$\int_{v_i, v_{-i} \in V_i \times V_{-i}} \left[ \log\left( \frac{g^i(b^i_{\mu_i}(v_i), v_i, \theta_i'')}{g^i(b^i_{\mu_i}(v_i), v_i, \theta_i')} \right) q^i(b^i_{\mu_i}(v_i), b^{-i}_{\mu_{-i}}(v_{-i})) + \right.$$
$$\left. \log\left( \frac{1 - g^i(b^i_{\mu_i}(v_i), v_i, \theta_i'')}{1 - g^i(b^i_{\mu_i}(v_i), v_i, \theta_i')} \right) \left(1 - q^i(b^i_{\mu_i}(v_i), b^{-i}_{\mu_{-i}}(v_{-i}))\right) \right] f_i(v_i) f_{-i}(v_{-i}) d(v_i, v_{-i}) \le -\kappa$$

**Lemma 7** (Componentwise Margin Transfer)**.** *Let $M \subseteq \Delta\Theta_1 \times \Delta\Theta_2$ and let $M^\times$ denote the set of product beliefs whose marginals $(\mu_1,\mu_2)$ lie in $M$. If $\theta_i' \succsim_M^{KL\text{-}C(\kappa)} \theta_i''$, then $(\theta_i', \theta_{-i}) \succsim_{M^\times}^{KL(\kappa)} (\theta_i'', \theta_{-i})$ for every $\theta_{-i} \in \Theta_{-i}$. If in addition $\theta_{-i}' \succsim_M^{KL\text{-}C(\kappa')} \theta_{-i}''$, then $(\theta_1', \theta_2') \succsim_{M^\times}^{KL(\kappa+\kappa')} (\theta_1'', \theta_2'')$.*

*Proof.* Fix $(\mu_1,\mu_2) \in M$ and selections $b^j_{\mu_j}(v_j) \in \beta^j(v_j,\mu_j)$ for $j = 1,2$, write $\boldsymbol{b} = (b^1_{\mu_1}(v_1), b^2_{\mu_2}(v_2))$, and let $p^*_{\boldsymbol{\mu}}$ and $\hat{p}_{\boldsymbol{\mu}}$ denote the objective and subjective outcome distributions for these selections. The densities $f_1(v_1) f_2(v_2)$ cancel in the likelihood ratio between any two parameter pairs $\boldsymbol{\theta}''$ and $\boldsymbol{\theta}'$, and the remaining factors of $\hat{p}_{\boldsymbol{\mu}}$ separate by agent, so pointwise in the outcome $(\boldsymbol{y}, \boldsymbol{v})$,

$$\log \frac{\hat{p}_{\boldsymbol{\mu}}(\boldsymbol{y}, \boldsymbol{v} \mid \boldsymbol{\theta}'')}{\hat{p}_{\boldsymbol{\mu}}(\boldsymbol{y}, \boldsymbol{v} \mid \boldsymbol{\theta}')} = \ell_1(y_1, v_1) + \ell_2(y_2, v_2), \qquad \ell_j(y_j, v_j) = \begin{cases} \log \dfrac{g^j(b^j_{\mu_j}(v_j), v_j, \theta_j'')}{g^j(b^j_{\mu_j}(v_j), v_j, \theta_j')} & \text{if } y_j = 1 \\ \log \dfrac{1 - g^j(b^j_{\mu_j}(v_j), v_j, \theta_j'')}{1 - g^j(b^j_{\mu_j}(v_j), v_j, \theta_j')} & \text{if } y_j = 0 \end{cases}$$

Integrating against $p^*_{\boldsymbol{\mu}}$,

$$\begin{aligned} \int \log \frac{\hat{p}_{\boldsymbol{\mu}}(\boldsymbol{y}, \boldsymbol{v} \mid \boldsymbol{\theta}'')}{\hat{p}_{\boldsymbol{\mu}}(\boldsymbol{y}, \boldsymbol{v} \mid \boldsymbol{\theta}')} \, dp^*_{\boldsymbol{\mu}} &= \sum_{j=1,2} \int_{V_1 \times V_2} \sum_{(y_1,y_2) \in \{0,1\}^2} \ell_j(y_j, v_j) \, q(y_1, y_2 \mid \boldsymbol{b}) \, f_1(v_1) f_2(v_2) \, d(v_1, v_2) \\ &= \sum_{j=1,2} \int_{V_1 \times V_2} \Big[ \ell_j(1, v_j) \, q^j(b^j_{\mu_j}(v_j), b^{-j}_{\mu_{-j}}(v_{-j})) \\ &\qquad + \ell_j(0, v_j) \left(1 - q^j(b^j_{\mu_j}(v_j), b^{-j}_{\mu_{-j}}(v_{-j}))\right) \Big] f_1(v_1) f_2(v_2) \, d(v_1, v_2) \end{aligned}$$

where the second equality groups the four outcomes by $y_j$ and uses

$$\sum_{y_{-j} \in \{0,1\}} q(y_1, y_2 \mid \boldsymbol{b}) = q^j(b^j_{\mu_j}(v_j), b^{-j}_{\mu_{-j}}(v_{-j})) \quad \text{at } y_j = 1$$

by the definitions of $q(\boldsymbol{y} \mid \boldsymbol{b})$ and $q^j$. The summand for agent $j$ is the left side of the display in Definition 14 for that agent. If $\theta_{-i}'' = \theta_{-i}'$ then $\ell_{-i}$ vanishes identically and the summand for agent $-i$ is zero, so the integral equals the summand for agent $i$, at most $-\kappa$ by hypothesis. Fix $\boldsymbol{\mu} \in M^\times$ with marginals $(\mu_1,\mu_2) \in M$. By the definition of $\mathbb{P}_{\boldsymbol{\mu}}$, every $p_{\boldsymbol{\mu}} \in \mathbb{P}_{\boldsymbol{\mu}}$ is the pair induced by some selection pair $(b^1_{\mu_1}, b^2_{\mu_2})$, and the bound above holds at every selection pair, so

$$\int \log \frac{\hat{p}_{\boldsymbol{\mu}}(\boldsymbol{y}, \boldsymbol{v} \mid (\theta_i'', \theta_{-i}))}{\hat{p}_{\boldsymbol{\mu}}(\boldsymbol{y}, \boldsymbol{v} \mid (\theta_i', \theta_{-i}))} \, dP^*_{\boldsymbol{\mu}} \le -\kappa \qquad \text{for all } p_{\boldsymbol{\mu}} \in \mathbb{P}_{\boldsymbol{\mu}}.$$

Since $\boldsymbol{\mu} \in M^{\times}$ was arbitrary, the bound holds for all $\boldsymbol{\mu} \in M^{\times}$ and all $p_{\boldsymbol{\mu}} \in \mathbb{P}_{\boldsymbol{\mu}}$, which is the statement $(\theta_i', \theta_{-i}) \succsim_{M^{\times}}^{KL(\kappa)} (\theta_i'', \theta_{-i})$ of Definition 9. When both hypotheses hold, the two summands are at most $-\kappa$ and $-\kappa'$, so their sum is at most $-(\kappa+\kappa')$, and the same two steps give $(\theta_1', \theta_2') \succsim_{M^{\times}}^{KL(\kappa+\kappa')} (\theta_1'', \theta_2'')$. □

### 7.2.2 Convergence of the Two-Buyer Process

*Proof of Proposition 1.* The two-agent process is embedded as a pseudo-single-agent process on $\boldsymbol{\Theta} = \Theta \times \Theta$ via the product model of the two-agent reduction above, with Lemma 6 supplying that the embedded posterior factors, so the set $\mathscr{P}$ of product beliefs is invariant. Bounded likelihood ratios and equicontinuous likelihoods on $\Delta\boldsymbol{\Theta}$ hold with the two-buyer constants of Remarks 3 and 4. By the factorization of the embedded likelihood ratios, margins for pairs reduce to margins for coordinates and margins for two-coordinate clips add by pointwise telescoping of log-ratios.

Fix a compact interval $\Theta'' = [\underline{\theta}'', \bar{\theta}''] \subseteq W$ with $\underline{\theta}'' < \theta^* < \bar{\theta}''$, and define the side margins $\kappa_a(\bar{\theta}''') = \inf\{Q(b \mid \bar{\theta}''') - h(b, \bar{\theta}''') : b \in \overline{\beta(V,W)}\}$ for $\bar{\theta}''' \in W$ above $\theta^*$ and $\kappa_b(\cdot)$ symmetrically below, with $\bar{\kappa}_a(c) = \inf\{\kappa_a(\theta) : \theta \geq \theta^* + c\}$ and $\bar{\kappa}_b(c) = \inf\{\kappa_b(\theta) : \theta \leq \theta^* - c\}$ their away-from-target minorants, positive for every $c > 0$ by parts (a) and (b) of Condition 1.

**Step 1: sandwich with margin on a belief neighbourhood.** Let $\omega_g$ be a modulus of continuity of $g$ in $\theta$, uniform over $(b, v)$ on the compact reachable domain. Write $c_a = \bar{\theta}'' - \theta^*$ and $c_b = \theta^* - \underline{\theta}''$, and choose $\varepsilon_1^a, \varepsilon_1^b > 0$ with

$$\omega_g(\varepsilon_1^a) \leq \frac{1}{2}\bar{\kappa}_a(c_a) \qquad \text{and} \qquad \omega_g(\varepsilon_1^b) \leq \frac{1}{2}\bar{\kappa}_b(c_b)$$

the right sides positive by the away-from-target condition. Each side's band width is chosen from its own minorant alone, so the upper width carries no dependence on the lower endpoint and vice versa. Since $W$ is relatively open and contains $\bar{\Theta}''$, there is $\varepsilon' \in (0, \min\{\varepsilon_1^a, \varepsilon_1^b\}/2]$ with $\Theta''' = [\underline{\theta}'' - \varepsilon', \bar{\theta}'' + \varepsilon'] \cap \Theta \subseteq W$ still straddling $\theta^*$. Since $\bar{\theta}''' \geq \bar{\theta}''$ and $\underline{\theta}''' \leq \underline{\theta}''$, the minorant bounds give $\omega_g(\varepsilon_1^a) \leq \kappa_a(\bar{\theta}''')/2$ and $\omega_g(\varepsilon_1^b) \leq \kappa_b(\underline{\theta}''')/2$ for every admissible enlargement. For any belief supported on $\bar{\Theta}''$, the first-order condition equates $v - b$ with $\int g\, d\mu_i \big/ \int g_b\, d\mu_i$, a ratio lying between the extremes of $g/g_b$ over the support since $g_b > 0$, and $g/g_b$ is decreasing in $\theta$ with the root of the decreasing-in-$b$ condition monotone in its level, so every selection at $\mu_i \in \Delta\bar{\Theta}''$ satisfies $b^i_{\mu_i}(v) \in [\beta(v, \underline{\theta}''), \beta(v, \bar{\theta}'')]$. Under (R1)–(R3) the bid function $\beta(v, \theta)$ is strictly increasing in $\theta$ and continuous, so this closed band lies, uniformly over the compact $V$, in the open tube $\{(v, b) : \beta(v, \underline{\theta}''') < b < \beta(v, \bar{\theta}''')\}$, and by Berge's theorem the bidding correspondences $\beta_i(v, \mu_i)$ are upper hemicontinuous in $\mu_i$, so the upper preimage of the tube contains a weakly open $M_i \supseteq \Delta\bar{\Theta}''$ on which every selection satisfies $b^i_{\mu_i}(v) \in [\beta(v, \underline{\theta}'''), \beta(v, \bar{\theta}''')]$ pointwise in $v$. Because $q$ is non-increasing in the opponent's bid, for every own bid $b \in \overline{\beta(V,W)}$ and every $\mu_2 \in M_2$:

$$Q(b \mid \mu_2) \geq Q(b \mid \bar{\theta}''') \geq h(b, \bar{\theta}''') + \kappa_a(\bar{\theta}''') \qquad \text{and} \qquad Q(b \mid \mu_2) \leq Q(b \mid \underline{\theta}''') \leq h(b, \underline{\theta}''') - \kappa_b(\underline{\theta}''')$$

by the definition of the side margins at $\Theta'''$. Write $\kappa_a' = \kappa_a(\bar{\theta}''')$ and $\kappa_b' = \kappa_b(\underline{\theta}''')$, and $M = \{\mu_1 \otimes \mu_2 : \mu_i \in M_i\}$, a relatively open subset of $\mathscr{P}$ containing $\Delta(\bar{\Theta}'' \times \bar{\Theta}'') \cap \mathscr{P}$.

**Step 2: drift slope on the band and the tail.** Let $c_g > 0$ bound $-g_\theta$ below and $c \leq g \leq 1 - c$ on the compact reachable domain. Fix any $\boldsymbol{\mu} \in M$ and any selection. At every realized $(\hat{b}, v)$ of agent 1, $\hat{b} \leq \beta(v, \bar{\theta}''')$, so $v \geq \beta^{-1}(\hat{b}, \bar{\theta}''')$ and, for every $\theta_1 \geq \bar{\theta}'''$, $g(\hat{b}, v, \theta_1) \leq g(\hat{b}, \beta^{-1}(\hat{b}, \bar{\theta}'''), \bar{\theta}''') =$

$h(\hat{b}, \bar{\theta}''')$, exactly as in the single-agent chain. With $\varepsilon_1^a$ as chosen in Step 1, $\omega_g(\varepsilon_1^a) \le \kappa_a'/2$, and for every $\theta_1 \ge \bar{\theta}''' - \varepsilon_1^a$ and every realized $(\hat{b}, v)$:

$$g(\hat{b}, v, \theta_1) \le g(\hat{b}, v, \bar{\theta}''') + \omega_g(\varepsilon_1^a) \le h(\hat{b}, \bar{\theta}''') + \frac{\kappa_a'}{2} \le Q(\hat{b} \mid \mu_2) - \frac{\kappa_a'}{2}$$

Substituting into the derivative of $K_1$ with respect to $\theta$, whose integrand is $-g_\theta \cdot [Q-g]/[g(1-g)]$ against the realized bid distribution, gives

$$\frac{\partial}{\partial \theta} K_1(\theta_1, b_{\boldsymbol{\mu}}(\cdot)) \ge 2 c_g \kappa_a' =: \kappa_a'' > 0 \qquad \forall\, \theta_1 \ge \bar{\theta}''' - \varepsilon_1^a$$

uniformly over $\boldsymbol{\mu} \in M$ and selections. Symmetrically, $\partial K_1/\partial \theta \le -\kappa_b'' := -2 c_g \kappa_b'$ for all $\theta_1 \le \underline{\theta}''' + \varepsilon_1^b$, and identically for agent 2.

Consequently, for any $\theta_1''' \in [\bar{\theta}''' - \varepsilon_1^a, \bar{\theta}'']$ and any $\hat{\theta}_1 \in (\theta_1''', \bar{\theta}]$, integrating the slope gives $K_1(\hat{\theta}_1) - K_1(\theta_1''') \ge (\hat{\theta}_1 - \theta_1''')\, \kappa_a''$ at every $\boldsymbol{\mu} \in M$ and selection, so $\hat{\theta}_1$ is KL-dominated by $\theta_1'''$ with margin $(\hat{\theta}_1 - \theta_1''')\kappa_a''$ on $M$, and likewise below with $\kappa_b''$.

**Step 3: margined strict B-NE relative to $\mathscr{P}$.** Let $\Theta^\circ \subseteq W$ be a relatively open interval with $\theta^* \in \Theta^\circ$ and $\bar{\Theta}^\circ \subseteq W$. Write $c_a^\circ = \bar{\theta}^\circ - \theta^*$ and $c_b^\circ = \theta^* - \underline{\theta}^\circ$ and fix $\hat{\varepsilon}^a, \hat{\varepsilon}^b > 0$ with $\omega_g(\hat{\varepsilon}^a) \le \bar{\kappa}_a(c_a^\circ)/2$ and $\omega_g(\hat{\varepsilon}^b) \le \bar{\kappa}_b(c_b^\circ)/2$, shrunk if necessary so that $\bar{\theta}^\circ - \hat{\varepsilon}^a/4$ and $\underline{\theta}^\circ + \hat{\varepsilon}^b/4$ lie in $\Theta^\circ$. Apply Steps 1 and 2 with $\Theta''$ a compact interval satisfying $\bar{\Theta}^\circ \subseteq \operatorname{int}\Theta''$, $\Theta'' \subseteq W$, $\bar{\theta}'' - \bar{\theta}^\circ \le \hat{\varepsilon}^a/4$, and $\underline{\theta}^\circ - \underline{\theta}'' \le \hat{\varepsilon}^b/4$, taking $\varepsilon_1^a = \hat{\varepsilon}^a$ and $\varepsilon_1^b = \hat{\varepsilon}^b$ in Step 1, admissible choices since $c_a \ge c_a^\circ$ and $c_b \ge c_b^\circ$ and the minorants are monotone. Then $\bar{\theta}''' - \varepsilon_1^a \le \bar{\theta}'' + \hat{\varepsilon}^a/2 - \hat{\varepsilon}^a \le \bar{\theta}^\circ - \hat{\varepsilon}^a/4$ and symmetrically below, so the points $\bar{\theta}^\circ - \hat{\varepsilon}^a/4$ and $\underline{\theta}^\circ + \hat{\varepsilon}^b/4$ lie in the dominator ranges of Step 2. For any $(\theta_1, \theta_2) \notin \Theta^\circ \times \Theta^\circ$, replace each coordinate lying in $\Theta \setminus \Theta^\circ$ above $\theta^*$ by $\bar{\theta}^\circ - \hat{\varepsilon}^a/4$ and each lying in $\Theta \setminus \Theta^\circ$ below $\theta^*$ by $\underline{\theta}^\circ + \hat{\varepsilon}^b/4$, leaving the remaining coordinates fixed. Each replaced coordinate is dominated with margin at least $(\hat{\varepsilon}^a/4)\kappa_a''$ or $(\hat{\varepsilon}^b/4)\kappa_b''$ on $M$ by Step 2, and two-coordinate replacements add margins by the telescoping noted above. The resulting witness has dominator in $\Theta^\circ \times \Theta^\circ$ and neighbourhood $M \supseteq \Delta(\bar{\Theta}^\circ \times \bar{\Theta}^\circ) \cap \mathscr{P}$, so $\Delta(\Theta^\circ \times \Theta^\circ)$ is a margined strict B-NE set relative to $\mathscr{P}$.

**Step 4: interval-rectangle scheme.** Fix a relatively open interval $\Theta^\circ$ with $\Theta' \subseteq \Theta^\circ$ and $\bar{\Theta}^\circ \subseteq W$, which exists since $\Theta'$ is compact, $W$ is a relatively open interval, and $\theta^*$ lies in the relative interior of $\Theta$, and which strictly straddles $\theta^*$. Set $R^0 = \bar{\Theta}^\circ \times \bar{\Theta}^\circ$ and iterate as follows. Given $R^k = [\ell_k, u_k]^2$ containing $(\theta^*, \theta^*)$, apply Steps 1 and 2 at $\Theta'' = [\ell_k, u_k]$ to obtain $\varepsilon_1^{a,k}, \varepsilon_1^{b,k}$ and $\kappa_a''^k, \kappa_b''^k > 0$ and set $u_{k+1} = \max\{\theta^* + (u_k - \theta^*)/2,\ u_k - \varepsilon_1^{a,k}/2\}$, and symmetrically $\ell_{k+1} = \min\{\theta^* - (\theta^* - \ell_k)/2,\ \ell_k + \varepsilon_1^{b,k}/2\}$; the geometric guards keep every stage interval strictly straddling $\theta^*$. Every point of $R^k \setminus R^{k+1}$ has a coordinate in $(u_{k+1}, u_k]$ or $[\ell_k, \ell_{k+1})$ and is dominated by a point of $R^k$ with a positive margin on the stage-$k$ neighbourhood $M^k \supseteq \Delta R^k \cap \mathscr{P}$. The scheme shrinks to the target: for any $c > 0$, as long as $u_k \ge \theta^* + c$ the choice $\varepsilon_1^{a,k} = \bar{\varepsilon}_1^a(c)$ is admissible with $\bar{\varepsilon}_1^a(c) > 0$ satisfying $\omega_g(\bar{\varepsilon}_1^a(c)) \le \bar{\kappa}_a(c)/2$ and $k$-free by monotonicity of the minorant, so the decrement $u_k - u_{k+1} \ge \min\{c/2,\ \bar{\varepsilon}_1^a(c)/2\} > 0$; hence $u_k \downarrow \theta^*$, symmetrically $\ell_k \uparrow \theta^*$, and $\bigcap_k R^k = \{(\theta^*, \theta^*)\}$.

**Step 5: assembly.** Each $R^k$ is a non-empty compact rectangle, the sequence is nested, and stages carry witnesses with internal dominators and simplex-containing relatively open neighbourhoods. By Step 3 applied to the interval $\Theta^\circ$ fixed in Step 4, $\Delta(\Theta^\circ \times \Theta^\circ)$ is a margined strict B-NE set relative to $\mathscr{P}$, so Corollary 2(ii), relativized per Remark 5, yields, for every $\gamma < 1$, a common basin, a neighbourhood of $\Delta(\Theta^\circ \times \Theta^\circ) \cap \mathscr{P}$, on which $\mathbb{P}(\inf_{v \in \Delta R^k} d(\mu_t, v) \to 0) \ge \gamma$ for every

$k \leq \infty$, in particular for $\Delta R^{\infty} = \Delta\{(\theta^*, \theta^*)\}$. Since $\Theta' \subseteq \Theta^{\circ}$, this basin contains a neighbourhood of $\Delta(\Theta' \times \Theta') \cap \mathscr{P}$. For product priors concentrated near $\Delta(\Theta' \times \Theta')$ whose factors assign positive mass to every relatively open set containing $\theta^*$, beliefs converge to $(\theta^*, \theta^*)$ with probability approaching one as the concentration increases, which is the conclusion of the proposition under Definition 1 on $\Theta' \times \Theta'$, for every adapted selection by Remark 6. □

**Remark 10.** *The margin is the invariant that each regime of interest delivers by its own means. If $Q(\cdot \mid \theta)$ has finitely many pieces in the own bid, as for the step mechanisms constructed in this paper, strict inequalities that extend to the one-sided limits at piece boundaries imply the margin, since the gap against each piece's value is then continuous and positive on the piece's compact closure, and the minimum over finitely many pieces is positive.*[17] *If $q$ is continuous in bids, the gap is continuous on the compact bid range and the margin follows from pointwise strictness by the extreme value theorem. Absent either structure, pointwise strictness does not compactify, and the margin must be hypothesized.*

**Remark 11.** *The single-buyer environment is the instance $q(b_1, b_2) = q(b_1)$. The rule is symmetric, constant and hence non-increasing in the opponent's bid, and satisfies $Q(b \mid \mu) = q(b)$ for every opponent belief $\mu$, so conditions (a) and (b) reduce to $q(b) - h(b, \bar{\theta}'') \geq \kappa_a(\bar{\theta}'')$ and $h(b, \underline{\theta}'') - q(b) \geq \kappa_b(\underline{\theta}'')$. The embedded rule generically violates two-buyer feasibility, since $q(b_1) + q(b_2) \leq 1$ fails for high bid pairs, and feasibility is not a requirement of Condition 1. Under the embedding, each buyer's outcome distribution does not depend on the opponent's bid, so each buyer's belief process coincides with the single-buyer process for every selection, and sustaining $(\theta^*, \theta^*)$ on $\Theta' \times \Theta'$ delivers the single-buyer conclusion on $\Theta'$.*

**Remark 12.** *Suppose $q$ is continuous in the own bid and, for some relatively open interval $W$ with $\Theta' \subseteq W$, the pointwise conditions $h(b, \underline{\theta}'') > q(b) > h(b, \bar{\theta}'')$ hold for all compact $[\underline{\theta}'', \bar{\theta}''] \subseteq W$ with $\underline{\theta}'' < \theta^* < \bar{\theta}''$ and all $b \in \overline{\beta(V, W)}$. Fix a relatively open interval $W'$ with $\Theta' \subseteq W'$ and $\bar{W}' \subseteq W$, which exists since $\Theta'$ is compact. The functions $q(b) - h(b, \theta)$ and $h(b, \theta) - q(b)$ are continuous on the compact set $\overline{\beta(V, W')} \times \bar{W}'$, since $h$ is continuous in $(b, \theta)$ under (R1)–(R3), and every point of $\bar{W}'$ on a given side of $\theta^*$ is an endpoint of a straddling compact contained in $W$, since $\bar{W}'$ is an interval containing $\theta^*$ in its relative interior. Setting $\kappa_a(\bar{\theta}'') = \min_{b \in \overline{\beta(V, W')}} [q(b) - h(b, \bar{\theta}'')]$ and $\kappa_b(\underline{\theta}'') = \min_{b \in \overline{\beta(V, W')}} [h(b, \underline{\theta}'') - q(b)]$ yields margins positive at every admissible endpoint and continuous in the endpoint by the extreme value theorem. For each $c > 0$ the infimum of $\kappa_a$ over endpoints of $W'$ at distance at least $c$ above $\theta^*$ is bounded below by its minimum over the compact set of such points in $\bar{W}'$, which is positive, and likewise for $\kappa_b$ below. Proposition 1 applied with $W'$ in place of $W$ then gives that $q$ sustains $\theta^*$ on $\Theta'$.*

### 7.2.3 Selling Mechanism Setting

*Proof of Corollary 1.* Let $W$ be the set provided by Proposition 1. Fix any $b \in \overline{\beta(V, \Theta')}$. Because $\Theta' \subseteq W$, $b \in \overline{\beta(V, W)}$. For any $\varepsilon > 0$ sufficiently small, $\Theta'' = [\theta^* - \varepsilon, \theta^* + \varepsilon] \subseteq W$ with $\underline{\theta}'' < \theta^* < \bar{\theta}''$, and conditions (a) and (b) give $h(b, \theta^* + \varepsilon) < Q(b \mid \mu) < h(b, \theta^* - \varepsilon)$ for all $\mu \in \Delta\Theta''$. Setting

[17] Pointwise strictness alone does not suffice, because the gap can vanish in the limit at a piece's open end while the mechanism jumps favourably there; the mechanisms constructed in this paper exclude this through slacks uniform on each piece.

$\mu = \delta_{\theta^*}$ yields $h(b, \theta^* + \varepsilon) < Q(b \mid \theta^*) < h(b, \theta^* - \varepsilon)$. Sending $\varepsilon \to 0$ and applying continuity of $h$ in $\theta$ gives $Q(b \mid \theta^*) = h(b, \theta^*)$. □

*Proof of Proposition 2.* The rule $q(b_1, b_2) = q(b_1)$ with $q(b) = h(b, \theta^*)$ is symmetric and constant in the opponent's bid, hence non-increasing in it, and satisfies $Q(b \mid \mu) = h(b, \theta^*)$ for every opponent belief $\mu$, so it lies within the scope of Proposition 1, and for it parts (a) and (b) of Condition 1 require, for every $c > 0$, a positive infimum of $h(b, \theta^*) - h(b, \theta)$ over $b \in \overline{\beta(V, W)}$ and $\theta \in W$ with $\theta \geq \theta^* + c$, and a positive infimum of $h(b, \theta) - h(b, \theta^*)$ over the symmetric set below the target. We construct such a $W$ from the hypotheses on $\Theta'$.

Write $\Theta' = [\underline{\theta}', \bar{\theta}']$, so that $\underline{\theta}' < \theta^* < \bar{\theta}'$ by the interior hypothesis. Under (R1)–(R3) the bid function $\beta(v, \theta)$ is strictly increasing in $(v, \theta)$ and continuous, so the closed bid range of any compact interval of parameters is the compact interval of bids between the values of $\beta$ at its corner points. Fix $\varepsilon_0 > 0$, set $W_0 = [\underline{\theta}' - \varepsilon_0, \bar{\theta}' + \varepsilon_0] \cap \Theta$ and $B_0 = \overline{\beta(V, W_0)}$. Every $b \in B_0$ outside $\overline{\beta(V, \Theta')}$ satisfies $b > \beta(\bar{v}, \bar{\theta}')$ or $b < \beta(\underline{v}, \underline{\theta}')$. In the first case $b$ lies above the range of $\beta(\cdot, \theta)$ for every $\theta \in \Theta'$, so $h(b, \theta) = g(b, \bar{v}, \theta)$ by the convention of (4), and $h(b, \theta^*) - h(b, \theta) = g(b, \bar{v}, \theta^*) - g(b, \bar{v}, \theta)$ is strictly positive for $\theta > \theta^*$ and strictly negative for $\theta < \theta^*$ by (R2); the second case is symmetric with $\underline{v}$ in place of $\bar{v}$. Combined with the hypotheses on $\overline{\beta(V, \Theta')}$, the gap $h(b, \theta^*) - h(b, \theta)$ is therefore strictly positive for every $b \in B_0$ and $\theta \in \Theta'$ with $\theta > \theta^*$, and strictly negative for $\theta < \theta^*$.

Since $h$ is continuous in $(b, \theta)$ under (R1)–(R3), for every $c > 0$ with $\{\theta \in \Theta' \mid \theta \geq \theta^* + c\}$ nonempty the minimum $m_a(c)$ of $h(b, \theta^*) - h(b, \theta)$ over the compact set $B_0 \times \{\theta \in \Theta' \mid \theta \geq \theta^* + c\}$ is attained and positive, and likewise $m_b(c)$ below the target. In particular $m^+ = m_a(\bar{\theta}' - \theta^*)$, the minimum over $B_0$ of $h(b, \theta^*) - h(b, \bar{\theta}')$, and the symmetric $m^-$ at $\underline{\theta}'$ are positive. Let $\omega$ be a modulus of continuity of $h$ on the compact set $B_0 \times W_0$ and choose $\varepsilon \in (0, \varepsilon_0]$ with $\omega(\varepsilon) \leq \min\{m^+, m^-\}/2$. Set $W = (\underline{\theta}' - \varepsilon, \bar{\theta}' + \varepsilon) \cap \Theta$, an interval open relative to $\Theta$ with $\Theta' \subseteq W \subseteq W_0$ and $\overline{\beta(V, W)} \subseteq B_0$. For $\theta \in W$ with $\theta > \bar{\theta}'$ and any $b \in B_0$, $h(b, \theta^*) - h(b, \theta) \geq h(b, \theta^*) - h(b, \bar{\theta}') - \omega(\varepsilon) \geq m^+/2$, and symmetrically $h(b, \theta) - h(b, \theta^*) \geq m^-/2$ for $\theta \in W$ with $\theta < \underline{\theta}'$; when an endpoint of $\Theta'$ is an endpoint of $\Theta$ the corresponding strip is empty and the bound is unneeded.

Fix $c > 0$ with $\{\theta \in W \mid \theta \geq \theta^* + c\}$ nonempty. Every such $\theta$ lies in $\Theta'$ or above $\bar{\theta}'$, so the infimum of $h(b, \theta^*) - h(b, \theta)$ over $b \in \overline{\beta(V, W)}$ and such $\theta$ is at least $\min\{m_a(c), m^+/2\} > 0$, and the symmetric infimum below the target is at least $\min\{m_b(c), m^-/2\} > 0$. Condition 1 therefore holds with this $W$. Proposition 1 gives that the mechanism sustains $(\theta^*, \theta^*)$ on $\Theta' \times \Theta'$. Under the embedding, each buyer's outcome distribution does not depend on the opponent's bid, so each buyer's belief process coincides with the single-buyer process for every selection, and the single-buyer conclusion on $\Theta'$ follows. □

*Proof of Lemma 1.* Fix $\theta$ and a bid $b$. If $b$ lies outside the range of $\beta(\cdot, \theta)$, then $\beta^{-1}(b, \theta)$ equals $\underline{v}$ or $\bar{v}$ by the convention of (4) and is locally constant in $\theta$, so $\partial_\theta h(b, \theta) = g_\theta < 0$ by (R2). For $b$ in the range, write $V(b, \theta) = \beta^{-1}(b, \theta)$, so $h(b, \theta) = g(b, V(b, \theta), \theta)$ and

$$h_\theta = g_\theta + g_v V_\theta$$

with all derivatives of $g$ evaluated at $(b, V(b, \theta), \theta)$. The optimal bid satisfies the first-order condition $(v - b)\, g_b = g$ at $v = V(b, \theta)$, identically in $(b, \theta)$. Differentiating the identity in $\theta$ and substituting $v - b = g/g_b$,

$$V_\theta = \frac{g_\theta - \frac{g}{g_b} g_{b\theta}}{D}, \qquad D = g_b - g_v + \frac{g}{g_b} g_{bv}.$$

Since $g/g_b$ is weakly decreasing in $v$ by (R3), $g\,g_{bv} \geq g_b\,g_v$, so $\frac{g}{g_b}\,g_{bv} \geq g_v$ and $D \geq g_b > 0$. Substituting $V_\theta$,

$$h_\theta = \frac{g_\theta\,g_b^2 + g\,g_\theta\,g_{bv} - g\,g_v\,g_{b\theta}}{g_b D}$$

whose denominator is positive. The product $g\,g_\theta\,g_v$ is positive, since $g > 0$ by (R1), $g_\theta < 0$ by (R2), and $g_v < 0$ by hypothesis, so dividing the numerator by it shows the numerator is negative exactly when

$$\frac{g_{b\theta}}{g_\theta} > \frac{g_b^2}{g\,g_v} + \frac{g_{bv}}{g_v}$$

and dividing through by $g_b > 0$ this is the displayed condition. The condition therefore gives $h_\theta < 0$ at every bid in the range. For fixed $b$, the parameter set splits into at most three intervals on which $b$ lies above, within, or below the range of $\beta(\cdot,\theta)$, since $\beta(\underline{v},\cdot)$ and $\beta(\bar{v},\cdot)$ are strictly increasing by (R3). On each interval $h_\theta < 0$, and $h(b,\cdot)$ is continuous, so $h(b,\theta)$ is strictly decreasing in $\theta$. □

## 7.3 Mechanism Construction

This subsection contains the proofs of the local stability results, followed by the discrete-valuation construction and its margin delivery.

### 7.3.1 Maximal Objective Responsiveness

Section 3.3.1 bounds the second buyer's effect by the self-confirming mechanism with the largest objective responsiveness. With $\alpha(b) = \Phi^{-1}(h(b,\theta^*) \mid \theta^*)$, it is

$$q_L(b_1,b_2) = \mathbf{1}\{b_2 < \alpha(b_1)\},$$

which awards the good with certainty when the opponent bids below $\alpha(b_1)$ and withholds it otherwise, $\alpha$ being chosen so that an opponent holding belief $\delta_{\theta^*}$ bids below it with probability exactly $h(b_1,\theta^*)$. The proof below shows that, under MLRP, every symmetric self-confirming mechanism non-increasing in $b_2$ wins weakly more often than $q_L$ against opponents above the target and weakly less often against opponents below it, so that its objective responsiveness is at most that of $q_L$. The mechanism $q_L$ need not be feasible.

**Lemma 8** (Maximal Objective Responsiveness)**.** *Suppose MLRP holds. If Condition 1 holds for $q_L$ on some $\Theta'$, then it holds on $\Theta'$ for every symmetric mechanism, non-increasing in $b_2$, that is self-confirming with respect to $\theta^*$ on some interval $W'$ open relative to $\Theta$ with $\Theta' \subseteq W'$.*

*Proof of Lemma 8.* Let $W$ be an interval open relative to $\Theta$ with $\Theta' \subseteq W$ witnessing conditions (a) and (b) for $q_L$ on $\Theta'$, with the canonical side margins $\kappa_a$ and $\kappa_b$ of the proof of Proposition 1, and let $q$ be a symmetric mechanism, non-increasing in $b_2$, self-confirming with respect to $\theta^*$ on an interval $W'$ open relative to $\Theta$ with $\Theta' \subseteq W'$. The set $W'' = W \cap W'$ is an interval open relative to $\Theta$ with $\Theta' \subseteq W''$, every compact $\Theta'' = [\underline{\theta}'', \bar{\theta}''] \subseteq W''$ with $\underline{\theta}'' < \theta^* < \bar{\theta}''$ is such a compact in $W$, and $\overline{\beta}(V,W'') \subseteq \overline{\beta}(V,W) \cap \overline{\beta}(V,W')$.

Fix $b \in \overline{\beta}(V,W'')$ and recall $\alpha(b) = \Phi^{-1}(h(b,\theta^*) \mid \theta^*)$. Under (R1), $h(b,\theta^*) \in (0,1)$, and since $f$ is continuous and positive and $\beta(\cdot,\theta^*)$ is strictly increasing, the distribution $\Phi(\cdot \mid \theta^*)$ is atomless

with density $\phi(\cdot \mid \theta^*)$ positive on the interior of $\beta(V,\theta^*)$, so $\Phi(\cdot \mid \theta^*)$ is strictly increasing on $\beta(V,\theta^*)$ and $\alpha(b)$ lies in its interior. The opponent bids below $\alpha(b)$ with probability $\Phi(\alpha(b) \mid \tilde{\theta})$ under any degenerate belief $\tilde{\theta}$, so $Q_L(b \mid \tilde{\theta}) = \Phi(\alpha(b) \mid \tilde{\theta})$ for every $\tilde{\theta}$, and in particular $Q_L(b \mid \theta^*) = h(b,\theta^*)$, so $q_L$ is self-confirming at $b$. Since $b \in \overline{\beta(V,W')}$, $q$ is self-confirming at $b$ as well. Writing $d(b_2) = q(b,b_2) - q_L(b,b_2)$,

$$\int d(b_2)\,\phi(b_2 \mid \theta^*)\,db_2 = Q(b \mid \theta^*) - Q_L(b \mid \theta^*) = 0.$$

Because $q_L(b,b_2) = 1$ for $b_2 < \alpha(b)$ and $q_L(b,b_2) = 0$ for $b_2 \geq \alpha(b)$, while $q(b,b_2) \in [0,1]$, the difference satisfies $d(b_2) \leq 0$ for $b_2 < \alpha(b)$ and $d(b_2) \geq 0$ for $b_2 > \alpha(b)$.

Fix $\tilde{\theta} \in W''$ with $\tilde{\theta} > \theta^*$. Since $\beta$ is strictly increasing in $\theta$, the support $\beta(V,\tilde{\theta})$ contains no bids below $\beta(\underline{v},\theta^*)$, and its portion above $\beta(V,\theta^*)$ is contained in $(\beta(\bar{v},\theta^*),\beta(\bar{v},\tilde{\theta})]$. Splitting the integral accordingly,

$$Q(b \mid \tilde{\theta}) - Q_L(b \mid \tilde{\theta}) = \int_{\beta(V,\theta^*)} d(b_2)\,\ell(b_2)\,\phi(b_2 \mid \theta^*)\,db_2 + \int_{(\beta(\bar{v},\theta^*),\,\beta(\bar{v},\tilde{\theta})]} d(b_2)\,\phi(b_2 \mid \tilde{\theta})\,db_2,$$

where $\ell(b_2) = \phi(b_2 \mid \tilde{\theta})/\phi(b_2 \mid \theta^*)$ on the interior of $\beta(V,\theta^*)$. The second integral is non-negative, since every bid in its range exceeds $\alpha(b)$ and $d \geq 0$ there. For the first, $\ell$ is increasing by MLRP, and $\ell(\alpha(b))$ is finite since $\phi(\alpha(b) \mid \theta^*) > 0$. For $b_2 < \alpha(b)$, $d(b_2) \leq 0$ and $\ell(b_2) \leq \ell(\alpha(b))$, so $d(b_2)\,\ell(b_2) \geq d(b_2)\,\ell(\alpha(b))$. For $b_2 > \alpha(b)$, $d(b_2) \geq 0$ and $\ell(b_2) \geq \ell(\alpha(b))$, so $d(b_2)\,\ell(b_2) \geq d(b_2)\,\ell(\alpha(b))$ again. Integrating the pointwise bound,

$$\int_{\beta(V,\theta^*)} d(b_2)\,\ell(b_2)\,\phi(b_2 \mid \theta^*)\,db_2 \geq \ell(\alpha(b)) \int_{\beta(V,\theta^*)} d(b_2)\,\phi(b_2 \mid \theta^*)\,db_2 = 0.$$

Therefore $Q(b \mid \tilde{\theta}) \geq Q_L(b \mid \tilde{\theta})$ for every $\tilde{\theta} \in W''$ with $\tilde{\theta} > \theta^*$.

For $\tilde{\theta} \in W''$ with $\tilde{\theta} < \theta^*$ the argument is symmetric. The support $\beta(V,\tilde{\theta})$ contains no bids above $\beta(\bar{v},\theta^*)$, and its portion below $\beta(V,\theta^*)$ lies below $\alpha(b)$, where $d \leq 0$, so that portion contributes a non-positive term. On $\beta(V,\theta^*)$ the ratio $\ell$ is decreasing by MLRP, so $d(b_2)\,\ell(b_2) \leq d(b_2)\,\ell(\alpha(b))$ on both sides of $\alpha(b)$, and the integral is at most $\ell(\alpha(b)) \cdot 0 = 0$. Therefore $Q(b \mid \tilde{\theta}) \leq Q_L(b \mid \tilde{\theta})$.

For any compact $\Theta'' \subseteq W''$ with $\underline{\theta}'' < \theta^* < \bar{\theta}''$ and any $b \in \overline{\beta(V,W'')}$, condition (a) for $q_L$ gives

$$Q(b \mid \bar{\theta}'') - h(b,\bar{\theta}'') \geq Q_L(b \mid \bar{\theta}'') - h(b,\bar{\theta}'') \geq \kappa_a(\bar{\theta}''),$$

and condition (b) gives $h(b,\underline{\theta}'') - Q(b \mid \underline{\theta}'') \geq h(b,\underline{\theta}'') - Q_L(b \mid \underline{\theta}'') \geq \kappa_b(\underline{\theta}'')$. The margins are those of $q_L$, each positive and a function of the indicated endpoint alone, and for every $c > 0$ the infimum of $\kappa_a$ over endpoints admissible in $W''$ at distance at least $c$ above $\theta^*$ is bounded below by the corresponding infimum over endpoints admissible in $W$, which is positive, and likewise for $\kappa_b$. Conditions (a) and (b) hold for $q$ on $\Theta'$ with $W''$ in place of $W$. □

### 7.3.2 Proof of Proposition 3

*Proof of Proposition 3.* Write $\underline{r} = \beta(\underline{v},\theta^*)$, $\bar{r} = \beta(\bar{v},\theta^*)$, and $R = [\underline{r},\bar{r}]$. At $b = \underline{r}$ and $b = \bar{r}$, $h_\theta(b,\theta^*)$ in the displayed condition denotes the limit of $h_\theta(\cdot,\theta^*)$ over interior bids, $h_\theta(\bar{r},\theta^*) =$

$\lim_{b\uparrow\bar{r}} h_\theta(b,\theta^*)$ and $h_\theta(\underline{r},\theta^*) = \lim_{b\downarrow\underline{r}} h_\theta(b,\theta^*)$, the limits existing by continuity of $h_\theta$ on the interior of $R$.

**Step 1: conditions (a) and (b) for $q_L$.** As in the proof of Lemma 8, $\alpha(b) = \Phi^{-1}(h(b,\theta^*) \mid \theta^*)$ is well defined at every bid and interior to $R$, and $Q_L(b \mid \tilde{\theta}) = \Phi(\alpha(b) \mid \tilde{\theta})$ for every degenerate belief $\tilde{\theta}$. Define $\Lambda(b,\tilde{\theta}) = \Phi(\alpha(b) \mid \tilde{\theta}) - h(b,\tilde{\theta})$, so that $\Lambda(b,\theta^*) = 0$ at every bid and $\Lambda_\theta(b,\tilde{\theta}) = \Phi_\theta(\alpha(b) \mid \tilde{\theta}) - h_\theta(b,\tilde{\theta})$ wherever the derivatives exist. At $\tilde{\theta} = \theta^*$ and $b \in R$, the displayed condition together with $h$ decreasing in $\theta$ at $\theta^*$ gives $h_\theta(b,\theta^*) < 0$, since a zero derivative would contradict the strict inequality, and therefore

$$\Lambda_\theta(b,\theta^*) \geq |h_\theta(b,\theta^*)| - |\Phi_\theta(\alpha(b) \mid \theta^*)| > 0.$$

For $\varepsilon > 0$ write $W_\varepsilon = (\theta^* - \varepsilon, \theta^* + \varepsilon)$, and fix $\varepsilon_0 > 0$ with $\bar{W}_{\varepsilon_0} \subseteq \Theta$, which exists since $\theta^*$ lies in the relative interior of $\Theta$. The function $h(\cdot,\theta^*)$ is continuous on $\overline{\beta(V,W_{\varepsilon_0})}$ with values in a compact subset of $(0,1)$, so $\alpha(\cdot)$ is continuous there with its image a compact subset of the interior of $R$. Since $\Phi(b' \mid \tilde{\theta}) = F(\beta^{-1}(b',\tilde{\theta}))$ with $f$ continuous and $\beta$ continuously differentiable, $\Phi_\theta(b' \mid \tilde{\theta})$ is continuous in $(b',\tilde{\theta})$ at bids interior to $\beta(V,\tilde{\theta})$, so, shrinking $\varepsilon_0$ if needed, $(b,\tilde{\theta}) \mapsto \Phi_\theta(\alpha(b) \mid \tilde{\theta})$ is continuous on $P_0 = \overline{\beta(V,W_{\varepsilon_0})} \times \bar{W}_{\varepsilon_0}$.

The convention following (4) partitions the definition of $h$ across three sets. Let $D = \{(b,\tilde{\theta}) \in P_0 \mid \beta(\underline{v},\tilde{\theta}) \leq b \leq \beta(\bar{v},\tilde{\theta})\}$, compact since $\beta$ is continuous. On $D$, $h(b,\tilde{\theta}) = g(b,\beta^{-1}(b,\tilde{\theta}),\tilde{\theta})$; where $b \geq \beta(\bar{v},\tilde{\theta})$, $h(b,\tilde{\theta}) = g(b,\bar{v},\tilde{\theta})$; where $b \leq \beta(\underline{v},\tilde{\theta})$, $h(b,\tilde{\theta}) = g(b,\underline{v},\tilde{\theta})$. On $P_0$ define

$$\chi_+(b,\tilde{\theta}) = \Phi_\theta(\alpha(b) \mid \tilde{\theta}) - g_\theta(b,\bar{v},\tilde{\theta}), \qquad \chi_-(b,\tilde{\theta}) = \Phi_\theta(\alpha(b) \mid \tilde{\theta}) - g_\theta(b,\underline{v},\tilde{\theta}),$$

and on the set $D$ define $\chi_0(b,\tilde{\theta}) = \Phi_\theta(\alpha(b) \mid \tilde{\theta}) - g_v\beta_\theta^{-1} - g_\theta$, with $g_v$ and $g_\theta$ evaluated at $(b,\beta^{-1}(b,\tilde{\theta}),\tilde{\theta})$ and $\beta_\theta^{-1} = -\beta_\theta/\beta_v$ evaluated at $(\beta^{-1}(b,\tilde{\theta}),\tilde{\theta})$, with $\beta_v$ bounded below on compacts under (R3), so that $\chi_0$ is continuous on all of $D$; $\chi_+$ and $\chi_-$ are continuous on $P_0$. On the interior of each of the three sets, $\Lambda(b,\cdot)$ is differentiable with $\Lambda_\theta$ equal to $\chi_0$, $\chi_+$, or $\chi_-$ respectively, one-sided where two sets meet, and under the endpoint convention above $\Lambda_\theta(b,\theta^*) = \chi_0(b,\theta^*)$ for every $b \in R$. Where $D$ meets $\{(b,\tilde{\theta}) \in P_0 \mid b \geq \beta(\bar{v},\tilde{\theta})\}$, $\beta^{-1}(b,\tilde{\theta}) = \bar{v}$ and $\chi_+ - \chi_0 = g_v\beta_\theta^{-1}$; by (R2), $g_v \leq 0$, and $\beta_\theta^{-1} < 0$ since $\beta$ is strictly increasing in $\theta$, so $\chi_+ \geq \chi_0$ there. Likewise $\chi_- \geq \chi_0$ where $D$ meets $\{(b,\tilde{\theta}) \in P_0 \mid b \leq \beta(\underline{v},\tilde{\theta})\}$.

Let $2m$ be the minimum of $\chi_0(\cdot,\theta^*)$ over $R \times \{\theta^*\} \subseteq D$, positive by the previous display and attained by continuity and compactness. Every point of $D$ with $|\tilde{\theta} - \theta^*| \leq \varepsilon$ lies within vanishing distance of $R \times \{\theta^*\}$ as $\varepsilon \to 0$, since the endpoints $\beta(\underline{v},\tilde{\theta})$ and $\beta(\bar{v},\tilde{\theta})$ converge to $\underline{r}$ and $\bar{r}$, so uniform continuity of $\chi_0$ on the compact $D$ gives $\varepsilon \in (0,\varepsilon_0]$ with $\chi_0 \geq m$ on $D \cap (\overline{\beta(V,W_\varepsilon)} \times \bar{W}_\varepsilon)$. The point $(\bar{r},\theta^*)$ lies in $D$ with $\bar{r} = \beta(\bar{v},\theta^*)$, so $\chi_+(\bar{r},\theta^*) \geq \chi_0(\bar{r},\theta^*) \geq 2m$, and every point of $\overline{\beta(V,W_\varepsilon)} \times \bar{W}_\varepsilon$ with $b \geq \beta(\bar{v},\tilde{\theta})$ lies within vanishing distance of $(\bar{r},\theta^*)$ as $\varepsilon \to 0$, since $\beta(\bar{v},\tilde{\theta}) \leq b \leq \beta(\bar{v},\theta^* + \varepsilon)$, so continuity of $\chi_+$ gives $\chi_+ \geq m$ at all such points, shrinking $\varepsilon$ if needed; likewise $\chi_- \geq m$ near $(\underline{r},\theta^*)$. For such $\varepsilon$, the derivative of $\Lambda(b,\cdot)$ is at least $m$ throughout $\overline{\beta(V,W_\varepsilon)} \times \bar{W}_\varepsilon$ wherever it exists.

For fixed $b \in \overline{\beta(V,W_\varepsilon)}$, the map $\tilde{\theta} \mapsto \Lambda(b,\tilde{\theta})$ is continuous on $\bar{W}_\varepsilon$ and continuously differentiable except where $\beta(\bar{v},\tilde{\theta}) = b$ or $\beta(\underline{v},\tilde{\theta}) = b$, each holding for at most one $\tilde{\theta}$ by strict monotonicity of $\beta$ in $\theta$. Integrating the derivative piecewise,

$$\Lambda(b,\tilde{\theta}) \geq m(\tilde{\theta} - \theta^*) \text{ for } \tilde{\theta} \in [\theta^*, \theta^* + \varepsilon], \qquad -\Lambda(b,\tilde{\theta}) \geq m(\theta^* - \tilde{\theta}) \text{ for } \tilde{\theta} \in [\theta^* - \varepsilon, \theta^*].$$

Set $W = W_\varepsilon$. Then $W$ is an interval open relative to $\Theta$ containing $\theta^*$ in its relative interior, the bounds above hold at every $b \in \overline{\beta(V,W)}$ and every $\tilde{\theta} \in W$, and setting $\kappa_a(\bar{\theta}'') = m(\bar{\theta}'' - \theta^*)$ and $\kappa_b(\underline{\theta}'') = m(\theta^* - \underline{\theta}'')$, conditions (a) and (b) hold for $q_L$ on $\{\theta^*\}$ with witness $W$, with the infimum of each margin over endpoints at distance at least $c$ from $\theta^*$ equal to $mc > 0$ for every $c > 0$.

**Step 2: a feasible self-confirming mechanism on $R$.** By Condition 2 and bid monotonicity, following the constructive approach of Maskin and Riley (1984) and Matthews (1984), there exists a feasible symmetric mechanism $q_0$, non-increasing in $b_2$, with $Q_0(b \mid \theta^*) = h(b, \theta^*)$ for all $b \in R$, where $Q_0$ denotes the objective win probability under $q_0$.

**Step 3: extension off $R$.** Define $q$ by $q(b_1,b_2) = q_0(b_1,b_2)$ for $b_1, b_2 \in R$; for $b_1 \in R$, $q(b_1,b_2) = q_0(b_1,\underline{r})$ when $b_2 < \underline{r}$ and $q(b_1,b_2) = 0$ when $b_2 > \bar{r}$; for $b_1 > \bar{r}$, $q(b_1,b_2) = h(b_1,\theta^*)$ when $b_2 \le \bar{r}$ and $q(b_1,b_2) = 0$ when $b_2 > \bar{r}$; and for $b_1 < \underline{r}$, $q(b_1,b_2) = \min\{q(\underline{r},b_2), c(b_1)\}$ at every $b_2$, where $q(\underline{r},\cdot)$ denotes the extended row. The map $c \mapsto \int_R \min\{q_0(\underline{r},b_2), c\}\,\phi(b_2 \mid \theta^*)\,db_2$ is continuous and non-decreasing, equal to 0 at $c = 0$ and to $h(\underline{r},\theta^*)$ at $c = 1$, and $h(b_1,\theta^*) \in (0, h(\underline{r},\theta^*))$ for $b_1 < \underline{r}$ by (R1) and bid monotonicity, so

$$c(b_1) = \inf\left\{ c \in [0,1] \,\middle|\, \int_R \min\{q_0(\underline{r},b_2), c\}\,\phi(b_2 \mid \theta^*)\,db_2 \ge h(b_1,\theta^*) \right\}$$

attains the self-confirming value and is non-decreasing in $b_1$, so $q$ is jointly measurable.

Each row of $q$ takes values in $[0,1]$ and is non-increasing in the opponent's bid, and $q$ is symmetric as a single rule applied to both buyers. Self-confirmation holds at every bid, since $\Phi(\cdot \mid \theta^*)$ has support $R$, where the on-range rows agree with $q_0$, the rows above $\bar{r}$ are constant at $h(b_1,\theta^*)$, and the rows below $\underline{r}$ are calibrated by $c(b_1)$. In particular $q$ is self-confirming with respect to $\theta^*$ on $W$.

Feasibility (1) holds profile by profile. For $b_1, b_2 \in R$ it is feasibility of $q_0$. For $b_1 < \underline{r}$ and $b_2 \in R$, $q(b_1,b_2) \le q_0(\underline{r},b_2)$ and $q(b_2,b_1) = q_0(b_2,\underline{r})$, so the sum is at most $q_0(\underline{r},b_2) + q_0(b_2,\underline{r}) \le 1$ by feasibility of $q_0$ at $(\underline{r},b_2)$. For $b_1 > \bar{r}$, the opposing row vanishes at every opponent bid above $\bar{r}$, whether $b_2 \in R$, $b_2 > \bar{r}$, or $b_2 < \underline{r}$, in the last case because the $\underline{r}$ row vanishes above $\bar{r}$, so the sum is at most $q(b_1,b_2) \le 1$. For $b_1, b_2 < \underline{r}$, both rows evaluate the $\underline{r}$ row at bids below $\underline{r}$, so the sum is at most $2q_0(\underline{r},\underline{r}) \le 1$ by feasibility of $q_0$ at $(\underline{r},\underline{r})$.

**Step 4: inheritance.** The mechanism $q$ is symmetric, non-increasing in $b_2$, and self-confirming with respect to $\theta^*$ on $W$, an interval open relative to $\Theta$ with $\{\theta^*\} \subseteq W$. By Step 1, $q_L$ satisfies conditions (a) and (b) on $\{\theta^*\}$. Lemma 8 applied with $W' = W$ gives that $q$ satisfies conditions (a) and (b) on $\{\theta^*\}$ with witness $W$ and the margins of Step 1.

**Step 5: conclusion.** The set $\{\theta^*\}$ is a compact interval containing $\theta^*$, and $\theta^*$ lies in the relative interior of $\Theta$. Proposition 1 applied with $\Theta' = \{\theta^*\}$ gives that $q$ sustains $\theta^*$ on $\{\theta^*\}$, so $\theta^*$ is locally stable under $q$, and $q$ is feasible, symmetric, self-confirming, and non-increasing in $b_2$. The same holds for every feasible mechanism in the maintained class that is self-confirming with respect to $\theta^*$ on an interval $W'$ open relative to $\Theta$ and containing $\theta^*$, since Lemma 8 applies with that $W'$ and the witness of Step 1. □

### 7.3.3 Discrete Valuation

Condition 1 for one-sided convergence, with $W = [\underline{\theta}, \theta^* + \varepsilon)$ for some $\varepsilon > 0$, has as its pointwise content four conditions on the mechanism $q$ over the bid range $\mathscr{B} = [\beta(\underline{v},\underline{\theta}), \beta(\bar{v},\theta^* + \varepsilon)]$:

(F) Feasibility: $q(b_1,b_2)+q(b_2,b_1)\le 1$ for all $(b_1,b_2)\in\mathscr{B}^2$

(SC) Self-Confirming: $\sum_{k=1}^{K} p_k q(b_1,\beta(v_k,\theta^*)) = h(b_1,\theta^*)$ for all $b_1\in\mathscr{B}$

(CF-O) Optimistic-Side Corrective Feedback: $\sum_{k=1}^{K} p_k q(b_1,\beta(v_k,\tilde{\theta})) < h(b_1,\tilde{\theta})$ for all $b_1\in\mathscr{B}$ and $\tilde{\theta}\in[\underline{\theta},\theta^*)$

(CF-P) Pessimistic-Side Corrective Feedback: $\sum_{k=1}^{K} p_k q(b_1,\beta(v_k,\tilde{\theta})) > h(b_1,\tilde{\theta})$ for all $b_1\in\mathscr{B}$ and $\tilde{\theta}\in(\theta^*,\theta^*+\varepsilon)$

Conditions (CF-O) and (CF-P) correspond to parts (b) and (a) of Condition 1 respectively. The conditions are stated on the full bid range $\mathscr{B}$ rather than the grid $B(\theta^*)$ alone because even with discrete valuations, a buyer whose belief is not exactly $\delta_{\theta^*}$ may submit bids outside $B(\theta^*)$, and the mechanism's response to these off-grid bids feeds back into the opponent's learning. For the same reason, (SC) is self-confirmation with respect to $\theta^*$ on $W$, holding at every bid in $\mathscr{B}$ rather than only at the bids induced by $\theta^*$.

Proposition 1 continues to apply, with the integral in (5) read as the finite sum $\sum_{k=1}^{K} p_k\, q(b,\beta(v_k,\tilde{\theta}))$. Nothing in the proposition or its proof requires the valuation distribution to admit a density. As there, strict inequalities alone carry no uniform pull over the continuum of off-grid bids and sub-target parameters, and one-sided convergence asks for (CF-O) and (CF-P) in margined form, with margins uniformly positive over parameters bounded away from the target.

Partition the bid space below $\beta(v_K,\theta^*)$ into $K$ regions $B_k=(\beta(v_{k-1},\theta^*),\beta(v_k,\theta^*)]$ for $k\ge 2$, with $B_1=(-\infty,\beta(v_1,\theta^*)]$. For any belief $\tilde{\theta}$, let $\pi_k(\tilde{\theta})=\Pr(\beta(v,\tilde{\theta})\in B_k)$ denote the probability that a buyer with belief $\tilde{\theta}$ bids into region $B_k$. The conditions that follow weight the mechanism's grid values by these probabilities.

For $\eta_1,\eta_2\ge 0$, let $LP(\eta_1,\eta_2)$ denote the linear program in variables $\{\hat{q}(\beta(v_j,\theta^*),\beta(v_k,\theta^*))\}_{j,k}$ taking values in $[0,1]$, with constraints (F′), (SC′), (MON′), and (CF-O′), where (F′) is the feasibility condition (F) at every pair of grid bids with the bound tightened to $1-\eta_1$, (SC′) is the self-confirming condition (SC) at every grid bid, and

(MON') $\hat{q}(\beta(v_j,\theta^*),\beta(v_k,\theta^*))\ge \hat{q}(\beta(v_j,\theta^*),\beta(v_{k+1},\theta^*))$ for all $j$ and $k<K$

(CF-O') $\sum_{k=1}^{K}\pi_k(\tilde{\theta})\,\hat{q}(\beta(v_j,\theta^*),\beta(v_k,\theta^*))\le h(\beta(v_j,\theta^*),\tilde{\theta})-\eta_2$ for all $j$ and $\tilde{\theta}\in\tilde{\Theta}$.

**Lemma 9** (Grid Program)**.** *Suppose that the model has fading sensitivity and adverse ordering. Whether some mechanism is feasible, self-confirming, and corrective on the optimistic side is decided by a finite linear program on the grid, in the sense that there exists a mechanism $q(b_1,b_2)$ satisfying, at every $b_1,b_2\le\beta(v_K,\theta^*)$, (F), (SC), and (CF-O) for every $\tilde{\theta}\in[\underline{\theta},\theta^*)$, if and only if $LP(0,\eta_2)$ is feasible for some $\eta_2>0$. The mechanism can moreover be taken to satisfy*

$(CE)_0$ *Constant Extension: for all $b_1$ and $b_2$,*

$$q(b_1,b_2)=q(b_1,\min\{\beta(v_k,\theta^*)\mid\beta(v_k,\theta^*)\ge b_2\})$$

*(TS) Threshold Structure: for all $b_1$, there exists $\zeta(b_1)\in[0,1]$ such that*

$$q(b_1,b_2)=\min\{\zeta(b_1),q(\bar{b}_{grid}(b_1),b_2)\}$$

*for all $b_2$, where $\bar{b}_{grid}(b_1)=\min\{\beta(v_k,\theta^*)\mid\beta(v_k,\theta^*)\ge b_1\}$ and $\zeta(b_1)=1$ at $b_1\in B(\theta^*)$*

The program $LP(0,\eta_2)$ has $K^2$ variables, $O(K^2)$ feasibility and monotonicity constraints, $K$ self-confirming constraints, and $O(K\cdot|\tilde{\Theta}|)$ (CF-O′) constraints, where $|\tilde{\Theta}|\leq K^2$. Constraints at the overlap beliefs suffice because the bin probabilities $\pi_k(\theta)$ are constant in $\theta$ between overlap beliefs while $h(b,\theta)$ rises as $\theta$ falls. Whether the program is feasible for some positive slack is itself decided by a linear program, obtained by maximising $\eta_2$ subject to the remaining constraints, so the existence of a mechanism in the maintained class satisfying (F), (SC), and (CF-O) on $[\underline{\theta},\theta^*)$ is decided in time polynomial in $K$. The grid values of any mechanism satisfying Condition 1 on $[\underline{\theta},\theta^*]$ are feasible for $LP(0,\eta_2)$ at some $\eta_2>0$, so where the program is infeasible for every positive slack, Condition 1 cannot be verified for one-sided convergence by any mechanism in the maintained class.

The forward direction is constructive. Own-bids on the grid take the solution values, extended to off-grid opponent bids by $(CE)_0$, which evaluates the mechanism at the grid bid at or above $b_2$ and makes it a step function in the opponent's bid with jumps only at grid bids. Off-grid own-bids take the win probabilities of the grid bid above them, capped at the constant $\zeta(b_1)$ that restores the self-confirming condition at $b_1$. The cap binds at low opponent bids, where the inherited win probability is high, and ceases to bind at high opponent bids, where the inherited win probability falls below it, so the win probability is flat in the opponent's bid up to a threshold and follows the inherited win probabilities beyond it. At a grid bid the cap does not bind and (TS) holds as an identity. Panel (a) of Figure 6 draws both cases.

The transformation preserves the three conditions and extends them to off-grid pairs of bids. Feasibility at a pair of bids follows from feasibility at the pair of grid bids at or above them. By the constant extension, replacing each bid by the grid bid at or above it leaves each buyer's win probability unchanged in the opponent's coordinate. By the threshold structure, the replacement weakly raises each buyer's win probability in his own coordinate. Therefore, the sum at the original pair is at most the grid-pair sum, which (F′) bounds by one. Self-confirmation holds at every own-bid by the choice of $\zeta(b_1)$. Condition (CF-O) at an off-grid own-bid follows from the slack $\eta_2$ at the own grid bid at or above it. At either bid, self-confirmation anchors the win probability at the opponent belief $\theta^*$, so (CF-O) compares the win probability gained when the opponent's belief falls below $\theta^*$ against the rise in $h$ over the same fall. By the threshold structure, the gain is weakly smaller at the off-grid own-bid than at the grid bid. By fading sensitivity, the rise in $h$ is weakly larger. Therefore, the gap of $\eta_2$ at the grid bid survives at the off-grid own-bid.

However, convergence asks more than the three conditions. An initial belief near $\Delta([\underline{\theta},\theta^*])$ may place mass slightly above $\theta^*$, so Proposition 1 requires $W$ to extend above the target, where local stability takes over. The mechanism of Lemma 9 need not be constant in the opponent's bid near the grid bids, but when the linear program is feasible with slack in its feasibility constraints, a perturbation of the construction makes it so, and local stability is obtained for free.

**Lemma 10** (One-Sided Construction)**.** *Suppose the model has fading sensitivity and adverse ordering. Some mechanism sustains $\theta^*$ on $[\underline{\theta},\theta^*]$ by generating corrective feedback if $LP(\eta_1,\eta_2)$ has a solution for some $\eta_1,\eta_2>0$. Given a solution, for every sufficiently small $\varepsilon>0$ there exist $\delta>0$ and such a mechanism $q$, satisfying Condition 1 with $W=[\underline{\theta},\theta^*+\varepsilon)$, with*

$(CE)_\delta$ *Constant Extension: for all $b_1\in\mathscr{B}$,*

$$q(b_1,b_2)=q(b_1,\min\{\beta(v_k,\theta^*)\mid\beta(v_k,\theta^*)+\delta\geq b_2\})$$

*(PS)* *Proportional Scaling: for every* $b_1 \in (\beta(v_k,\theta^*), \beta(v_k,\theta^*)+\delta)$ *and all* $b_2$,

$$q(b_1,b_2) = \frac{h(b_1,\theta^*)}{h(\beta(v_k,\theta^*),\theta^*)}\, q(\beta(v_k,\theta^*),b_2)$$

*(TS)* *Threshold Structure: at every* $b_1 \in \mathscr{B}$ *outside* $\bigcup_k (\beta(v_k,\theta^*), \beta(v_k,\theta^*)+\delta)$, *there exists* $\zeta(b_1) \in [0,1]$ *such that*

$$q(b_1,b_2) = \min\{\zeta(b_1), q(\bar{b}_{grid}(b_1),b_2)\}$$

*for all* $b_2$, *with* $\zeta(b_1) = 1$ *at* $b_1 \in B(\theta^*)$

The construction extends the grid solution to a mechanism on all bids, preserving feasibility and stability. Property $(CE)_\delta$ moves each step boundary upward by $\delta$, making the mechanism constant in the opponent's bid near each grid bid at $\theta^*$, and local stability is obtained for free. The shift also moves each overlap belief. A bid crosses the moved boundary at the belief at which it falls to the grid bid plus $\delta$, above the belief at which it falls to the grid bid itself, and we call that belief the shifted overlap belief. At $\delta = 0$ it is the overlap belief. Property (PS) assigns own-bids in the windows the win probabilities of the grid bid below, rescaled so that self-confirmation holds exactly at the own-bid, and the threshold structure of Lemma 9 carries over outside the windows. The slacks $\eta_1$ and $\eta_2$ absorb the perturbations the shift induces in feasibility and in (CF-O). Panel (b) of Figure 6 draws the mechanism under the shifted boundaries at the example of panel (a).

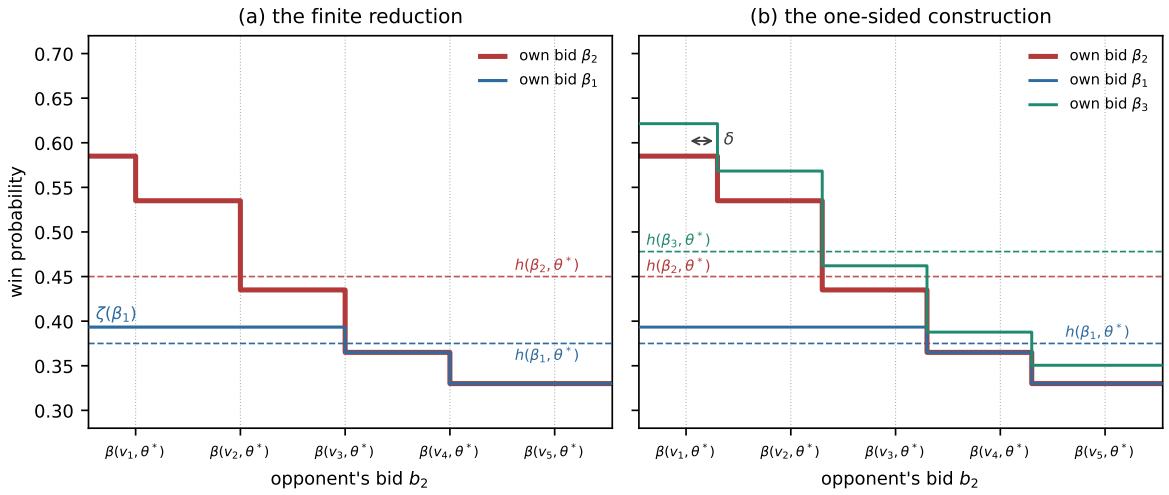



Figure 6: The constructions of Lemmas 9 and 10 at an illustrative five-value example with uniform type probabilities. Throughout, $\beta_2$ denotes the grid bid $\beta(v_4,\theta^*)$, $\beta_1$ an own-bid in $(\beta(v_3,\theta^*),\beta_2)$, and $\beta_3$ an own-bid in the window $(\beta_2,\beta_2+\delta)$. Panel (a), cross-sections in the opponent's bid at own-bids $\beta_2$ and $\beta_1$ under the extension of Lemma 9. The mechanism is constant in the opponent's bid between grid bids by $(CE)_0$, and at $\beta_1$ the win probabilities inherited from $\beta_2$ are capped at $\zeta(\beta_1)$ by (TS), flat up to a threshold and following the inherited values beyond it. Dashed lines mark the level each cross-section averages to across the opponent's grid bids at $\theta^*$. Panel (b), the same cross-sections under the construction of Lemma 10 at shift $\delta$, the step boundaries moved upward by $\delta$ by $(CE)_\delta$, together with the cross-section at $\beta_3$, the $\beta_2$ values rescaled by $h(\beta_3,\theta^*)/h(\beta_2,\theta^*)$ by (PS).

Throughout the construction, subscripted constraints denote margined forms. For $\eta > 0$, $(F)_\eta$ at a pair $(b_1, b_2)$ denotes $q(b_1,b_2)+q(b_2,b_1) \le 1-\eta$, and (CF-O)$_\eta$ at a bid $b_1$ and parameter $\tilde{\theta}$ denotes $\sum_{k=1}^{K} p_k q(b_1, \beta(v_k,\tilde{\theta})) \le h(b_1,\tilde{\theta}) - \eta$. The primed grid constraints carry the same notation, so $(F')_\eta$ at $(j,k)$ denotes $\hat{q}(\beta(v_j,\theta^*),\beta(v_k,\theta^*)) + \hat{q}(\beta(v_k,\theta^*),\beta(v_j,\theta^*)) \le 1-\eta$, and (CF-O$'$)$_\eta$ at $j$ and $\tilde{\theta} \in \tilde{\Theta}$ denotes $\sum_{k=1}^{K} \pi_k(\tilde{\theta})\,\hat{q}(\beta(v_j,\theta^*),\beta(v_k,\theta^*)) \le h(\beta(v_j,\theta^*),\tilde{\theta}) - \eta$. Where a statement asserts a subscripted constraint over a range of arguments, the inequality holds at every argument in the range with the subscripted slack. $(CE)_0$ denotes $(CE)_\delta$ with $\delta = 0$, and $(F)_0$ and $(F')_0$ denote the unmargined forms (F) and (F'). For any bid $b$, $\underline{b}_{grid}(b) = \max\{\beta(v_k,\theta^*) \,|\, \beta(v_k,\theta^*) < b\}$ denotes the largest grid bid strictly below $b$, with $\underline{b}_{grid}(b) = -\infty$ for $b \le \beta(v_1,\theta^*)$, so that every bid in $(-\infty, \beta(v_1,\theta^*)]$ satisfies $b \ge \underline{b}_{grid}(b) + \delta$ for every $\delta > 0$. To prove Proposition 9, we must first establish the following lemmas:

**Lemma 11** (Off-Grid Extension)**.** *Suppose the model has bid monotonicity and fading sensitivity. If there exists a symmetric mechanism $q(b_1,b_2)$, non-increasing in $b_2$, satisfying $(CE)_0$ and, at every $b_1, b_2 \in B(\theta^*)$, (SC), $(F)_{\eta_1}$, and $(CF\text{-}O)_{\eta_2}$ for every $\tilde{\theta} \in \tilde{\Theta}$, then there exists a symmetric mechanism $\hat{q}(b_1,b_2)$, non-increasing in $b_2$, satisfying $(CE)_0$ and, at every $b_1, b_2 \le \beta(v_K,\theta^*)$, (SC), $(F)_{\eta_1}$, and $(CF\text{-}O)_{\eta_2}$ for every $\tilde{\theta} \in \tilde{\Theta}$, with $\hat{q}(b_1,b_2) = \min\{\zeta(b_1), \hat{q}(\bar{b}_{grid}(b_1), b_2)\}$ at every $b_1 \notin B(\theta^*)$, where $\bar{b}_{grid}(b_1) = \min\{\beta(v_k,\theta^*) | \beta(v_k,\theta^*) \ge b_1\}$ and $\zeta(b_1) \in [0,1]$.*

*Proof.* Define $\hat{q}(b_1,b_2) = q(b_1,b_2)$ for $b_1 \in B(\theta^*)$, so that the conditions hold at grid own bids by hypothesis. For $b_1 \notin B(\theta^*)$, let $\bar{b} = \bar{b}_{grid}(b_1)$ and

$$\hat{q}(b_1,b_2) = \min\{\zeta(b_1), q(\bar{b},b_2)\}$$

where $\zeta(b_1)$ is such that

$$\sum_{k=1}^{K} p_k \hat{q}(b_1, \beta(v_k,\theta^*)) = h(b_1,\theta^*)$$

Such $\zeta(b_1)$ exists by the intermediate value theorem, because the sum is continuous and weakly increasing in $\zeta(b_1)$, equals 0 at $\zeta(b_1) = 0$, and equals $h(\bar{b},\theta^*) \ge h(b_1,\theta^*)$ at $\zeta(b_1) = 1$ by (SC) at $\bar{b}$ and bid monotonicity. Therefore $\hat{q}$ satisfies (SC) at every $b_1 \le \beta(v_K,\theta^*)$.

$\hat{q}(b_1,b_2)$ is non-increasing in $b_2$ and satisfies $(CE)_0$, because $q(\bar{b},b_2)$ is non-increasing in $b_2$ and satisfies $(CE)_0$ while $\zeta(b_1)$ is constant in $b_2$.

For feasibility, at every $b_1, b_2 \le \beta(v_K,\theta^*)$,

$$\hat{q}(b_1,b_2)+\hat{q}(b_2,b_1) \le q(\bar{b}_{grid}(b_1),b_2)+q(\bar{b}_{grid}(b_2),b_1) = q(\bar{b}_{grid}(b_1),\bar{b}_{grid}(b_2))+q(\bar{b}_{grid}(b_2),\bar{b}_{grid}(b_1)) \le 1-\eta_1$$

where the equality is $(CE)_0$ and the last inequality is $(F)_{\eta_1}$ on $B(\theta^*) \times B(\theta^*)$, with $\bar{b}_{grid}(b) = b$ for $b \in B(\theta^*)$.

For (CF-O), fix $b_1 \notin B(\theta^*)$ with $\bar{b} = \bar{b}_{grid}(b_1)$ and $\tilde{\theta} \in \tilde{\Theta}$. Because $q(\bar{b},b_2)$ is non-increasing in $b_2$ and $\beta(v_k,\tilde{\theta}) \le \beta(v_k,\theta^*)$, for every $k$

$$\min\{\zeta(b_1), q(\bar{b},\beta(v_k,\tilde{\theta}))\} - \min\{\zeta(b_1), q(\bar{b},\beta(v_k,\theta^*))\} \le q(\bar{b},\beta(v_k,\tilde{\theta})) - q(\bar{b},\beta(v_k,\theta^*))$$

Summing over $k$ with weights $p_k$, and using the definition of $\zeta(b_1)$ and (SC) at $\bar{b}$,

$$\sum_{k=1}^{K} p_k \hat{q}(b_1,\beta(v_k,\tilde{\theta})) \le h(b_1,\theta^*) + \sum_{k=1}^{K} p_k q(\bar{b},\beta(v_k,\tilde{\theta})) - h(\bar{b},\theta^*) < h(b_1,\theta^*) + h(\bar{b},\tilde{\theta}) - \eta_2 - h(\bar{b},\theta^*) \le h(b_1,\tilde{\theta}) -$$

where the strict inequality is $(\text{CF-O})_{\eta_2}$ at $\bar{b}$, and the final inequality follows from fading sensitivity because

$$h(\bar{b},\tilde{\theta})-h(\bar{b},\theta^*)-h(b_1,\tilde{\theta})+h(b_1,\theta^*)=\int_{b_1}^{\bar{b}}\left[\frac{\partial h}{\partial b}(b,\tilde{\theta})-\frac{\partial h}{\partial b}(b,\theta^*)\right]db\leq 0$$

□

**Lemma 12** (Optimistic-Side Extension)**.** *Suppose $h(b,\theta)$ is decreasing in $\theta$. Let $q(b_1,b_2)$ satisfy (SC) and $(CE)_0$ for all $(b_1,b_2)$. If $(CF\text{-}O)_{\eta_2}$ holds at every $b_1\leq\beta(v_K,\theta^*)$ and every $\tilde{\theta}\in\tilde{\Theta}$, then (CF-O) holds at every $b_1\leq\beta(v_K,\theta^*)$ and every $\tilde{\theta}\in[\underline{\theta},\theta^*)$, with $(CF\text{-}O)_{\eta_2}$ at every such $b_1$ and every $\tilde{\theta}$ for which $\tilde{\Theta}\cap[\tilde{\theta},\theta^*)\neq\emptyset$.*

*Proof.* First observe that because $V$ is finite, the set:

$$\tilde{\Theta}=\{\theta<\theta^*|B(\theta^*)\cap B(\theta)\neq\emptyset\}$$

is a finite set. Therefore, define $\tilde{\Theta}=\{\tilde{\theta}_1,\tilde{\theta}_2,...,\tilde{\theta}_M\}$, and assume without loss of generality that $\theta^*>\tilde{\theta}_1>\tilde{\theta}_2>...>\tilde{\theta}_M$. Now consider some arbitrary $\tilde{\theta}<\theta^*$ with $\tilde{\Theta}\cap[\tilde{\theta},\theta^*)\neq\emptyset$ and define

$$\tilde{\theta}_m=\min\{\theta\in\tilde{\Theta}|\theta\geq\tilde{\theta}\}$$

Because (CE) with $\delta=0$ is satisfied, $\forall k=1,...,K$:

$$q(b_1,\beta(v_k,\tilde{\theta}))=q(b_1,\beta(v_k,\tilde{\theta}_m))$$

Therefore:

$$\sum_{k=1}^{K}p_kq(b_1,\beta(v_k,\tilde{\theta}))=\sum_{k=1}^{K}p_kq(b_1,\beta(v_k,\tilde{\theta}_m))<h(b_1,\tilde{\theta}_m)-\eta_2<h(b_1,\tilde{\theta})-\eta_2$$

where the last inequality follows by the monotonicity of $h(b,\theta)$.

Now consider $\tilde{\theta}<\theta^*$ with $\tilde{\Theta}\cap[\tilde{\theta},\theta^*)=\emptyset$. For every $k$, no element of $B(\theta^*)$ lies in $[\beta(v_k,\tilde{\theta}),\beta(v_k,\theta^*))$, since $\beta(v_k,\theta)$ is continuous and increasing in $\theta$, so any such element would equal $\beta(v_k,\theta)$ for some $\theta\in[\tilde{\theta},\theta^*)$, contradicting $\tilde{\Theta}\cap[\tilde{\theta},\theta^*)=\emptyset$. Therefore $(CE)_0$ gives $q(b_1,\beta(v_k,\tilde{\theta}))=q(b_1,\beta(v_k,\theta^*))$ for every $k$, and

$$\sum_{k=1}^{K}p_kq(b_1,\beta(v_k,\tilde{\theta}))=\sum_{k=1}^{K}p_kq(b_1,\beta(v_k,\theta^*))=h(b_1,\theta^*)<h(b_1,\tilde{\theta})$$

where the equality uses (SC) and the inequality the monotonicity of $h(b,\theta)$. □

**Lemma 13** (Boundary Shift)**.** *Suppose the model has bid monotonicity. Let $q(b_1,b_2)$ be a symmetric mechanism, non-increasing in $b_2$, satisfying, at every $b_1,b_2 \le \beta(v_K,\theta^*)$, (SC), $(F)_{\eta_1}$, (CF-O) for every $\tilde{\theta} \in [\underline{\theta},\theta^*)$, and $(CF\text{-}O)_{\eta_2}$ for every $\tilde{\theta} \in \tilde{\Theta}$, together with $(CE)_0$, and with $q(b_1,b_2) = \min\{\zeta(b_1), q(\bar{b}_{grid}(b_1),b_2)\}$ at every $b_1 \notin B(\theta^*)$ for some $\zeta(b_1) \in [0,1]$. Then for every sufficiently small $\varepsilon > 0$ there exist $\delta > 0$ and a mechanism $\hat{q}$ satisfying (F), (SC), (CF-O), and (CF-P) on $\mathscr{B}^2 \times [\underline{\theta},\theta^*+\varepsilon)$ together with $(CE)_\delta$, with $\hat{q}(b_1,b_2) = \min\{\zeta(b_1), \hat{q}(\bar{b}_{grid}(b_1),b_2)\}$ at every $b_1 \in \mathscr{B} \setminus B(\theta^*)$ lying outside $\bigcup_k (\beta(v_k,\theta^*), \beta(v_k,\theta^*)+\delta)$, and with $\hat{q}(b_1,b_2) = \frac{h(b_1,\theta^*)}{h(\beta(v_k,\theta^*),\theta^*)}\hat{q}(\beta(v_k,\theta^*),b_2)$ for every $b_1 \in (\beta(v_k,\theta^*), \beta(v_k,\theta^*)+\delta)$ and all $b_2$.*

*Proof.* Given mechanism $q$ satisfying (SC), (F) and (CF-O) with positive slack, (CE) with $\delta = 0$, and the threshold form of the hypothesis at every $b_1 \notin B(\theta^*)$. Define a $\delta-$shifted mechanism $\hat{q}_\delta$ as:

$$\hat{q}_\delta(b_1,b_2) = \begin{cases} q(b_1,b_2) & \text{if } b_2 > \underline{b}_{grid}(b_2)+\delta, b_1 \ge \underline{b}_{grid}(b_1)+\delta \\ \hat{q}_\delta(b_1, \max\{\beta(v_k,\theta^*) | \beta(v_k,\theta^*) < b_2\}) & \text{if } b_2 \le \underline{b}_{grid}(b_2)+\delta \\ \frac{h(b_1,\theta^*)}{h(\underline{b}_{grid}(b_1),\theta^*)} q(\underline{b}_{grid}(b_1),b_2) & \text{if } b_1 < \underline{b}_{grid}(b_1)+\delta \end{cases}$$

For $\delta < \min_{k=2,\dots,K} \beta(v_k,\theta^*) - \beta(v_{k-1},\theta^*)$, $\hat{q}_\delta$ is well defined, with the cases applied in the listed order. We prove that $\hat{q}_\delta$ satisfies all the listed conditions. First observe that by construction $\hat{q}$ satisfies the threshold form with the shifted opponent bins, (CE) with $\delta > 0$, as well as SC. Furthermore, for $\varepsilon$ sufficiently small:

$$\beta(v_k,\theta^*+\varepsilon) < \beta(v_k,\theta^*)+\delta \ \ \forall v_k \in V$$

and

$$Q(b|\theta) = Q(b|\theta^*) > h(b,\theta) \ \forall \theta \in (\theta^*, \theta^*+\varepsilon]$$

Therefore, (CF-P) is also trivially satisfied. We now focus on (CF-O) and (F).

**Feasibility**: Because $q$ satisfies feasibility everywhere with slack $\eta > 0$ and has (CE) with $\delta = 0$, $\hat{q}_\delta$ satisfies feasibility everywhere if and only if it satisfies feasibility for all $b_2$ such that $b_2 \le \underline{b}_{grid}(b_2)+\delta$. Now consider one such $b_2$:

$$\begin{aligned} \hat{q}_\delta(b_1,b_2) + \hat{q}_\delta(b_2,b_1) &= \hat{q}_\delta(b_1,\underline{b}_{grid}(b_2)) + \hat{q}_\delta(b_2,b_1) \\ &= q(b_1,\underline{b}_{grid}(b_2)) + \frac{h(b_2,\theta^*)}{h(\underline{b}_{grid}(b_2),\theta^*)} q(\underline{b}_{grid}(b_2),b_1) \\ &= q(b_1,\underline{b}_{grid}(b_2)) + q(\underline{b}_{grid}(b_2),b_1) + q(\underline{b}_{grid}(b_2),b_1)\frac{h(b_2,\theta^*) - h(\underline{b}_{grid}(b_2),\theta^*)}{h(\underline{b}_{grid}(b_2),\theta^*)} \\ &\le 1-\eta + \frac{h(b_2,\theta^*) - h(\underline{b}_{grid}(b_2),\theta^*)}{h(\underline{b}_{grid}(b_2),\theta^*)} \\ &\to 1-\eta \text{ as } \delta \to 0 \end{aligned}$$

Finally, because $\underline{b}_{grid}(b_2) \in B(\theta^*)$ can only take $K$ different values, the convergence is uniform across $b_2$, and there exists $\delta$ sufficiently small such that $\hat{q}_\delta$ satisfies feasibility.

**Optimistic-Side Corrective Feedback**: For (CF-O), let $\tilde{\Theta} = \{\tilde{\theta} < \theta^* | B(\tilde{\theta}) \cap B(\theta^*) \neq \emptyset\}$ and also define $\hat{\Theta} = \{\hat{\theta} < \theta^* | B(\hat{\theta}) \cap (B(\theta^*)+\delta) \neq \emptyset\}$. Observe that $\hat{\Theta}$ has the same dimension as

$\tilde{\Theta}$ and that each element of $\hat{\Theta}$ is an upward shift of $\tilde{\Theta}$.[18] i.e., let $\tilde{\theta}_n$ and $\hat{\theta}_n$ be the $n-th$ largest element of $\tilde{\Theta}$ and $\hat{\Theta}$ respectively, then there exists sequence of positive numbers $\xi_n$ s.t.:

$$\hat{\theta}_n = \tilde{\theta}_n + \xi_n$$

For every $b_1$, the shifted bin assignment of each $\beta(v_k, \theta)$ changes only at elements of $\hat{\Theta}$, and each bin is closed on the right, so $\hat{Q}_\delta(b_1|\theta)$ is constant on each interval of $[\underline{\theta}, \theta^*)$ between consecutive elements of $\hat{\Theta}$, taking its value at the interval's upper endpoint. Because $h(b_1, \theta)$ is decreasing in $\theta$, (CF-O) on $[\underline{\theta}, \theta^*)$ therefore holds if it holds at every $\theta \in \hat{\Theta}$ and at every $\theta$ with $\hat{\Theta} \cap [\theta, \theta^*) = \emptyset$. For $\theta$ with $\hat{\Theta} \cap [\theta, \theta^*) = \emptyset$, no element of $B(\theta^*) + \delta$ lies in $[\beta(v_k, \theta), \beta(v_k, \theta^*)]$ for any $k$, since $\beta(v_k, \theta)$ is continuous and increasing in $\theta$ and any such element would place a point of $\hat{\Theta}$ in $[\theta, \theta^*)$, so $\hat{q}_\delta(b_1, \beta(v_k, \theta)) = \hat{q}_\delta(b_1, \beta(v_k, \theta^*))$ for every $k$, and $\hat{Q}_\delta(b_1|\theta) = \hat{Q}_\delta(b_1|\theta^*) = h(b_1, \theta^*) < h(b_1, \theta)$ by (SC) and the monotonicity of $h(b, \theta)$.

Now consider a $b_1 \geq \underline{b}_{grid}(b_1) + \delta$ and any $\hat{\theta}_n \in \hat{\Theta}$. Then

$$\hat{Q}_\delta(b_1|\hat{\theta}_n) = Q(b_1|\tilde{\theta}_n) < h(b_1, \tilde{\theta}_n) - \eta_2 < h(b_1, \hat{\theta}_n) \text{ as } \delta \to 0$$

The equality holds because $\hat{q}_\delta(b_1, \beta(v_k, \hat{\theta}_n)) = q(b_1, \beta(v_k, \tilde{\theta}_n))$ for every $k$ once $\delta$ is sufficiently small, and the convergence is uniform because $h(b_1, \theta)$ and $\beta(v, \theta)$ are both continuous in $b_1$ and the set of bids and valuations are both compact. Therefore, there exists $\delta$ sufficiently small such that (CF-O) holds for all $b_1 \geq \underline{b}_{grid}(b_1) + \delta$.

Now consider arbitrary $b_1 < \underline{b}_{grid}(b_1) + \delta$. First observe that any $b_1 \in B(\theta^*)$ does not fall under this category. Furthermore, by the preceding argument and the finiteness of $B(\theta^*)$ and $\hat{\Theta}$, (CF-O) holds with a positive uniform slack $\hat{\eta}_2$ under $\hat{q}_\delta$ for all $b_1 \in B(\theta^*)$ and $\hat{\theta}_n \in \hat{\Theta}$. Therefore, for any $b_1 < \underline{b}_{grid}(b_1) + \delta$ and $\hat{\theta}_n \in \hat{\Theta}$:

$$\begin{aligned}
\hat{Q}_\delta(b_1|\hat{\theta}_n) &= \hat{Q}_\delta(\underline{b}_{grid}(b_1)|\hat{\theta}_n) + \hat{Q}_\delta(\underline{b}_{grid}(b_1)|\hat{\theta}_n) \frac{h(b_1, \theta^*) - h(\underline{b}_{grid}(b_1), \theta^*)}{h(\underline{b}_{grid}(b_1), \theta^*)} \\
&< \left(h(\underline{b}_{grid}(b_1), \hat{\theta}_n) - \hat{\eta}_2\right) \frac{h(b_1, \theta^*)}{h(\underline{b}_{grid}(b_1), \theta^*)} \\
&< h(b_1, \hat{\theta}_n) \text{ as } \delta \to 0
\end{aligned}$$

Where once again the convergence is uniform because of the continuity of $h(b, \theta)$, compactness of $\Theta$, and compactness of the set of bids. We therefore conclude that there exists $\delta$ sufficiently small such that (CF-O) holds for all $b_1$ under $\hat{q}_\delta$.

Taking $\delta$ to be the smaller of the one needed for (F) and (CF-O), while keeping $\delta$ above $\max_k \beta(v_k, \theta^* + \varepsilon) - \beta(v_k, \theta^*)$, a window that is nonempty for $\varepsilon$ sufficiently small, we conclude that for every sufficiently small $\varepsilon > 0$ there exists $\delta > 0$ such that $\hat{q}_\delta$ is well defined and satisfies (F), (SC), (CF-O), and (CF-P) on $\mathscr{B}^2 \times [\underline{\theta}, \theta^* + \varepsilon)$ together with $(CE)_\delta$, with the threshold form of the statement at every $b_1 \in \mathscr{B} \setminus B(\theta^*)$ lying outside $\bigcup_k (\beta(v_k, \theta^*), \beta(v_k, \theta^*) + \delta)$, and with the proportional scaling form on those intervals. □

[18] This takes each crossing to be simple, one bid pair per element of $\tilde{\Theta}$; simultaneous crossings of two pairs at a common $\tilde{\theta}$ require an exact coincidence among finitely many equations in the primitives and are excluded as knife-edge.

Let $q^{pre}$ denote a mechanism satisfying the hypothesis of Lemma 13, obtained from a grid solution with slacks $\eta_1, \eta_2 > 0$ as in Lemma 10, and let $\hat{q}_\delta$ denote the shifted mechanism its proof constructs from $q^{pre}$ at shift $\delta$. Write $Q(b \mid \tilde{\theta}) = \sum_{k=1}^{K} p_k \hat{q}_\delta(b, \beta(v_k, \tilde{\theta}))$ and $Q^{pre}(b \mid \tilde{\theta}) = \sum_{k=1}^{K} p_k q^{pre}(b, \beta(v_k, \tilde{\theta}))$. Let $\hat{\Theta} = \{\hat{\theta} < \theta^* \mid B(\hat{\theta}) \cap (B(\theta^*) + \delta) \neq \emptyset\} = \{\hat{\theta}_1 > \cdots > \hat{\theta}_M\}$ denote the shifted crossing set, and write $\xi_n = \xi_n(\delta) = \hat{\theta}_n - \tilde{\theta}_n$. Each $\xi_n$ is positive, increasing and continuous in $\delta$, and vanishes as $\delta \to 0$, since $\hat{\theta}_n$ solves $\beta(v_k, \hat{\theta}_n) = \beta(v_j, \theta^*) + \delta$ for its crossing pair and $\beta$ is strictly increasing and continuous in $\theta$. Let $\omega_h(\cdot)$ denote a modulus of continuity of $h$ in $\theta$, uniform over the compact bid range.

**Lemma 14** (Margin Delivery)**.** *Under the hypotheses of Lemma 10 with $h(b, \theta)$ strictly decreasing in $\theta$, there exists $\bar{\delta} > 0$ such that for every $\delta \in (0, \bar{\delta}]$ and every $\varepsilon > 0$ with $\max_k [\beta(v_k, \theta^* + \varepsilon) - \beta(v_k, \theta^*)] < \delta$, with $W = [\underline{\theta}, \theta^* + \varepsilon)$, the mechanism $\hat{q}_\delta$ satisfies parts (a) and (b) of Condition 1 for $W$, the defining gaps bounded below, at every compact $\Theta'' = [\underline{\theta}'', \bar{\theta}''] \subseteq W$ with $\underline{\theta}'' < \theta^* < \bar{\theta}''$, by margins*

$$\kappa_a(\bar{\theta}'') = \min_{b \in \overline{\beta(V,W)}} \left[ h(b, \theta^*) - h(b, \bar{\theta}'') \right]$$

*and*

$$\kappa_b(\underline{\theta}'') = \bar{\eta}_2 \ \text{ if } \ \underline{\theta}'' \leq \hat{\theta}_1, \qquad \kappa_b(\underline{\theta}'') = \min_{b \in \overline{\beta(V,W)}} \left[ h(b, \underline{\theta}'') - h(b, \theta^*) \right] \ \text{ if } \ \underline{\theta}'' \in (\hat{\theta}_1, \theta^*),$$

*where $\bar{\eta}_2 = \eta_2/2$. Each margin is positive and depends only on its own endpoint. For every $c > 0$, the infimum of $\kappa_a$ over $\bar{\theta}'' \in [\theta^* + c, \theta^* + \varepsilon)$ equals $\kappa_a(\theta^* + c)$, and the infimum of $\kappa_b$ over $\underline{\theta}'' \in [\underline{\theta}, \theta^* - c]$ is at least $\min\{\bar{\eta}_2, \min_{b \in \overline{\beta(V,W)}} [h(b, \theta^* - c) - h(b, \theta^*)]\}$.*

*Proof.* Assume $\bar{\delta} > 0$ is small enough that for every $\delta \in (0, \bar{\delta}]$ the conclusions of Lemma 13 hold for $\hat{q}_\delta$, together with the crossing-point identity established in its proof, $Q(b_1 \mid \hat{\theta}_n) = Q^{pre}(b_1 \mid \tilde{\theta}_n)$ for every $n$ and every $b_1 \geq \underline{b}_{grid}(b_1) + \delta$, and that additionally $\omega_h(\max_n \xi_n(\delta)) \leq \eta_2/4$, with $\xi_n(\delta) = \hat{\theta}_n - \tilde{\theta}_n$ the upward shift of the $n$th crossing point, and $C\delta \leq \eta_2/4$, where $C = L_h/\underline{h}_*$, $L_h$ bounds $\partial h/\partial b(\cdot, \theta^*)$ on $\mathscr{B}$, finite under (R1) and (R3) by the argument of Lemma 1 applied in the bid, and $\underline{h}_* = \min_{b \in \mathscr{B}} h(b, \theta^*) > 0$. Such $\bar{\delta}$ exists. The first requirement holds below the thresholds of that proof, which are uniform in $\delta$; the second since each $\xi_n(\delta)$ is increasing in $\delta$ and vanishes as $\delta \to 0$; the third directly. Fix $\delta \in (0, \bar{\delta}]$ and $\varepsilon$ as in the statement. Both displayed minima in the statement are over continuous functions on the compact set $\overline{\beta(V,W)}$, positive pointwise by strict monotonicity of $h$ in $\theta$, and hence positive by the extreme value theorem.

**Condition (a).** Fix $\bar{\theta}'' \in (\theta^*, \theta^* + \varepsilon)$. By the choice of $\varepsilon$, $\beta(v_k, \theta) < \beta(v_k, \theta^*) + \delta$ for every $k$ and $\theta \in (\theta^*, \theta^* + \varepsilon)$, so under $(CE)_\delta$ every opponent bid at belief $\bar{\theta}''$ falls in the same shifted bin as the corresponding bid at $\theta^*$, and $Q(b \mid \bar{\theta}'') = Q(b \mid \theta^*) = h(b, \theta^*)$ for all $b \in \overline{\beta(V,W)}$, the second equality by (SC), which the construction delivers at every $b \in \mathscr{B} \supseteq \overline{\beta(V,W)}$. Hence

$$Q(b \mid \bar{\theta}'') - h(b, \bar{\theta}'') = h(b, \theta^*) - h(b, \bar{\theta}'') \geq \kappa_a(\bar{\theta}'').$$

**Condition (b).** By the piecewise structure of Lemma 13, for each $b$ the map $\theta \mapsto Q(b \mid \theta)$ is constant on each interval into which $\hat{\Theta}$ cuts $[\underline{\theta}, \theta^*)$, each interval closed at its upper end and taking its value there. If $\underline{\theta}'' \in (\hat{\theta}_1, \theta^*)$, no element of $\hat{\Theta}$ lies in $[\underline{\theta}'', \theta^*)$, so $Q(b \mid \underline{\theta}'') = Q(b \mid \theta^*) = h(b, \theta^*)$ and the gap is $h(b, \underline{\theta}'') - h(b, \theta^*) \geq \kappa_b(\underline{\theta}'')$.

Suppose $\underline{\theta}'' \le \hat{\theta}_1$. We first bound the gap at every crossing point $\hat{\theta}_n$ and every bid, using that $q^{pre}$ carries $(\text{CF-O})_{\eta_2}$ at every $\tilde{\theta} \in \tilde{\Theta}$ and that the belief shift from $\tilde{\theta}_n$ to $\hat{\theta}_n$ costs at most a modulus of $h$. At bids $b_1 \ge \underline{b}_{grid}(b_1) + \delta$, the crossing-point identity gives $Q(b_1 \mid \hat{\theta}_n) = Q^{pre}(b_1 \mid \tilde{\theta}_n)$, and with $\hat{\theta}_n = \tilde{\theta}_n + \xi_n$,

$$Q(b_1 \mid \hat{\theta}_n) \le h(b_1, \tilde{\theta}_n) - \eta_2 \le h(b_1, \hat{\theta}_n) + \omega_h(\xi_n) - \eta_2,$$

so the gap at such bids exceeds $\eta_2 - \omega_h(\max_n \xi_n) \ge 3\eta_2/4$ by the choice of $\bar{\delta}$; grid bids fall under this case since $\delta$ lies below the grid spacing, as Lemma 13 requires. At bids $b_1 \in (\beta(v_k, \theta^*), \beta(v_k, \theta^*) + \delta)$, writing $b_g = \underline{b}_{grid}(b_1) = \beta(v_k, \theta^*)$ and $r = h(b_1, \theta^*)/h(b_g, \theta^*)$, the proportional scaling form gives $Q(b_1 \mid \hat{\theta}_n) = r\, Q(b_g \mid \hat{\theta}_n)$, and by bid monotonicity,

$$h(b_1, \hat{\theta}_n) - Q(b_1 \mid \hat{\theta}_n) \;\ge\; h(b_g, \hat{\theta}_n) - Q(b_g \mid \hat{\theta}_n) - (r-1)\, Q(b_g \mid \hat{\theta}_n) \;\ge\; \frac{3\eta_2}{4} - (r-1).$$

The excess $r - 1 = [h(b_1, \theta^*) - h(b_g, \theta^*)]/h(b_g, \theta^*)$ is at most $C\delta$, so the gap is at least $3\eta_2/4 - C\delta \ge \eta_2/2 = \bar{\eta}_2$ by the choice of $\bar{\delta}$. Finally, $\underline{\theta}''$ lies in a piece whose upper endpoint $\hat{\theta}_n$ belongs to $\hat{\Theta}$ and at which $Q(b \mid \cdot)$ takes its value on the piece, so $Q(b \mid \underline{\theta}'') = Q(b \mid \hat{\theta}_n)$; since $h$ is decreasing in $\theta$, $h(b, \underline{\theta}'') \ge h(b, \hat{\theta}_n)$, and the gap at $\underline{\theta}''$ is bounded below by the gap at $\hat{\theta}_n$, hence by $\bar{\eta}_2$.

The infimum bounds in the statement follow from the monotonicity of the $h$-gap expressions in their endpoints. □

**Remark 13.** *The h-gap expressions for $\kappa_a$ and $\kappa_b$ vanish as the corresponding endpoint approaches $\theta^*$, and Condition 1 permits this, its away-from-target infima constraining only endpoints at a fixed distance. The slack level $\bar{\eta}_2$ does not vanish.*

*Proof of Lemma 9.* ($\Leftarrow$) Let $\eta_2 > 0$ be such that $LP(0, \eta_2)$ is feasible and let $\{\hat{q}(\beta(v_j, \theta^*), \beta(v_k, \theta^*))\}_{j,k}$ be a solution, so that $(F')_0$, (MON'), (SC'), and $(\text{CF-O}')_{\eta_2}$ hold. Define a symmetric mechanism $q_0$ for $b_1 \in B(\theta^*)$ and $b_2 \le \beta(v_K, \theta^*)$ by

$$q_0(b_1, b_2) = \hat{q}\,(b_1, \min\{\beta(v_k, \theta^*) | \beta(v_k, \theta^*) \ge b_2\})$$

Then $q_0$ satisfies $(CE)_0$ by construction, is non-increasing in $b_2$ by (MON'), and satisfies $(F)_0$ and (SC) on $B(\theta^*) \times B(\theta^*)$ by $(F')_0$ and (SC'). For every $b_1 \in B(\theta^*)$ and $\tilde{\theta} \in \tilde{\Theta}$, grouping the valuations by the bin their bids fall into gives

$$\sum_{k=1}^{K} p_k q_0(b_1, \beta(v_k, \tilde{\theta})) = \sum_{k=1}^{K} \pi_k(\tilde{\theta})\, \hat{q}(b_1, \beta(v_k, \theta^*)) < h(b_1, \tilde{\theta}) - \eta_2$$

where the equality holds because $q_0(b_1, \beta(v_k, \tilde{\theta})) = \hat{q}(b_1, \beta(v_m, \theta^*))$ exactly when $\beta(v_k, \tilde{\theta}) \in B_m$, and the inequality is $(\text{CF-O}')_{\eta_2}$. Therefore $q_0$ satisfies $(\text{CF-O})_{\eta_2}$ on $B(\theta^*) \times B(\theta^*) \times \tilde{\Theta}$.

By Lemma 11 at $\eta_1 = 0$, using bid monotonicity and fading sensitivity, there exists a symmetric mechanism $q_2$, non-increasing in $b_2$, satisfying $(CE)_0$ and, at every $b_1, b_2 \le \beta(v_K, \theta^*)$, (SC), $(F)_0$, and $(\text{CF-O})_{\eta_2}$ for every $\tilde{\theta} \in \tilde{\Theta}$, with $q_2(b_1, b_2) = \min\{\zeta(b_1), q_2(\bar{b}_{grid}(b_1), b_2)\}$ at every $b_1 \notin B(\theta^*)$. By Lemma 12, using that $h(b, \theta)$ is decreasing in $\theta$, $q_2$ satisfies (CF-O) at every $b_1 \le \beta(v_K, \theta^*)$ and every $\tilde{\theta} \in [\underline{\theta}, \theta^*)$. Setting $\zeta(b_1) = 1$ at $b_1 \in B(\theta^*)$, the threshold form holds there as an identity, and $q_2$ satisfies (TS) at every $b_1 \le \beta(v_K, \theta^*)$.

($\Rightarrow$) Let $q$ be a symmetric mechanism, non-increasing in $b_2$, satisfying (F), (SC), and (CF-O) as in the statement, and define $\hat{q}(\beta(v_j,\theta^*),\beta(v_k,\theta^*)) = q(\beta(v_j,\theta^*),\beta(v_k,\theta^*))$ for all $j,k$. Feasibility of $q$ gives $\hat{q}(\beta(v_j,\theta^*),\beta(v_k,\theta^*)) + \hat{q}(\beta(v_k,\theta^*),\beta(v_j,\theta^*)) \leq 1$, and monotonicity in $b_2$ gives (MON'). Restricting (SC) to $b_1 \in B(\theta^*)$ gives (SC').

For (CF-O'), fix $j$ and $\tilde{\theta} \in \tilde{\Theta}$. Grouping the valuations by the bin their bids fall into, and using that $q$ is non-increasing in $b_2$ with each $\beta(v_k,\tilde{\theta})$ lying weakly below the grid bid of its bin,

$$\sum_{k=1}^{K} \pi_k(\tilde{\theta})\, \hat{q}(\beta(v_j,\theta^*),\beta(v_k,\theta^*)) \leq \sum_{k=1}^{K} p_k q(\beta(v_j,\theta^*),\beta(v_k,\tilde{\theta})) < h(\beta(v_j,\theta^*),\tilde{\theta})$$

where the last inequality is (CF-O) at $\tilde{\theta}$. The (CF-O') inequalities are strict and finitely many, since $|\tilde{\Theta}| \leq K^2$, so they hold with a common slack $\eta_2 > 0$, and $LP(0,\eta_2)$ is feasible. □

*Proof of Lemma 10.* Let grid values $\{\hat{q}(\beta(v_j,\theta^*),\beta(v_k,\theta^*))\}_{j,k}$ satisfy $(F')_{\eta_1}$, (MON'), (SC'), and $(\text{CF-O}')_{\eta_2}$ with $\eta_1,\eta_2 > 0$. The construction in the proof of Lemma 9, applied to this solution with $(F')_{\eta_1}$ in place of $(F')_0$, yields a symmetric mechanism $q_2$, non-increasing in $b_2$, satisfying $(CE)_0$ and, at every $b_1,b_2 \leq \beta(v_K,\theta^*)$, (SC), $(F)_{\eta_1}$, $(\text{CF-O})_{\eta_2}$ for every $\tilde{\theta} \in \tilde{\Theta}$, and (CF-O) for every $\tilde{\theta} \in [\underline{\theta},\theta^*)$, with the threshold form at every $b_1 \notin B(\theta^*)$. Then $q_2$ satisfies the hypothesis of Lemma 13, which yields, for every sufficiently small $\varepsilon > 0$, a $\delta > 0$ and a mechanism $\hat{q}_\delta$ satisfying (F), (SC), (CF-O), and (CF-P) on $\mathscr{B}^2 \times [\underline{\theta},\theta^* + \varepsilon)$ together with $(CE)_\delta$, with the threshold form at every $b_1 \in \mathscr{B} \setminus B(\theta^*)$ lying outside $\bigcup_k (\beta(v_k,\theta^*),\beta(v_k,\theta^*)+\delta)$, and with the proportional scaling form (PS) on those intervals.

Conditions (CF-O) and (CF-P) on $\mathscr{B} \supseteq \overline{\beta(V,W)}$ supply the pointwise content of parts (b) and (a) of Condition 1 for $W = [\underline{\theta},\theta^* + \varepsilon)$, and $\hat{q}_\delta$ is symmetric, non-increasing in $b_2$, and feasible by (F). Lemma 14 supplies the margins $\kappa_a$ and $\kappa_b$ together with their away-from-target infima. Therefore $\hat{q}_\delta$ satisfies Condition 1 for $W = [\underline{\theta},\theta^* + \varepsilon)$, and taking $\Theta' = [\underline{\theta},\theta^*]$ in the proposition, $\hat{q}_\delta$ sustains $\theta^*$ on $[\underline{\theta},\theta^*]$, with properties $(CE)_\delta$, (TS) outside the shifted intervals, and (PS) on them. □

*Proof of Proposition 4.* The values of a grid mechanism as in the statement satisfy (F′), (SC′), (MON′), and (CF-O′) at every pair of grid bids and every overlap belief, the inequalities strictly. These are finitely many strict inequalities, since $|\tilde{\Theta}| \leq K^2$, so they hold with common slacks $\eta_1,\eta_2 > 0$, and the values solve $LP(\eta_1,\eta_2)$. Lemma 10 then gives a mechanism that sustains $\theta^*$ on $[\underline{\theta},\theta^*]$. □

## 7.4 Applications and Discussion

### 7.4.1 Auction with an Unknown Fixed Reserve

Consider a buyer who believes he is in a first-price auction (FPA) with $N-1$ other bidders. He knows the distribution of the highest competing valuation is given by $F(\cdot)$, but does not know the reserve price $\theta$. He believes that all other $N-1$ bidders are aware of the reserve price and bid according to the symmetric Nash equilibrium with $N$ participants. The equilibrium bidding strategy with $N$ bidders and reserve price $\theta$ is

$$\beta(v,\theta) = v - \frac{1}{F(v)} \int_{\theta}^{v} F(x)\, dx$$

In the standard FPA, the buyer's win probability includes an indicator $\mathbb{I}\{b \geq \theta\}$ for clearing the reserve, which violates (R1) by placing $g$ on the boundary of $[0,1]$ for certain bids. To ensure $g \in (0,1)$ strictly, we introduce a mean-zero noise term $\varepsilon$ with CDF $\Phi_\varepsilon$ satisfying $\Phi_\varepsilon \in (0,1)$ everywhere, replacing the indicator with $\Phi_\varepsilon(b-\theta)$. The buyer's model becomes

$$g(b,v,\theta) = \Phi_\varepsilon(b-\theta) \cdot F(\beta^{-1}(b,\theta))$$

As the variance of $\varepsilon$ vanishes, the model reduces to the standard FPA with reserve price $\theta$. Since both $\Phi_\varepsilon(b-\theta)$ and $F(\beta^{-1}(b,\theta))$ are increasing in $b$ and decreasing in $\theta$, the direct monotonicities required by (R1) and (R2) are satisfied. [Conditions on $\Phi_\varepsilon$ and $F$ ensuring that $g/g_b$ satisfies (R3) are to be verified.]

### 7.4.2 Correlated Reserve Price

**Proposition 7** (Correlated Reserve Price)**.** *If $G(\cdot)$ is log-concave and $\psi(v,\theta)$ is increasing in $v,\theta$, then the correlated reserve price model $g(b,v,\theta) = G(b-\psi(v,\theta))$ satisfies conditions (R1)-(R3), and is adversely ordered.*

*Proof of Proposition 7.* As has already been demonstrated in the text, conditions (R1)-(R3) are satisfied under maintained assumptions. Because

$$\frac{\partial}{\partial\theta}h(b,\theta) = -G'(b-\psi(\beta^{-1}(b,\theta),\theta))\frac{\partial}{\partial\theta}\psi(\beta^{-1}(b,\theta),\theta)$$

which is negative if and only if $\frac{\partial}{\partial\theta}\psi(\beta^{-1}(b,\theta),\theta) > 0$. Note that this is trivially true for $b \notin [\beta(\underline{v},\theta),\beta(\bar{v},\theta)]$. For any $b \in [\beta(\underline{v},\theta),\beta(\bar{v},\theta)]$, because $\beta(v,\theta)$ is the optimal bidding behaviour under belief $\theta$, $\beta^{-1}(b,\theta)$ must satisfy the FOC:

$$\beta^{-1}(b,\theta) - b = \frac{G(b-\psi(\beta^{-1}(b,\theta),\theta))}{G'(b-\psi(\beta^{-1}(b,\theta),\theta))}$$

Taking the derivative with respect to $\theta$ on both sides and rearranging the terms, we obtain:

$$\frac{\partial}{\partial\theta}\beta^{-1}(b,\theta)\left[\frac{\partial}{\partial x}\frac{G(x)}{G'(x)}\Big|_{x=b-\psi(\beta^{-1}(b,\theta),\theta)}\right]^{-1} = -\frac{\partial}{\partial\theta}\psi(\beta^{-1}(b,\theta),\theta)$$

As the LHS is negative under (R3) and log-concavity of $G$, we obtain $\frac{\partial}{\partial\theta}\psi(\beta^{-1}(b,\theta),\theta) > 0$, and Proposition 2 applies. □

### 7.4.3 Taste Projection

The previous examples concern uncertainty about the mechanism. This one concerns uncertainty about the value distribution of the other buyer.

Consider a buyer who believes he is in an FPA with one other participant whose valuation $\tilde{v}$ is drawn according to some unknown $\hat{F}(\cdot)$, which he assumes his competitor knows and bids against at the symmetric equilibrium. He treats his own private value $v$ as informative about $\hat{F}(\cdot)$, with the relation governed by $\theta$, so that his posterior over opponent value distributions is $F(\cdot \mid v,\theta)$. The

dependence on $v$ captures taste projection, where a buyer with a high value believes others are likely to have high values as well. The buyer's model $g(b,v,\theta)$ is then determined by the equilibrium bidding strategy of the imagined FPA. Proposition 8 shows that the regularity conditions (R1)-(R3) and the conditions of Proposition 2 are satisfied whenever $F(\tilde{v} \mid v,\theta)$ and $\frac{F(\tilde{v}|v,\theta)}{f(\tilde{v}|v,\theta)}$ are both increasing in $\tilde{v}$ and decreasing in $v$ and $\theta$, and $F(v \mid v,\theta)$ is increasing in $v$. These conditions are readily satisfied, holding for instance for any location family $F(\tilde{v} \mid v,\theta) = F_0(\tilde{v} - m(v,\theta))$ with a log-concave $F_0$, such as the normal, and a shift $m$ increasing in $v$ and $\theta$ but rising less than one for one in $v$. The last requirement prevents a higher valuation from lowering the buyer's perceived chance of holding the highest value.

**Proposition 8** (Taste Projection)**.** *If $F(\tilde{v}|v,\theta)$ is twice continuously differentiable in $\tilde{v},v,\theta$, FOSD increasing in $v,\theta$, $\frac{F(\tilde{v}|v,\theta)}{f(\tilde{v}|v,\theta)}$ is increasing in $\tilde{v}$, decreasing in $(v,\theta)$, and $F(v|v,\theta)$ is increasing in $v$, then the taste projection model $g(b,v,\theta) = F(\hat{\beta}^{-1}_{\theta,v}(b)|v,\theta)$ satisfies conditions (R1)-(R3), and the model is adversely ordered.*

*Proof.* By standard results from FPA, an agent with valuation $v$ and degenerate belief $\theta$ believes his opponent is bidding with bidding strategy

$$\hat{\beta}_{\theta,v}(\tilde{v}) = \frac{\int_{\underline{v}}^{\tilde{v}} x f(x|v,\theta)\partial x}{F(\tilde{v}|v,\theta)}$$

Therefore, the agent's model is given by:

$$g(b,v,\theta) = F(\hat{\beta}^{-1}_{\theta,v}(b)|v,\theta)$$

The agent's optimal bidding strategy for any given $\theta$ is to follow the symmetric Nash equilibrium bidding strategy, i.e.,

$$\beta(v,\theta) = \hat{\beta}_{\theta,v}(v) = \frac{\int_{\underline{v}}^{v} x f(x|v,\theta)dx}{F(v|v,\theta)}$$

and $h(b,\theta)$ is given by:

$$h(b,\theta) = \begin{cases} F(\hat{\beta}^{-1}_{\theta,\underline{v}}(b)|\underline{v},\theta) & \text{if } b < \beta(\underline{v},\theta) \\ F(\beta^{-1}(b,\theta)|\beta^{-1}(b,\theta),\theta) & \text{if } b \in [\beta(\underline{v},\theta),\beta(\bar{v},\theta)] \\ F(\hat{\beta}^{-1}_{\theta,\bar{v}}(b)|\bar{v},\theta) & \text{if } b > \beta(\bar{v},\theta) \end{cases}$$

First note that this implies that the monotonicity of $h(b,\theta)$ in $\theta$ can be rewritten as:

$$\begin{aligned} h(b,\theta) &= g(\beta(\beta^{-1}(b,\theta),\theta),\beta^{-1}(b,\theta),\theta) \\ &= \mathbb{P}(\beta(\beta^{-1}(b,\theta),\theta) > \hat{\beta}_{\theta,\beta^{-1}(b,\theta)}(\tilde{v})|\beta^{-1}(b,\theta),\theta) \\ &= \mathbb{P}(\tilde{v} < \beta^{-1}(b,\theta)|\beta^{-1}(b,\theta),\theta) \\ &= F(\beta^{-1}(b,\theta)|\beta^{-1}(b,\theta),\theta) \text{ is decreasing in } \theta \ \forall b \end{aligned}$$

where the third equality follows from the fact that $\hat{\beta}_{\theta,v}(\tilde{v})$ is monotonically increasing in $\tilde{v}$. Because $F(x|v,\theta)$ is decreasing in $\theta$ and $F(v|v,\theta)$ is increasing in $v$, the monotonicity condition is satisfied.

To verify the regularity conditions, we first note that monotonicity conditions can be easily established by directly studying the properties of $\hat{\beta}_{\theta,v}(\tilde{v})$. Because $\frac{f(x|v,\theta)}{F(x|v,\theta)}$ is increasing in $\theta, v$, by properties of reverse hazard rate dominance, we know that $\forall(x,\tilde{v})$ s.t. $x<\tilde{v}$, $\frac{f(x|v,\theta)}{F(\tilde{v}|v,\theta)}$ is weakly increasing in $(\theta,v)$, and:

$$\hat{\beta}_{\theta,v}(\tilde{v}) = \frac{\int_{\underline{v}}^{\tilde{v}} xf(x|v,\theta)dx}{F(\tilde{v}|v,\theta)} = \int_{\underline{v}}^{\tilde{v}} \frac{xf(x|v,\theta)}{F(\tilde{v}|v,\theta)}dx$$

is increasing in $(v,\theta)$. Therefore, $\hat{\beta}^{-1}_{\theta,v}(b)$ is decreasing in $(v,\theta)$, and:

$$\begin{aligned} g(b,v,\theta) &= \mathbb{P}(\hat{\beta}_{\theta,v}(\tilde{v}) < b) \\ &= F(\hat{\beta}^{-1}_{\theta,v}(b)|v,\theta) \end{aligned}$$

is decreasing in $(v,\theta)$ and increasing in $b$. Finally, for conditions on $\frac{g_b}{g}$, we compute $\frac{g_b(b,v,\theta)}{g(b,v,\theta)}$ directly:

$$\begin{aligned} \frac{g_b(b,v,\theta)}{g(b,v,\theta)} &= \frac{f(\hat{\beta}^{-1}_{\theta,v}(b)|v,\theta)}{F(\hat{\beta}^{-1}_{\theta,v}(b)|v,\theta)} \frac{\partial \hat{\beta}^{-1}_{\theta,v}(b)}{\partial b} \\ &= \frac{F(\hat{\beta}^{-1}_{\theta,v}(b)|v,\theta)}{F(\hat{\beta}^{-1}_{\theta,v}(b)|v,\theta)\hat{\beta}^{-1}_{\theta,v}(b) - \int_{\underline{v}}^{\hat{\beta}^{-1}_{\theta,v}(b)} xf(x|v,\theta)dx} \\ &= \left(\hat{\beta}^{-1}_{\theta,v}(b) - \frac{\int_{\underline{v}}^{\hat{\beta}^{-1}_{\theta,v}(b)} xf(x|v,\theta)dx}{F(\hat{\beta}^{-1}_{\theta,v}(b)|v,\theta)}\right)^{-1} \end{aligned}$$

Therefore, the relevant conditions are satisfied if

$$\hat{\beta}^{-1}_{\theta,v}(b) - \frac{\int_{\underline{v}}^{\hat{\beta}^{-1}_{\theta,v}(b)} xf(x|v,\theta)dx}{F(\hat{\beta}^{-1}_{\theta,v}(b)|v,\theta)}$$

is increasing in $b$ and decreasing in $v$ and $\theta$. Now, letting $x = \hat{\beta}^{-1}_{\theta,v}(b)$, consider the function:

$$J(x) = x - \frac{\int_{\underline{v}}^{x} yf(y|v,\theta)dy}{F(x|v,\theta)}$$

Because $\frac{f(x|v,\theta)}{F(x|v,\theta)}$ is decreasing in $x$, $J(x)$ is always increasing in $x$. Then, because $\hat{\beta}^{-1}_{\theta,v}(b)$ is increasing in $b$, we obtain that $\frac{g}{g_b}$ is increasing in $b$. Furthermore, because $\hat{\beta}^{-1}_{\theta,v}(b)$ is decreasing in $\theta$ and

$$\frac{\int_{\underline{v}}^{\tilde{v}} xf(x|v,\theta)dx}{F(\tilde{v}|v,\theta)}$$

is increasing in $\theta$ and $v$ for any fixed $\tilde{v}$ by the reverse hazard rate ordering assumption, we also obtain that $\frac{g}{g_b}$ is decreasing in $\theta$ and $v$. □

### 7.4.4 Uncertain Bidder Count

A related model, which reduces to the same form, arises when the buyer believes he faces $N$ competing bidders with i.i.d. values drawn from $F(\cdot)$ but is uncertain about $N$, taking $N$ to be Poisson with rate $\lambda(v,\theta)$ increasing in both arguments. As only the perceived distribution of the highest competing valuation enters his win probability, a Poisson number of opponents drawn from $F$ is equivalent to a single opponent drawn from

$$F(\tilde{v} \mid v,\theta) = e^{-\lambda(v,\theta)(1-F(\tilde{v}))}$$

and uncertainty about the number of competitors plays the role that uncertainty about their values plays above. Proposition 9 shows that the associated $g(b,v,\theta)$ satisfies (R1)–(R3) whenever $F$ is concave and $\lambda(v,\theta)$ is increasing in $v$ and $\theta$.

To verify that the uncertain bidder count example satisfies conditions (R1)-(R3) and has an $h(b,\theta)$ that is decreasing in $\theta$, we first prove the following lemma.

**Lemma 15** (Separable Models)**.** *Suppose regularity assumptions are satisfied, then $h(b,\theta)$ is decreasing in $\theta$ if $g(b,v,\theta) = \tilde{g}(b,s(v,\theta))$ for some function $\tilde{g}(\cdot,\cdot)$ and $s(\cdot,\cdot)$.*

*Proof.* Suppose $g(b,v,\theta) = \tilde{g}(b,s(v,\theta))$ for some $\tilde{g}(\cdot,\cdot)$ and $s(\cdot,\cdot)$ and consider the function

$$\tilde{h}(b,\theta) = \tilde{g}(b,s(\beta^{-1}(b,\theta),\theta))$$

First note that because optimal bidding behaviour under any degenerate belief $\theta$ is characterised by:

$$v - \beta(v,\theta) = \frac{g(\beta(v,\theta),v,\theta)}{g_b(\beta(v,\theta),v,\theta)}$$

Conditions (R1)-(R3) mean that the RHS is strictly decreasing in $\theta$, weakly decreasing in $v$, and strictly increasing in $b$ while the LHS is decreasing in $b$, increasing in $v$, and constant in $\theta$. Therefore, an increase in $\theta$ leads to an increase in the optimal bid, and we obtain that $\beta(v,\theta)$ is increasing in $(\theta,v)$. Additionally, substituting in $g(b,v,\theta) = \tilde{g}(b,s(v,\theta))$ and taking derivative with respect to $\theta$ on both sides of the FOC condition, we obtain:

$$\psi_s(b,s(\beta^{-1}(b,\theta),\theta))\frac{\partial}{\partial\theta}s(\beta^{-1}(b,\theta),\theta) = \frac{\partial}{\partial\theta}\beta^{-1}(b,\theta) < 0$$

where $\psi(b,s(v,\theta)) = \frac{\tilde{g}(b,s(v,\theta))}{\tilde{g}_b(b,s(v,\theta))}$ Therefore:

$$\begin{aligned}\frac{\partial\tilde{h}(b,\theta)}{\partial\theta} &= \tilde{g}_s(b,s(\beta^{-1}(b,\theta),\theta))\frac{\partial s(\beta^{-1}(b,\theta),\theta)}{\partial\theta} \\ &= \frac{\tilde{g}_s(b,s(\beta^{-1}(b,\theta),\theta))}{\psi_s(b,s(\beta^{-1}(b,\theta),\theta))}\frac{\partial}{\partial\theta}\beta^{-1}(b,\theta)\end{aligned}$$

Because $\dfrac{\partial}{\partial\theta}\beta^{-1}(b,\theta) < 0$, we only need to show that $\dfrac{\tilde{g}_s(b,s(\beta^{-1}(b,\theta),\theta))}{\psi_s(b,s(\beta^{-1}(b,\theta),\theta))}$ is positive. Note that by conditions (R2) and (R3), we have:

$$\begin{aligned}\frac{\partial}{\partial\theta}\psi(b,s(v,\theta)) &= \psi_s(b,s(v,\theta))s_\theta(v,\theta) < 0 \\ \frac{\partial}{\partial\theta}g(b,s(v,\theta)) &= g_s(b,s(v,\theta))s_\theta(v,\theta) < 0\end{aligned}$$

Therefore:

$$\frac{\tilde{g}_s(b,s(\beta^{-1}(b,\theta),\theta))}{\psi_s(b,s(\beta^{-1}(b,\theta),\theta))} = \frac{\tilde{g}_s(b,s(\beta^{-1}(b,\theta),\theta))s_\theta(\beta^{-1}(b,\theta),\theta)}{\psi_s(b,s(\beta^{-1}(b,\theta),\theta))s_\theta(\beta^{-1}(b,\theta),\theta)} > 0$$

which means that

$$\tilde{h}(b,\theta) = \tilde{g}(b,s(\beta_\theta^{-1}(b),\theta)) = g(b,\beta_\theta^{-1}(b),\theta) = h(b,\theta)$$

is decreasing in $\theta$. □

**Proposition 9** (Uncertain Bidder Count)**.** *If $F(v)$ and $\lambda(v,\theta)$ are twice continuously differentiable, $F$ is concave, and $\lambda(v,\theta)$ is increasing in $v,\theta$, then the uncertain bidder count model $g$ satisfies conditions (R1)-(R3), and the associated $h(b,\theta)$ is strictly decreasing in $\theta$.*

*Proof.* The uncertain bidder count model can be treated as a special case of a taste projection model where the bidder believes he's facing a single opponent with a perceived value distribution of:

$$F(\tilde{v}|v,\theta) = \sum_{N=0}^{\infty} \frac{\lambda(v,\theta)^N e^{-\lambda(v,\theta)}}{N!} F(\tilde{v})^N = e^{-\lambda(v,\theta)} \sum_{N=0}^{\infty} \frac{(\lambda(v,\theta)F(\tilde{v}))^N}{N!} = e^{-\lambda(v,\theta)(1-F(\tilde{v}))}$$

Direct verification shows that $F(\tilde{v}|v,\theta)$ is decreasing in $v,\theta$. Furthermore, because $\lambda(v,\theta)$ is increasing in $(v,\theta)$ and $F$ is concave,

$$\frac{F(\tilde{v}|v,\theta)}{f(\tilde{v}|v,\theta)} = \frac{1}{\lambda(v,\theta)f(\tilde{v})}$$

is increasing in $\tilde{v}$ and decreasing in $v,\theta$. Therefore, by the proof of Proposition 8, we obtain that the associated $g(b,v,\theta)$ satisfies conditions (R1)-(R3). Finally, although $F(v|v,\theta)$ is generally not monotonically increasing in $v$, because $(v,\theta)$ only appear in $\lambda(v,\theta)$, Lemma 15 applies and $h(b,\theta)$ is decreasing in $\theta$. Therefore, Proposition 2 applies. □

### 7.4.5 Proof of Proposition 5

*Proof of Proposition 5.* Under value-independence, $g(b,v,\theta) = \tilde{g}(b,\theta)$ for all $v$. Since $h(b,\theta) = g(b,\beta^{-1}(b,\theta),\theta)$ and $g$ does not depend on $v$, we have $h(b,\theta) = \tilde{g}(b,\theta)$ for all $(b,\theta)$. If the mechanism is self-confirming with respect to $\theta^*$, then

$$Q(b \mid \theta^*) = h(b,\theta^*) = \tilde{g}(b,\theta^*)$$

for all $b \in [\beta(\underline{v},\theta^*),\beta(\bar{v},\theta^*)]$, where $Q(b \mid \theta^*) = \int q(b,\beta(v_2,\theta^*))f(v_2)\,dv_2$ is the objective interim win probability when the opponent holds belief $\theta^*$. Each buyer with value $v$ and belief $\theta^*$ optimizes against his subjective model, so

$$\beta(v,\theta^*) \in \arg\max_b\ (v-b)\,\tilde{g}(b,\theta^*) = \arg\max_b\ (v-b)\,Q(b \mid \theta^*)$$

for all $v \in V$. Define the reduced-form allocation $\hat{Q}(v) = Q(\beta(v,\theta^*) \mid \theta^*)$. The optimality condition states that each buyer, knowing $\hat{Q}$ exactly, finds it optimal to bid $\beta(v,\theta^*)$ when his value is $v$. This is the Bayesian incentive compatibility condition for the reduced-form allocation $\hat{Q}$. Since $\hat{Q}$ arises from a feasible mechanism satisfying $q(b_1,b_2)+q(b_2,b_1) \le 1$, the revenue equivalence theorem Myerson (1981) applies: the seller's expected revenue under any incentive-compatible allocation rule is pinned down by the allocation rule and the lowest type's surplus, and cannot exceed the revenue from the optimal auction. □

### 7.4.6 Proof of Proposition 6

The proof proceeds through a sequence of lemmas. Throughout, fix $g(b,v,\theta)$ satisfying (R1)-(R3) with $h(b,\theta)$ strictly decreasing in $\theta$, a target $\theta^* \in \text{int}(\Theta)$, and the self-confirming allocation rule $q_{\theta^*}(b) = h(b,\theta^*)$. Let $\tilde{g}(b,v,\theta)$ be a continuous perturbed model with $\|\tilde{g} - g\|_\infty \leq \varepsilon$, let $\tilde{\beta}(v,\mu)$ denote the optimal bidding correspondence under $\tilde{g}$, and let $\tilde{\lambda}_\mu$ denote the joint distribution over bids and valuations induced by a selection $\tilde{b}_\mu(v) \in \tilde{\beta}(v,\mu)$. Throughout this proof, we focus on contraction from above ($\theta > \theta^*$). The other half of the argument follows analogously. Every threshold on $\varepsilon$ below is computed from $g$ and its regularity constants before $\tilde{g}$ is fixed.

**Remark 14.** *By (R1) there is $c > 0$ with $c \leq g \leq 1-c$ on $[0,\bar{v}] \times V \times \Theta$. For $\varepsilon \leq c/2$, $\tilde{g}$ takes values in $[c/2, 1-c/2]$ on the same domain, so likelihood ratios under $\tilde{g}$ are bounded by $L = \log \frac{1-c/2}{c/2}$ and Remark 2 applies to the perturbed environment. Because $\tilde{g}$ is continuous on a compact domain it is uniformly continuous, and* log *is Lipschitz with constant $2/c$ on $[c/2, 1-c/2]$, so likelihoods under $\tilde{g}$ are equicontinuous on $\Delta\Theta$ in the sense of Definition 11. The modulus depends on $\tilde{g}$ and enters only after $\tilde{g}$ is fixed, in the application of the framework results; no threshold on $\varepsilon$ is computed from it.*

**Lemma 16** (Containment)**.** *(i) For every $v \in V$ and every $\mu \in \Delta\Theta$ supported on a compact interval $[\underline{\theta}'', \bar{\theta}''] \subseteq \Theta$, every selection $b_\mu(v) \in \beta(v,\mu)$ satisfies $b_\mu(v) \in [\beta(v,\underline{\theta}''), \beta(v,\bar{\theta}'')]$. (ii) Let $\Theta'' = [\underline{\theta}'', \bar{\theta}''] \subseteq \Theta$ be compact with $\bar{\theta}'' + \varepsilon' \leq \bar{\theta}$ for some $\varepsilon' > 0$. There exist $\bar{\varepsilon} > 0$ and a weakly open $M \supseteq \Delta\Theta''$ such that $\|\tilde{g} - g\|_\infty \leq \bar{\varepsilon}$ implies $\tilde{b}_\mu(v) \leq \beta(v, \bar{\theta}'' + \varepsilon')$ for every $\mu \in M$, every selection $\tilde{b}_\mu(v) \in \tilde{\beta}(v,\mu)$, and every $v \in V$. The symmetric statement bounds selections below by $\beta(v, \underline{\theta}'' - \varepsilon')$ when $\underline{\theta}'' - \varepsilon' \geq \underline{\theta}$.*

*Proof.* (i) The first-order condition equates $v - b$ with $\int g\, d\mu \big/ \int g_b\, d\mu$, a ratio lying between the extremes of $g/g_b$ over the support since $g_b > 0$, and $g/g_b$ is decreasing in $\theta$ with the root of the decreasing-in-$b$ condition monotone in its level, so every selection lies in the stated band, as in Step 1 of the proof of Proposition 1.

(ii) Bids above the value are dominated, so the feasible bid set may be taken to be $[0,\bar{v}]$. The map $(\tilde{g}, b, v, \mu) \mapsto (v-b)\int \tilde{g}(b,v,\theta)\, d\mu(\theta)$ is jointly continuous on $\mathscr{G} \times [0,\bar{v}] \times V \times \Delta\Theta$, with $\mathscr{G}$ under the sup norm and $\Delta\Theta$ under weak convergence, so by Berge's Maximum Theorem the correspondence $(\tilde{g}, v, \mu) \mapsto \tilde{\beta}(v,\mu)$ is upper hemicontinuous, and by compactness of $V$ so is the graph correspondence $(\tilde{g}, \mu) \mapsto \{(v,b) \mid v \in V,\ b \in \tilde{\beta}(v,\mu)\}$, with compact values. By (i), every $g$-selection at $\mu \in \Delta\Theta''$ satisfies $b_\mu(v) \leq \beta(v, \bar{\theta}'')$, and by strict monotonicity and continuity of $\beta$ in $\theta$ the set $\{(v,b) \mid b \leq \beta(v,\bar{\theta}'')\}$ lies in the open tube $T = \{(v,b) \mid b < \beta(v, \bar{\theta}'' + \varepsilon')\}$ with a gap uniform over the compact $V$. The upper preimage of $T$ under the graph correspondence is open in $\mathscr{G} \times \Delta\Theta$ and contains the compact set $\{g\} \times \Delta\Theta''$, so it contains $B_{\bar{\varepsilon}}(g) \times M$ for some $\bar{\varepsilon} > 0$ and weakly open $M \supseteq \Delta\Theta''$. The symmetric statement follows from the symmetric tube. □

For a joint distribution $\lambda$ over bids and valuations, define

$$K_g(\theta \mid \lambda) = \int \left[ q_{\theta^*}(b) \log \frac{q_{\theta^*}(b)}{g(b,v,\theta)} + (1 - q_{\theta^*}(b)) \log \frac{1 - q_{\theta^*}(b)}{1 - g(b,v,\theta)} \right] d\lambda(b,v)$$

with $K_{\tilde{g}}$ defined by replacing the model $g$ with $\tilde{g}$, and write $K_g(\theta_1, \theta_2 \mid \lambda) = K_g(\theta_2 \mid \lambda) - K_g(\theta_1 \mid \lambda)$ for the divergence gap. The valuation marginal and the entropy of the outcome cancel from

likelihood ratios, so $\theta_1$ KL-dominates $\theta_2$ with margin $\kappa$ on $M$ in the perturbed environment, in the sense of Definition 9, exactly when $K_{\tilde{g}}(\theta_1, \theta_2 \mid \tilde{\lambda}_\mu) \geq \kappa$ for every $\mu \in M$ and every selection.

For $\delta > 0$ with $\theta^* + \delta < \bar{\theta}$, let

$$\bar{\kappa}_a(\delta) = \min_{b \in [0, \bar{v}]} [h(b, \theta^*) - h(b, \theta^* + \delta)]$$

positive because $h$ is continuous and strictly decreasing in $\theta$ on a compact domain, and since $h$ is decreasing in $\theta$, $h(b, \theta^*) - h(b, \theta) \geq \bar{\kappa}_a(\delta)$ for every $\theta \geq \theta^* + \delta$ and every bid. Let $\omega_g$ denote a modulus of continuity of $g$ in $\theta$, uniform over $(b, v)$ on the compact domain, let $c_g > 0$ bound $-g_\theta$ below there, and fix $\varepsilon_1^a(\delta) > 0$ with $\omega_g(\varepsilon_1^a(\delta)) \leq \bar{\kappa}_a(\delta)/2$ and $\varepsilon_1^a(\delta) \leq \delta/2$. The lower-side constants $\bar{\kappa}_b(\delta)$ and $\varepsilon_1^b(\delta)$ are defined symmetrically.

**Lemma 17** (Drift Bounds)**.** *Let $\delta > 0$ with $\theta^* + \delta < \bar{\theta}$, and let $\Theta'' = [\underline{\theta}'', \bar{\theta}''] \subseteq \Theta$ be compact with $\bar{\theta}'' \in [\theta^* + \delta,\ \bar{\theta} - \varepsilon']$, where $\varepsilon' = \varepsilon_1^a(\delta)/4$. Write $\bar{\theta}''' = \bar{\theta}'' + \varepsilon'$ and $d = \bar{\theta}''' - \varepsilon_1^a(\delta)$. There exist $\bar{\varepsilon} > 0$ and a weakly open $M \supseteq \Delta\Theta''$ such that $\|\tilde{g} - g\|_\infty \leq \bar{\varepsilon}$ implies, for every $\mu \in M$ and every selection $\tilde{b}_\mu$,*

$$\frac{\partial}{\partial \theta} K_g(\theta_1 \mid \tilde{\lambda}_\mu) \geq c_g \bar{\kappa}_a(\delta) \quad \forall\, \theta_1 \geq d, \qquad \textit{hence} \qquad K_g(d, \theta_1 \mid \tilde{\lambda}_\mu) \geq (\theta_1 - d)\, c_g \bar{\kappa}_a(\delta) \quad \forall\, \theta_1 \in (d, \bar{\theta}].$$

*Proof.* Apply Lemma 16(ii) at $\Theta''$ with enlargement $\varepsilon'$ to obtain $\bar{\varepsilon}$ and $M$. For $\mu \in M$ and any selection, every realized $(\tilde{b}, v)$ satisfies $\tilde{b} \leq \beta(v, \bar{\theta}''')$, so $v \geq \beta^{-1}(\tilde{b}, \bar{\theta}''')$ and, for every $\theta_1 \geq \bar{\theta}'''$, $g(\tilde{b}, v, \theta_1) \leq g(\tilde{b}, \beta^{-1}(\tilde{b}, \bar{\theta}'''), \bar{\theta}''') = h(\tilde{b}, \bar{\theta}''')$, as in Step 2 of the proof of Proposition 1. For $\theta_1 \in [d, \bar{\theta}''')$ the modulus gives $g(\tilde{b}, v, \theta_1) \leq h(\tilde{b}, \bar{\theta}''') + \omega_g(\varepsilon_1^a(\delta))$. Since $\bar{\theta}''' \geq \theta^* + \delta$, at every realized $(\tilde{b}, v)$ and every $\theta_1 \geq d$,

$$q_{\theta^*}(\tilde{b}) - g(\tilde{b}, v, \theta_1) \geq \big[h(\tilde{b}, \theta^*) - h(\tilde{b}, \bar{\theta}''')\big] - \omega_g(\varepsilon_1^a(\delta)) \geq \bar{\kappa}_a(\delta) - \frac{\bar{\kappa}_a(\delta)}{2} = \frac{\bar{\kappa}_a(\delta)}{2}.$$

Differentiating $K_g$ under the integral,

$$\frac{\partial}{\partial \theta} K_g(\theta_1 \mid \tilde{\lambda}_\mu) = \int \frac{-g_\theta(b, v, \theta_1)}{g(b, v, \theta_1)\,(1 - g(b, v, \theta_1))} \,[q_{\theta^*}(b) - g(b, v, \theta_1)]\, d\tilde{\lambda}_\mu(b, v) \geq 2c_g \cdot \frac{\bar{\kappa}_a(\delta)}{2} = c_g \bar{\kappa}_a(\delta)$$

using $-g_\theta \geq c_g$ and $g(1 - g) \leq 1/4$. Integrating the slope from $d$ to $\theta_1$ gives the gap bound. □

**Lemma 18** (KL Transfer)**.** *Suppose $\varepsilon \leq c/2$. For every joint distribution $\lambda$ over $[0, \bar{v}] \times V$ and all $\theta_1, \theta_2 \in \Theta$,*

$$\big|K_{\tilde{g}}(\theta_1, \theta_2 \mid \lambda) - K_g(\theta_1, \theta_2 \mid \lambda)\big| \leq C\varepsilon, \qquad C = \frac{8}{c}.$$

*Proof.* Both $g$ and $\tilde{g}$ take values in $[c/2, 1 - c/2]$ on the domain, where log is Lipschitz with constant $2/c$, so $|\log \tilde{g} - \log g| \leq (2/c)\varepsilon$ and $|\log(1 - \tilde{g}) - \log(1 - g)| \leq (2/c)\varepsilon$ pointwise. The gap is the integral of

$$q_{\theta^*}(b) \log \frac{g(b, v, \theta_1)}{g(b, v, \theta_2)} + (1 - q_{\theta^*}(b)) \log \frac{1 - g(b, v, \theta_1)}{1 - g(b, v, \theta_2)}$$

against $\lambda$, so replacing $g$ by $\tilde{g}$ moves the integrand by at most four terms of $(2/c)\varepsilon$ each. □

**Lemma 19** (Approximation Scheme)**.** *For any $\delta > 0$ there exists $\bar{\varepsilon} > 0$, computed from $g$, its regularity constants, and $\delta$, such that $\|\tilde{g} - g\|_\infty \le \bar{\varepsilon}$ implies the following. There is a finite nested sequence of compact intervals $\Theta = R^0 \supset R^1 \supset \cdots \supset R^K = [\theta^* - \delta, \theta^* + \delta] \cap \Theta$ and a margin $\kappa_w(\delta) > 0$ such that for every $k < K$, every $\theta \in R^k \setminus R^{k+1}$ is KL-dominated in the perturbed environment, with margin $\kappa_w(\delta)$, by a point of $R^{k+1}$, on a weakly open $M^k \supseteq \Delta R^k$.*

*Proof.* If $\theta^* + \delta \ge \bar{\theta}$ the upper side contributes no stages; suppose $\theta^* + \delta < \bar{\theta}$ and treat the upper side, the lower side being symmetric with its own constants. Set $u_0 = \bar{\theta}$ and $u_{k+1} = \max\{u_k - \varepsilon_1^a(\delta)/2,\ \theta^* + \delta\}$, so that $u_k$ reaches $\theta^* + \delta$ after finitely many stages, and let $R^k$ have upper endpoint $u_k$ and lower endpoint given by the symmetric lower recursion. The stage sets and every constant below are determined by $g$ and $\delta$ alone.

At any stage with $u_k \in (\theta^* + \delta, \bar{\theta} - \varepsilon']$, Lemma 17 applied at $\Theta'' = R^k$ yields $\bar{\varepsilon}_k > 0$ and weakly open $M_a^k \supseteq \Delta R^k$ with, for every $\mu \in M_a^k$ and every selection, $K_g(d_k, \theta_1 \mid \tilde{\lambda}_\mu) \ge (\theta_1 - d_k)\, c_g \bar{\kappa}_a(\delta)$ for all $\theta_1 > d_k$, where $d_k = u_k - 3\varepsilon_1^a(\delta)/4$. Every $\theta_1 \in (u_{k+1}, u_k]$ satisfies $\theta_1 - d_k \ge u_{k+1} - d_k \ge \varepsilon_1^a(\delta)/4$, and $d_k$ lies in $R^{k+1}$, since $d_k < u_{k+1}$ and $d_k \ge \theta^* + \delta - 3\varepsilon_1^a(\delta)/4 > \theta^*$. The gap under $g$ is therefore at least $(\varepsilon_1^a(\delta)/4)\, c_g \bar{\kappa}_a(\delta)$.

At the stage $u_0 = \bar{\theta}$ no enlargement exists and the containment carries a vanishing allowance instead. By Lemma 16(i), every $g$-selection at every $\mu \in \Delta\Theta$ satisfies $b_\mu(v) \le \beta(v, \bar{\theta})$, so for any $\eta > 0$ the upper preimage of the open tube $\{(v, b) \mid b < \beta(v, \bar{\theta}) + \eta\}$ under the graph correspondence of the proof of Lemma 16(ii) contains $B_{\bar{\varepsilon}_0(\eta)}(g) \times \Delta\Theta$, and every realized $\tilde{b}$ satisfies $\tilde{b} \le \beta(v, \bar{\theta}) + \eta$, with $M^0 = \Delta\Theta$. Let $\omega_g^b$ and $\omega_h^b$ be moduli of continuity in the bid of $g$ and of $h(\cdot, \bar{\theta})$, and fix $\bar{\eta}(\delta) > 0$ with $\omega_g^b(\bar{\eta}(\delta)) + \omega_h^b(\bar{\eta}(\delta)) \le \bar{\kappa}_a(\delta)/4$. Writing $\hat{b} = \min\{\tilde{b}, \beta(v, \bar{\theta})\}$, for every $\theta_1 \ge \bar{\theta} - \varepsilon_1^a(\delta)$,

$$\begin{aligned} g(\tilde{b}, v, \theta_1) \le g(\hat{b}, v, \theta_1) + \omega_g^b(\bar{\eta}(\delta)) &\le h(\hat{b}, \bar{\theta}) + \omega_g(\varepsilon_1^a(\delta)) + \omega_g^b(\bar{\eta}(\delta)) \\ &\le h(\tilde{b}, \bar{\theta}) + \omega_g(\varepsilon_1^a(\delta)) + \omega_g^b(\bar{\eta}(\delta)) + \omega_h^b(\bar{\eta}(\delta)), \end{aligned}$$

so $q_{\theta^*}(\tilde{b}) - g(\tilde{b}, v, \theta_1) \ge \bar{\kappa}_a(\delta) - \bar{\kappa}_a(\delta)/2 - \bar{\kappa}_a(\delta)/4 = \bar{\kappa}_a(\delta)/4$, and the argument of Lemma 17 delivers the slope $c_g \bar{\kappa}_a(\delta)/2$ and, for $\theta_1 \in (u_1, \bar{\theta}]$ with dominator $d_0 = \bar{\theta} - 3\varepsilon_1^a(\delta)/4 \in R^1$, a gap under $g$ of at least $(\varepsilon_1^a(\delta)/4)\, c_g \bar{\kappa}_a(\delta)/2$.

Let $2\kappa_w(\delta)$ be the smaller of $(\varepsilon_1^a(\delta)/4)\, c_g \bar{\kappa}_a(\delta)/2$ and its lower-side counterpart, and let $M^k$ be the intersection of the two sides' neighbourhoods at stage $k$. Every gap established above is then at least $2\kappa_w(\delta)$ under $g$. Whenever $\|\tilde{g} - g\|_\infty \le \min\{c/2,\ \kappa_w(\delta)/C\}$, Lemma 18 applies with loss at most $\kappa_w(\delta)$, so each gap survives the passage to $K_{\tilde{g}}$ with at least $\kappa_w(\delta)$ remaining. Hence, for every $\mu \in M^k$ and every selection, every point of $R^k \setminus R^{k+1}$ is KL-dominated in the perturbed environment, with margin $\kappa_w(\delta)$, by the stage dominator in $R^{k+1}$ ($d_k$ on the upper side). Taking $\bar{\varepsilon}$ to be the minimum of $c/2$, $\kappa_w(\delta)/C$, $\bar{\varepsilon}_0(\bar{\eta}(\delta))$, and the finitely many $\bar{\varepsilon}_k$ completes the proof; every threshold in this minimum is computed from $g$ and $\delta$ before $\tilde{g}$ is chosen. □

Finally, we assemble the previous lemmas to prove Proposition 6.

*Proof of Proposition 6.* Fix $\delta > 0$ and an open $B \subseteq \Delta\Theta$ with $\Delta([\theta^* - \delta, \theta^* + \delta] \cap \Theta) \subseteq B$, and let $\bar{\varepsilon}$ be the threshold of Lemma 19, so that $\varepsilon \le \bar{\varepsilon}$ implies $\varepsilon \le c/2$ and the scheme $R^0 \supset \cdots \supset R^K = [\theta^* - \delta, \theta^* + \delta] \cap \Theta$ exists with its witnesses. By Remark 14, likelihood ratios are bounded and likelihoods are equicontinuous on $\Delta\Theta$ in the perturbed environment. The witnesses of Lemma 19

hold for every $\mu \in M^k$ and every selection, with $\kappa_w(\delta)$ and $M^k$ fixed independently of the process, so they apply at every history of the belief process under every $\mathbb{P}_{\mu_0}$; the conclusions below therefore hold for every adapted selection and hence for every $\gamma_t \in \tilde{\Gamma}_t(\mu_0, q_{\theta^*}(\cdot))$. The stages of Lemma 19, extended constantly by $R^k = R^K$ for $k > K$, form an elimination scheme in the sense of Corollary 2 with $\mathscr{P} = \Delta\Theta$, closed and trivially invariant: every $\theta \in R^{k-1} \setminus R^k$ admits the witness $(\varphi(\theta), \kappa_w(\delta), M^{k-1})$, where $\varphi(\theta)$ is the stage dominator, $\varphi(\theta) \in R^k \subseteq R^{k-1}$, $M^{k-1} \supseteq \Delta R^{k-1}$, and $\varphi(\theta) \succsim_{M^{k-1}}^{KL(\kappa_w(\delta))} \theta$. The scheme extended constantly has $\bigcap_k R^k = R^K$, so Corollary 2(i) gives, for every $\mu_0 \in \Delta\Theta$,

$$\mathbb{P}_{\mu_0}\left(\inf_{\nu \in \Delta R^K} d(\mu_t, \nu) \to 0\right) = 1.$$

$\Delta R^K$ is compact, $\Delta R^K \subseteq \Delta([\theta^* - \delta, \theta^* + \delta] \cap \Theta) \subseteq B$, and $B$ is weakly open, so the distance from $\Delta R^K$ to the complement of $B$ is positive, and every belief within that distance of $\Delta R^K$ lies in $B$. Therefore $\mathbb{1}\{\mu_t \in B\} \to 1$ almost surely, and

$$\lim_{t \to \infty} \gamma_t(B \mid \mu_0, q_{\theta^*}(\cdot)) = 1 \quad \forall\, \gamma_t \in \tilde{\Gamma}_t(\mu_0, q_{\theta^*}(\cdot))$$

for every $\mu_0 \in \Delta\Theta$. □

# 8 Appendix B: Simulations

## 8.1 Two-Value Inequalities

Notation as in Section 4. For any given $p$, Border's condition for the pairs $(h_1, h_2)$ consists of

$$ph_1 + (1-p)h_2 \le \frac{1}{2}, \qquad h_2 \le \frac{1+p}{2}, \qquad h_1 \le 1 - \frac{p}{2},$$

obtained by taking $\hat{B} = \{b_1, b_2\}$, $\{b_2\}$, and $\{b_1\}$ in Condition 2. The Fourier-Motzkin elimination of the four mechanism variables under (F'), (SC'), (CF-O'), and (MON') yields

$$h_1 \le \frac{1}{2}, \qquad h_2 \le p\tilde{h}_2 + \frac{1-p}{2},$$
$$ph_1 + (1-p)h_2 \le \frac{1}{2}, \qquad ph_1 + (1-p)h_2 \le p^2\tilde{h}_1 + \frac{1-p^2}{2}.$$

Of the four, the third coincides with the full-set Border inequality, the first follows from it for admissible pairs ($h_1 < h_2$), the second is the cap on $h_2$ described in Section 4, and the fourth is tighter than the third exactly when $\tilde{h}_1 < 1/2$.

## 8.2 The Two Instances

Both instances of Section 4 are value-independent, so the model is the function $h(b, \theta)$ alone, and both are specified by closed forms rather than by numerical construction. We give each in the coordinates in which it is closed, together with the pinned values and the recorded verification margins. Throughout, $m(b, \theta) = h(b, \theta)/h_b(b, \theta)$ is the markup implied by the model, so that the buyer's first-order condition reads $v - b = m(b, \theta)$, and the belief enters the formulas through its offset from the target.

**The first instance.** The parameters are the value pair and the pins, $(v_1, v_2, h_1, h_2) = (10, 11, 0.43, 0.51)$, together with the shape dials $(r, \tau, \sigma, \varepsilon_1) = (0.1,\ 0,\ 1.75,\ 0.044049)$, on the belief space $[0.10,\ 0.80]$ with $\theta^* = 0.70$. The last of these is fixed by the requirement that the responsiveness at $b_2$ equal it identically, so it is an input to the build rather than a free parameter. Everything else is arithmetic on these. Writing $M_1$ for the markup at the lower pin and $\Delta$ for the bid gap,

$$M_1 = \frac{r\,(v_2 - v_1)}{(1+r)\left[(h_2/h_1)^r - 1\right]}, \qquad b_1 = v_1 - M_1, \qquad \Delta = \frac{v_2 - v_1}{1+r}, \qquad b_2 = b_1 + \Delta,$$
$$m_1(\theta^* - \theta) = M_1 + \sigma\,(\theta^* - \theta), \quad R(\theta^* - \theta) = r + \tau\,(\theta^* - \theta), \qquad m(b, \theta) = m_1 + R\,(b - b_1),$$

and the model itself is

$$h(b, \theta) \;=\; h_2\, e^{\varepsilon_1 (\theta^* - \theta)} \left[\frac{m(b, \theta)}{m(b_2, \theta)}\right]^{1/R(\theta^* - \theta)} . \tag{9}$$

This gives $M_1 = 5.2827$, grid bids $b_1 = 4.7173$ and $b_2 = 5.6264$, and the pins $h(b_1, \theta^*) = 0.43$ and $h(b_2, \theta^*) = 0.51$ by construction, the first through the definition of $M_1$ and the second by inspection. The overlap belief, at which the high type's bid falls to $b_1$, is $\tilde{\theta} = 0.1286$, and the subjective win probabilities there are $\tilde{h}_1 = 0.4530$ and $\tilde{h}_2 = 0.5230$. The program of Proposition 4 has a strictly feasible solution at these values, with a slack of 0.0015 in its tightest constraint. Over the belief space $h$ ranges in $[0.3525,\ 0.5368]$, so it is interior. It is increasing in $b$ and decreasing in $\theta$ throughout, and the recorded minima of the responsiveness and fading margins are 0.0338 and 0.1551. Against the build code, (9) agrees to $1.2 \times 10^{-7}$, which is that code's own quadrature error, and both pins reproduce to machine precision.

**The second instance.** Here the parameters are $(p, r, \kappa, \varepsilon, \tilde{d}, \gamma, M_1, b_1) = (1/2,\ 0.22,\ 15,\ 0.001,\ 0.3,\ 1,\ 1,\ 2)$, on the belief space $[0.20,\ 0.70]$, and the values are constructed rather than chosen, the instance being built so that the corner grid is self-confirming. With $\rho = (1+p)/p$,

$$M_2 = M_1 \rho^r, \qquad \Delta = (M_2 - M_1)/r, \qquad b_2 = b_1 + \Delta, \qquad v_1 = b_1 + M_1, \qquad v_2 = b_2 + M_2,$$

and the markup along the lower gridline follows the belief path $m_1(\theta^* - \theta) = M_1\big[1 + (\rho^r - 1)\frac{1+r}{r}\big((\theta^* - \theta)/\tilde{d}\big)^{\gamma}\big]$, with $m_2(\theta^* - \theta) = m_1(\theta^* - \theta) + r\Delta$ along the upper one. The model is explicit on the whole bid line,

$$h(b, \theta) = \frac{p}{2}\left[\frac{m_1(\theta^* - \theta)\, M_2}{M_1\, m_2(\theta^* - \theta)}\right]^{\frac{1+\varepsilon}{r}} \times \begin{cases} \left(1 + \dfrac{r\,(b - b_1)}{m_1(\theta^* - \theta)}\right)^{1/r}, & b_1 \le b \le b_2, \\[2ex] \exp\!\left[\dfrac{1 - e^{-\kappa(b - b_1)}}{\kappa\, m_1(\theta^* - \theta)}\right], & b < b_1, \end{cases} \tag{10}$$

the two branches integrating $h_b/h = 1/m$ above and below $b_1$ and the prefactor carrying the anchor. This gives $M_2 = 1.2734$, $\Delta = 1.2428$, grid bids $b_1 = 2$ and $b_2 = 3.2428$, and values $v_1 = 3$ and $v_2 = 4.5162$. The pins are the Border-corner win probabilities of the symmetric auction, the low type winning only the split tie against a low opponent and the high type beating low opponents and splitting ties with high ones, so $h(b_1, \theta^*) = p/2 = 0.25$ and $h(b_2, \theta^*) = (1+p)/2 = 0.75$. The overlap belief is $\tilde{\theta} = 0.40$ and the subjective win probabilities there are $\tilde{h}_1 = 0.4696$ and

$\tilde{h}_2 = 0.7505$. At the overlap belief the opponent's high type bids $b_1$ and his low type bids below it, so a mechanism non-increasing in the opponent's bid wins at least $\hat{q}_{j1}$ at own bid $b_j$, while corrective feedback requires at most $\tilde{h}_j$. The requirement therefore fails at both bids, since $\hat{q}_{11} = 1/2 > \tilde{h}_1$ and $\hat{q}_{21} = 1 > \tilde{h}_2$. The program of Proposition 4 has no solution at these values even with its inequalities weakened. Feasibility also binds exactly here, since $\hat{q}_{12} + \hat{q}_{21} = 1$, placing the mechanism outside the strict feasibility that Proposition 4 requires. That requirement supports the extension of stability to beliefs just above $\theta^*$, which is vacuous at this instance because $\theta^*$ is the top of the belief space. Over the belief space $h$ ranges in $[0.2134, 0.7506]$, and it is increasing in $b$ and decreasing in $\theta$ throughout. Against the build machinery, (10) agrees to a maximum relative deviation of $8.5 \times 10^{-10}$.

## 8.3 Simulation Protocol

The runs reported in Section 4 share a common structure. Beliefs are carried as a distribution on a finite grid over the belief space. In each period every buyer's valuation is drawn, each bids the maximiser of $(v - b)$ times the win probability his model predicts averaged under his current posterior, the allocation is settled against the mechanism, and each posterior is multiplied by the model's predicted win probability at his own bid if he won and by its complement if he lost, then renormalised. Outcomes are therefore generated by the mechanism and likelihoods by the model, which is where the misspecification enters. At two buyers a single uniform draw settles each pair, which is admissible because every extended mechanism satisfies $\hat{q}(b, b') + \hat{q}(b', b) \leq 1$.

**The first instance.** The bid grid carries 241 nodes and the belief grid 101 over $[0.10, 0.80]$. Sixty mechanisms are drawn uniformly from the constraint set of $LP(\eta_1, \eta_2)$ with the (CF-O′) constraints dropped, at $\eta_1 = 0.002$ and with (MON′) imposed, and each is extended to all bids by $(CE)_\delta$, (PS) and (TS) at $\delta = 0.05$, reduced for nine draws to the largest value at which the extended allocation remains feasible. Each is simulated for 125,000 periods from two priors, Gaussian on the grid with standard deviation 0.005 centred at 0.120 and at 0.148, with four seeds each, and counts as failing if beliefs fail to converge from any one of those prior and seed pairs. A prior at or above a mechanism's own shifted overlap belief is not run for that mechanism.

**The second instance.** The bid grid carries 4802 nodes and the belief grid 601 over $[0.20, 0.70]$. The single feasible self-confirming grid mechanism is extended by $(CE)_0$ and (TS) and simulated for 150,000 periods with eight seeds, from Gaussian priors of standard deviation 0.02 centred at 0.340, 0.370, 0.380, 0.390, 0.410 and 0.500 and from the uniform prior. The single-buyer runs use the same horizon and grids at two seeds per prior, with the mechanism $q(b) = h(b, \theta^*)$ of Proposition 2. Converging chains settle at 0.6990 rather than exactly at $\theta^*$ under two buyers and under one, owing to the finite horizon and the posterior's residual spread.

The variant of the second instance's closed form discussed in Appendix 8.4 takes $\kappa = 1$ in place of 15 in (10), all else unchanged. Its grid values and the subjective win probabilities at the overlap belief are those of the second instance, so (CF-O) fails at the grid by the same amounts, and it satisfies (R1) to (R3), adverse ordering and fading sensitivity on the realised set, with recorded minimum margins of $4.7 \times 10^{-4}$ for the responsiveness and 0.132 for fading. Its low type bids 1.490 at the overlap belief against 1.942 under the second instance. It is run under the same

extension, horizon and seeds as the two-buyer runs of the second instance, on a bid grid of 1,201 nodes and a belief grid of 151 nodes, from the same six priors and the uniform prior, and Figure 7 reports the runs.

**Classifying a run.** A run that has left its trapped zone by the horizon has converged. A run still inside the zone is harder to call, since at these horizons many posteriors have moved very little, and the classification must not depend on the sign of that movement. We therefore count a run as converging if it has left the zone or if its posterior mean has moved at least 0.0045 from its prior, and as failing otherwise.

The threshold is calibrated rather than chosen. Taking fifteen mechanisms spanning the sampled range and running all their prior and seed pairs to 200,000 periods, 79 of the 120 runs leave their zone and the remaining 41 are still at their priors, with no run finishing between the two groups. Against that outcome, a displacement threshold of 0.0045 evaluated at 125,000 periods misclassifies two of the 120, both of them runs that do eventually leave the zone, and none in the opposite direction. Terminal displacement, maximum displacement and the mean over the final fifth of the run all perform alike. The horizon matters more than the statistic, the same rule misclassifying six of 120 when evaluated at 50,000 periods and two at 125,000.

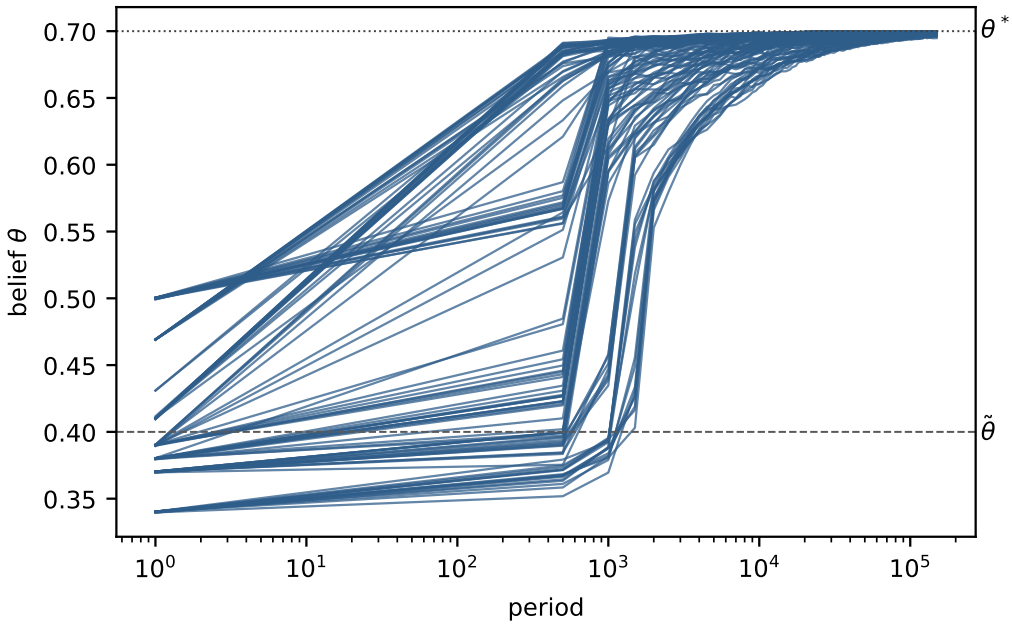



Figure 7: The variant of the second instance with a slower collapse of the markup below $b_1$, from the six priors of Figure 4 and the uniform prior, eight seed pairs each. Beliefs converge from every prior, including the three that trap under the second instance. Time is on a logarithmic scale.

## 8.4 Two Features of the Runs

Figure 8 places the second instance in the plane of the two buyers' beliefs. Of the five trajectories shown, the four with at least one belief above the overlap belief reach the target and the one with

both below it is trapped, and the field of arrows says the same throughout the plane. Against an opponent whose belief lies above the overlap belief, the opponent's two bids bracket $b_1$, and because the mechanism used in the runs is self-confirming at every bid, the buyer then wins with probability $h(b, \theta^*)$ whatever his own belief. He is then placed much as the single buyer of Section 4.2 is, and with $h$ decreasing in $\theta$ his prediction errors point toward the target. A buyer faces positive objective responsiveness only once his opponent's belief falls below the overlap belief, and by the same token a buyer whose own belief stays above it creates none for his opponent. The boundary in the figure therefore runs along each buyer's own overlap belief.

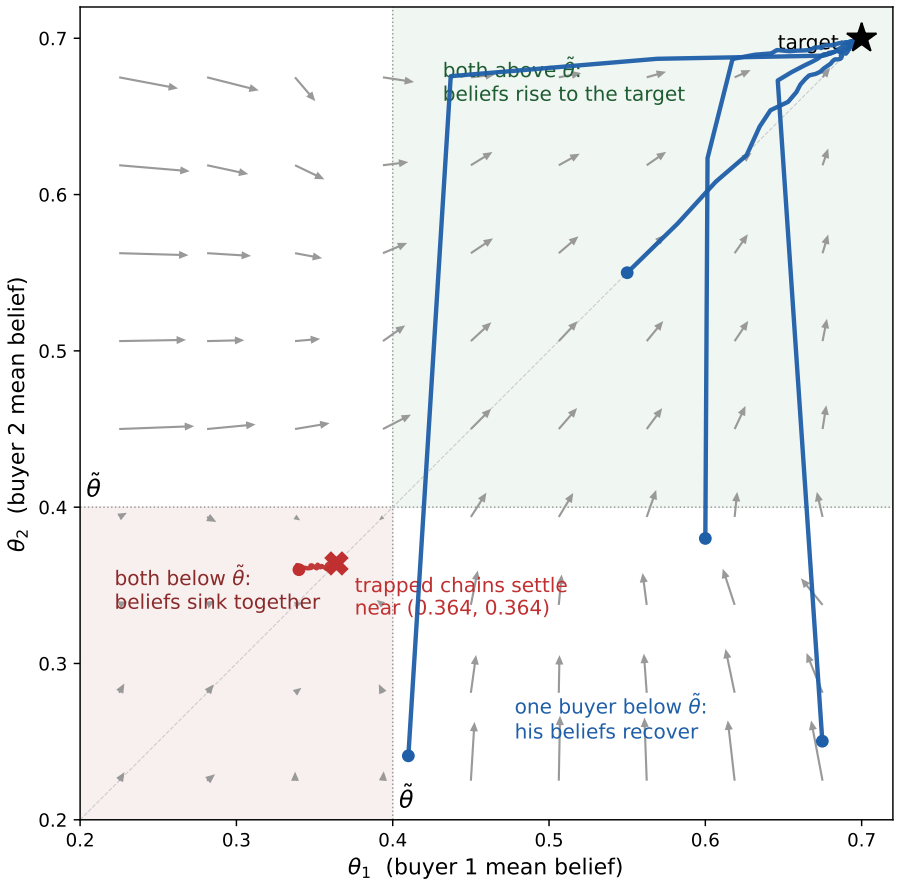



Figure 8: The second instance in the plane of the two buyers' mean beliefs. Each arrow is the simulated change in the two mean beliefs over 150 periods from priors concentrated at its base, averaged over 24 seed pairs, with length scaled by the square root of the displacement. The five paths are seed-mean trajectories over 60,000 periods from the marked starts, eight seed pairs each. Dotted lines mark the overlap belief $\tilde{\theta}$ on each axis. Beliefs reach the target from every start with at least one belief above $\tilde{\theta}$ and settle near $(0.364, 0.364)$ from the start with both below it.

The same geometry explains why the outcome at the first instance is ordered by $\hat{q}_{11}$ alone, although every draw also fails to generate corrective feedback at $b_2$. A buyer bids above $b_1$ only while his belief lies above the overlap belief, so the failure at $b_2$ matters only against an opponent whose belief lies below it, and such an opponent recovers. At the first instance the relevant bound-

ary is more precisely the shifted overlap belief of Appendix 7.3.3, which the construction places slightly above the overlap belief. The argument relies on there being a single overlap belief, as there is with two valuations.

The closed form of the second instance is built so that beliefs are trapped, and the way the trapped chains settle says what that takes. At the grid, corrective feedback fails at $b_2$ on the grid values alone. At $b_1$ it fails only if the model's prediction there is still below $\hat{q}_{11} = 1/2$ at the overlap belief. Below the grid, where a trapped buyer bids, the failure has to persist, and fading sensitivity works against that, since it makes subjective responsiveness larger at lower bids. The model meets these demands in two ways. It holds the buyer's bids close to $b_1$ as his belief falls, so the failure established at $b_1$ still governs the bids he makes, and it keeps his subjective responsiveness at those bids small, so that the win probability $h(b, \theta^*)/p$ he faces below the grid reads as optimism over a long stretch of belief. Two buyers whose beliefs both lie in that stretch each read their wins as evidence of a lower $\theta$ still, with $h$ decreasing in $\theta$, and the stretch stands between the level at which they settle and the target as a wall beliefs do not climb.

The wall holds because both types bid inside it. Were corrective feedback to fail only on a sliver of bids below $b_1$, or were the failure to give way quickly as the bid falls, the low type, who bids below the high type, could bid past the failing bids into the range where fading sensitivity makes his subjective responsiveness large, and his evidence would be corrective. Beliefs then climb through the wall if his corrective pull outweighs the high type's push the other way. The variant of Appendix 8.3 keeps every grid value and every maintained condition, but its low type bids far below $b_1$ at the overlap belief, and it converges from every prior of Figure 4, as Figure 7 shows.

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